Ordinal Googology

The mathematics behind the strength of googological systems

hemidemisemipresent

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0 Contents

1 Introduction

When I was learning Googology, I was never really interested in naming specific large numbers, but instead learning about *how* people generated large numbers. I tried understanding notation like B.E.A.F. and BAN but the rules were complex and they felt too clunky and inelegant to deal with. I then (re-)discovered about the Fast-Growing Hierarchy and how it can systematically categorize the growth of sequences, and it let me on a rabbit hole to finally be able to grasp the growth rate of $\mathrm{TREE}(n)$. However, when I tried digging into the formal definitions of the things I was writing, I realized that there was a lot that I wasn't sure of myself. If I want to compute something like $f_{\Gamma_0}(3)$, what exactly am I supposed to do? There's a Geometry Dash level where I have to click $f_{\psi_0(\Omega_\omega)}(520)$ times, what does that mean? So I tried learning Googology more formally with a more solid math foundation, and started taking down notes. Eventually, I compiled these notes into something resembling a book, though many of the later chapters were much more of a work-in-progress. Generally, I introduce new concepts more informally, then go back around to formalize them, though I feel in some sections I might have frontloaded the formality too much.

Q: Why write out so much when you can contribute to the wiki?
A: I don't trust myself to give accurate information, I have already went back to fix errors in earlier chapters many times when learning more. Also because I can't go back to latex after using Typst omfg its so much easier to do everything. That and wikis in general suffer from "Wikipedia Math" problem where it may serve as a useful reference if you already know the thing, but it's terrible for learning, so I'm writing this more explicitly as a ladder to go from power towers to the Buchholz Ordinal.

Feel free to contribute! Like correcting any mistakes or re-working some sections. The best way is probably a pull request on github, but you can just ping/dm me on discord `@hemidemisemipresent`.

References:
- Googology wiki(s)
- The googology discord (too many random scraps of messages)
- An Introduction to Proof Theory: Normalization, Cut-elimination, and Consistency Proofs by Paolo Mancosu, Richard Zach, and Sergio Galvan
- Zongshu Wu PrSS document on google drive
- Buchholz, W. (1986). "A new system of proof-theoretic ordinal functions". Annals of Pure and Applied Logic. 32: 195–207. doi:10.1016/0168-0072(86)90052-7.
- Patcail's first answer on a codegolf challenge

1.1 Prerequisites

- Set theory
  - Notations $\cup, \cap, \in, \subset, \setminus, \emptyset, \times$ (cartesian product)
  - Notation $\{x \mid P(x)\}$ where $P(x)$ is a statement about x
  - Set vs class
- The Natural Numbers $\mathbb{N}$ (in this book we include $0 \in \mathbb{N}$)
- Knowledge on functions

2 Hyperoperators

2.1 Addition, Multiplication, and Exponentiation

Before we get to large numbers with powerful operations, let's first start from concepts we are more familiar with. Let's first start with the very basic **Successor Function**.

Definition 2.1.

The **Successor Function** is the function $S(x) =$ the least natural number greater than x, or more commonly, $S(x) = x + 1$

In this case, **Addition** is actually just the Successor Function applied repeatedly:

Definition 2.2.

Addition can be defined recursively as:

1. $a + 0 = a$
2. $a + S(b) = S(a + b)$

Alternatively, it can also be seen as the Successor Function applied repeatedly:

$a + b = \underbrace{S(S(...S(a)))}_{S() \text{ applied } b \text{ dot}} = a + \underbrace{1 + 1 + ... + 1}_{b \text{ number of } +1}$

Example 2.3.

Evaluate $2 + 3$ with the recursive definition:

$$
\begin{aligned}
&2 + 0 = 2 \\
&2 + 1 = 2 + S(0) = S(2 + 0 = 2) = 3 \\
&2 + 2 = 2 + S(1) = S(2 + 1 = 3) = 4 \\
&2 + 3 = 2 + S(2) = S(2 + 2 = 4) = 5
\end{aligned}
$$

Note that each line requires the previous $2 + n$ to be known before the Successor Function $S(x)$ can be applied onto it

We next move onto **Multiplication**, which, as you have learnt in school, is repeated addition:

Definition 2.4.

Multiplication can be defined recursively as:

1. $a \cdot 0 = 0$
2. $a \cdot (b + 1) = (a \cdot b) + a$

or alternatively, as just repeated addition:

$a \cdot b = \underbrace{a + a + ... + a}_{a \text{ is here } b \text{ times}}$

Example 2.5.

Evaluate $2 \cdot 3$

- With the recursive definition:

$$
\begin{aligned}
2 \cdot 0 &= 0 \\
2 \cdot 1 &= (2 \cdot 0) + 2 = 2 \\
2 \cdot 2 &= (2 \cdot 1) + 2 = 4 \\
2 \cdot 3 &= (2 \cdot 2) + 2 = 6
\end{aligned}
$$

- or just applying addition repeatedly:

$$2 \cdot 3 = \underbrace{2 + 2 + 2}_{2 \text{ is here } 3 \text{ times}} = 6$$

Now let's move on to **Exponentiation**, which is repeated multiplication:

Definition 2.6.

Exponentiation can be defined recursively as such:

$$a^b := \begin{cases} 1 & (b = 0) \\ a^{b-1} \cdot a & (b > 0) \end{cases}$$

or alternatively, as just repeated multiplication: $a^b = \underbrace{a \cdot a \cdot a \cdot \ldots \cdot a}_{a \text{ is here } b \text{ times}}$

Exponentiation may also be denoted using Donald Knuth's **up-arrow notation** as $a \uparrow b = a^b$

Example 2.7.

Evaluate 2^3

- With the recursive definition:

$$
\begin{aligned}
2^0 &= 1 \\
2^1 &= (2^0) \cdot 2 = 2 \\
2^2 &= (2^1) \cdot 2 = 4 \\
2^3 &= (2^2) \cdot 2 = 8
\end{aligned}
$$

- or just applying multiplication repeatedly:

$$2 \cdot 3 = \underbrace{2 \cdot 2 \cdot 2}_{2 \text{ is here } 3 \text{ times}} = 8$$

We can start to see a pattern here, where each new, more powerful operator is defined as repeatedly applying the previous operator. So what happens when we repeatedly do this for exponentiation?

2.2 Tetration and Beyond

This is where we start venturing beyond the realm of useful numbers and into the realm of googology:

Definition 2.8.

Tetration is defined as repeated exponentiation. This can be expressed recursively:

$$
\begin{aligned}
a \uparrow\uparrow 1 &= a \\
a \uparrow\uparrow b &= a^{a \uparrow\uparrow (b-1)} = a \uparrow (a \uparrow\uparrow (b-1))
\end{aligned}
$$

or simply as repeated exponentiation, but computed from right to left:

$$a \uparrow\uparrow b = \underbrace{a^{a^{a^{\cdot^{\cdot^{\cdot}}}}}}_{a \text{ is here } b \text{ times}} = \underbrace{a \uparrow a \uparrow \ldots \uparrow a}_{a \text{ is here } b \text{ times}}$$

Example 2.9.

Evaluate $2 \uparrow\uparrow 3$

- With the recursive definition:

$$
\begin{aligned}
2 \uparrow\uparrow 1 &= 2 \\
2 \uparrow\uparrow 2 &= 2 \uparrow (2 \uparrow\uparrow 1) = 2^2 = 4 \\
2 \uparrow\uparrow 3 &= 2 \uparrow (2 \uparrow\uparrow 2) = 2^4 = 16
\end{aligned}
$$

- or just applying exponentiation repeatedly:

$$2 \uparrow\uparrow 3 = \underbrace{2 \uparrow 2 \uparrow 2}_{2 \text{ is here } 3 \text{ times}} = 2^{2^2} = 2^{(2^2)} = 2^4 = 16$$

This might not seem like a lot, but try something like $10 \uparrow\uparrow 10$:

$$10 \uparrow\uparrow 10 = 10 \uparrow 10 \uparrow 10 \uparrow 10 \uparrow 10 \uparrow 10 \uparrow 10 \uparrow 10 \uparrow 10 \uparrow 10 = 10^{10^{10^{10^{10^{10^{10^{10^{10^{10}}}}}}}}}$$

We can similarly define **Pentation**:

Definition 2.10.

Pentation is repeated Tetrations like so:

$$a \uparrow\uparrow\uparrow b = \underbrace{a \uparrow\uparrow a \uparrow\uparrow \dots \uparrow\uparrow a}_{a \text{ is here } b \text{ times}}$$

Example 2.11.

$$
\begin{aligned}
2 \uparrow\uparrow\uparrow 3 = &\underbrace{2 \uparrow\uparrow 2 \uparrow\uparrow 2}_{2 \text{ is here } 3 \text{ times}} \\
&= 2 \uparrow\uparrow (2 \uparrow\uparrow 2) \\
&= 2 \uparrow\uparrow (2^2) \\
&= 2 \uparrow\uparrow 4 \\
&= 2 \uparrow 2 \uparrow 2 \uparrow 2 \\
&= 2 \uparrow 2 \uparrow (2^2) \\
&= 2 \uparrow 2 \uparrow 4 \\
&= 2 \uparrow (2^4) \\
&= 2 \uparrow 16 \\
&= 65536
\end{aligned}
$$

Now, 65536 might not look like much, but when you put in slightly larger numbers like $3 \uparrow\uparrow\uparrow 3$, you will get monstrous results:

Example 2.12.

$$
\begin{aligned}
3 \uparrow\uparrow\uparrow 3 &= 3 \uparrow\uparrow 3 \uparrow\uparrow 3 \\
&= 3 \uparrow\uparrow \left(3^{3^3}\right) \\
&= 3 \uparrow\uparrow (\sim 7.63 \times 10^{12}) \\
&= \underbrace{3 \uparrow 3 \uparrow \dots \uparrow 3}_{\sim 7.63 \times 10^{12} \text{ arrows}}
\end{aligned}
$$

We can generalize the definition of increasingly powerful hyperoperators as such:

Definition 2.13.

Let $a \uparrow^n b = a \underbrace{\uparrow\uparrow \dots \uparrow}_{n} b$

$a \uparrow^{n+1} b = \underbrace{a \uparrow^n a \uparrow^n \dots \uparrow^n a}_{a \text{ is here } b \text{ times}}$

With up-arrow notation, we can already make enormously big numbers like Graham's number:

Example 2.14.

$g_0 = 4$

$g_{n+1} = 3 \uparrow^{g_n} 3 = 3 \underbrace{\uparrow\uparrow\uparrow \dots \uparrow}_{g_n \text{ times}} 3$

So for example:

$g_1 = 3 \underbrace{\uparrow\uparrow\uparrow\uparrow}_{4} 3$

$g_2 = 3 \underbrace{\uparrow\uparrow\uparrow \dots \uparrow}_{3\uparrow\uparrow\uparrow\uparrow 3 \text{ times}} 3$

Graham's number is defined as g_{64}

This seems like quite the huge number, but Graham's number barely scratches the surface of what googology has to offer.

3 Introduction to Ordinals

3.1 Informal definition

The easiest way to think about ordinals is as an extension of the natural numbers $\mathbb{N} = \{0, 1, 2, 3, ...\}$ (We include 0 in the natural numbers for this book).

Definition 3.1.

We define ω as the **first ordinal larger than all the natural numbers**, or **the first transfinite ordinal**.

We can continue by having an $\omega + 1, \omega + 2, ...$, so we have something like $\{0, 1, 2, ..., \omega, \omega + 1, \omega + 2, ...\}$

We then have $\omega + \omega = \omega \cdot 2, \omega \cdot 3, \omega \cdot 4, ...$ until we get $\omega \cdot \omega = \omega^2, \omega^3, \omega^\omega, \omega^{\omega+1}, \omega^{\omega^\omega}, ...$. Eventually, we reach $\omega^{\omega^{\cdot^{\cdot^{\cdot}}}} = \varepsilon_0$. The special property of ε_0 is that $\omega^{\varepsilon_0} = \varepsilon_0$, as

$$\omega^{\varepsilon_0} = \omega^{\left(\omega^{\omega^{\cdot^{\cdot^{\cdot}}}}\right)} = \omega^{\omega^{\omega^{\cdot^{\cdot^{\cdot}}}}} = \varepsilon_0$$

We note that there are many intermediate ordinals that, while not mentioned explicitly, are still present, like $\omega^{\omega^3} < \omega^{\omega^3 + \omega^2 \cdot 3 + \omega \cdot 4 + 5} + 6 < \omega^{\omega^4}$.

3.2 Von Neumann definition of ordinals

Definition 3.2.

An ordinal is defined as **the set of all ordinals lesser than that ordinal**.

Let's look at some examples of how this plays out:

- There are no ordinals smaller than 0, so $0 = \emptyset = \{\}$, the empty set.
- For the ordinal 1, 0 is the only ordinal smaller than 1, so $1 = \{0\} = \{\emptyset\} = \{\{\}\}$.
- Similarly, we have $2 = \{0, 1\} = \{\emptyset, \{\emptyset\}\}$, $3 = \{0, 1, 2\} = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$, etc...
- In this definition, ω is defined as the set of all ordinals less than ω, i.e. all natural numbers:
$$\omega = \{0, 1, 2, ...\} = \mathbb{N}$$
- $\omega + 1$ can similarly be defined as all the natural numbers + ω:
$$\omega + 1 = \{0, 1, 2, ..., \omega\} = \mathbb{N} \cup \{\omega\} = \omega \cup \{\omega\}$$
- $\omega \cdot 2 = \omega + \omega$ can be defined as the ordinal larger than all $\omega + n$:
$$\omega \cdot 2 = \{0, 1, 2, ..., \omega, \omega + 1, \omega + 2, ...\}$$

We can therefore see from this that **for any set of ordinals**, there exists another ordinal greater than all of them. Now, this can lead to a contradiction - what about the set of all ordinals? (this is Burali-Forti paradox). In modern set theory, however, it is not possible to construct "the set of all ordinals" which resolves this paradox. We only have the ***proper class* of all ordinals**, denoted **Ord**, **On**, or less commonly, **ON**.

We can see from this definition of ordinals that for ordinals α and β, $\alpha < \beta$ can be defined as $\alpha \in \beta$. Since α also contains all ordinals smaller than itself, $\alpha < \beta$ can be written as $\alpha \subsetneq \beta$ ($\alpha \subset \beta$ and $\alpha \neq \beta$)

3.3 Ordinal Arithmetic

We previously defined the successor function $S(n)$ to be the least natural number greater than n, or just $n + 1$. Using the von Neumann definition of ordinals, we can also write the successor function as:

Definition 3.3.

The **successor function** $S(\alpha) = \alpha \cup \{\alpha\} = \alpha + 1$

Definition 3.4.

A **Successor Ordinal** β is an ordinal which can be expressed as the **successor of another ordinal**.

A **Limit Ordinal** is an ordinal that is **not a successor ordinal**.

Example 3.5.

Categorize the following ordinals into whether they are limit or successor ordinals:

1. ω
2. $\omega + 5$
3. ω^2

Explanations:

1. ω is bigger than all natural numbers, so there is no natural number whose successor is ω. Therefore ω is a **limit ordinal**.
2. $\omega + 5$ is the successor of $\omega + 4$. Therefore $\omega + 5$ is a **successor ordinal**.
3. ω^2 is the ordinal bigger than all $\omega \cdot n$. Since there is no greatest $\omega \cdot n$, ω^2 can't be a successor to any of them. Therefore ω^2 is a **limit ordinal**.

We previously defined addition recursively here. Let's add a new case to accommodate limit ordinals:

Definition 3.6.

Addition can be defined recursively as:

1. $\alpha + 0 = a$
2. $\alpha + S(\beta) = S(\alpha + \beta)$
3. $\alpha + \beta = \bigcup_{\gamma < \beta} (\alpha + \gamma)$ when β is a limit ordinal

Example 3.7.

Evaluate $\omega + 1$ and $1 + \omega$. Are they the same?

$$\begin{aligned} \omega + 1 &= \omega + S(0) \\ &= S(\omega + 0) \ \text{(Rule 2)} \\ &= S(\omega) \ \text{(Rule 1)} \\ &= \omega + 1 \\ 1 + \omega &= \bigcup_{\delta < \omega} (1 + \delta) \ \text{(Rule 3)} \\ &= 1 \cup 2 \cup 3 \cup \ldots \\ &= \{0\} \cup \{0, 1\} \cup \{0, 1, 2\} \cup \ldots \\ &= \{0, 1, 2, \ldots\} \\ &= \omega \end{aligned}$$

We see that addition is not commutative (i.e. $a + b$ is not always equal to $b + a$)

We can similarly accommodate limit ordinals into our definition of multiplication and exponentiation too:

Definition 3.8.

Multiplication can be defined recursively as:

1. $\alpha \cdot 0 = 0$
2. $\alpha \cdot S(\beta) = (\alpha \cdot \beta) + \beta$
3. $\alpha \cdot \beta = \bigcup_{\gamma < \beta} (\alpha \cdot \gamma)$ when β is a limit ordinal

Example 3.9.

Evaluate $\omega \cdot 2$ and $2 \cdot \omega$

$$\begin{aligned} \omega \cdot 2 &= \omega \cdot S(1) \\ &= (\omega \cdot 1) + \omega \\ &= ((\omega \cdot 0) + \omega) + \omega \\ &= \omega + \omega \end{aligned}$$

$$\begin{aligned} 2 \cdot \omega &= \bigcup_{\gamma < \omega} (2 \cdot \gamma) \\ &= 0 \cup 2 \cup 4 \cup \ldots \\ &= \{\} \cup \{0, 1\} \cup \{0, 1, 2, 3\} \cup \ldots \\ &= \{0, 1, 2, 3, \ldots\} \\ &= \omega \end{aligned}$$

We see that multiplication is not commutative either (i.e. $a \cdot b$ is not always equal to $b \cdot a$)

Multiplication can also be denoted with the $\times$ symbol (e.g. $\omega \times 2$), or even without any symbol, like in algebra (e.g. $\omega 2$)

Definition 3.10.

Exponentiation can be defined recursively as:

1. $\alpha^0 = 1$
2. $\alpha^{S(\beta)} = \alpha^\beta \cdot \alpha$
3. $\alpha^\beta = \bigcup_{\gamma < \beta} (\alpha^\gamma)$ when β is a limit ordinal

The usual exponential rules apply, e.g. $\alpha^{\beta + \gamma} = \alpha^\beta \cdot \alpha^\gamma$, and $\alpha^{\beta \cdot \gamma} = \left(\alpha^\beta\right)^\gamma$.

Distributivity holds on the left, but not on the right:

$$\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$$

$$(\alpha + \beta) \cdot \gamma \neq \alpha \cdot \gamma + \beta \cdot \gamma$$

As such we can also “cancel” terms on the left, but no on the right:

$$\alpha \cdot \beta = \alpha \cdot \gamma \Rightarrow \beta = \gamma$$

$$\beta \cdot \alpha = \gamma \cdot \alpha \nRightarrow \beta = \gamma$$

If you recall, $1 + \omega = \omega$. We can show a more general version of this:

Proposition 3.11.

If $\alpha < \beta$, $\omega^\alpha + \omega^\beta = \omega^\beta$

Proof. Let $\beta = \alpha + \gamma$ for some ordinal γ. Then:

$$\begin{aligned} \omega^\alpha + \omega^\beta &= \omega^\alpha + \omega^{\alpha+\gamma} \\ &= \omega^\alpha + \omega^\alpha \cdot \omega^\gamma \\ &= \omega^\alpha \cdot (1 + \omega^\gamma) \\ &= \omega^\alpha \cdot \omega^\gamma \\ &= \omega^{\alpha+\gamma} \\ &= \omega^\beta \end{aligned}$$

■

Higher hyperoperators are tricky to define for transfinite ordinals, so we will only allow addition, multiplication and exponentiation on transfinite ordinals.

3.4 Ordinals versus Cardinals

The natural numbers $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ can be used in 2 ways: denoting the **position** of an element in a sequence, or denoting the **size** of a set.

Denoting the position leads to **Ordinal numbers**, while denoting the size leads to **Cardinal numbers**. For finite sets it both are the same, but when it comes to infinite sets and sequences, there are differences.

Definition 3.12.

Two sets are said to have the same **cardinality** if there is a **one-to-one correspondence (bijection)** between them.

Note that since we have defined ordinals as sets, we can measured the cardinality of ordinals.

Example 3.13.

Show that ω and $\omega 2$ have the same cardinality.

We can construct a bijection f from ω to $\omega 2$ as such:

$$\begin{aligned} &0 \to 0 \\ &1 \to \omega \\ &2 \to 1 \\ &3 \to \omega + 1 \\ &\text{and in general:} \\ &2n \to n \\ &2n + 1 \to \omega + n \end{aligned}$$

To be precise, to show that f is a bijection, we need to show that:

1. f is total: $f(n)$ is defined for all $n \in \omega$
2. f is injective: if $n_1 \neq n_2$ then $f(n_1) \neq f(n_2)$
3. f is surjective: for every $\alpha \in \omega 2$ there exists an $n \in \omega$ such that $f(n) = \alpha$

Every natural number is a cardinal. For infinite cardinals, they are labelled as $\aleph_0, \aleph_1, \aleph_2, \ldots$. We set $\aleph_0$ to be the cardinality of ω (and from the above example, this is the cardinality of $\omega 2$ too).

But how do we know that there are cardinalities above $\aleph_0$? We know there are cardinalities larger than $\aleph_0$, as the cardinality of the set of all real numbers $\mathbb{R}$ is $2^{\aleph_0}$, which is greater than $\aleph_0$. But what is $2^{\aleph_0}$

in terms of $\aleph_n$? Is $2^{\aleph_0} = \aleph_1$? (This is known as the Continuum Hypothesis). It turns out that we can set $2^{\aleph_0} = \aleph_n$ for *any* finite n, and it will be consistent with ZFC.

In ZFC, we assign each cardinal to an ordinal via **von Neumann cardinal assignment**:

> **Definition 3.14.**
>
> Von Neumann cardinal assignment defines the **cardinality of a set** as the **smallest ordinal that has a bijection with the set**.

Therefore by this definition, **cardinals are ordinals** and we can assign cardinals to ordinals like this: $\aleph_0 = \omega, \aleph_1 = \omega_1, \aleph_2 = \omega_2, \ldots$.

> **Definition 3.15.**
>
> We say a set, and by extension ordinal, is **countable**, if its cardinality is less than ω_1 (or $\aleph_1$).

4 The Fast-Growing Hierarchy

Definition 4.1.

The **fast-growing hierarchy** (FGH) is defined as follows:

1. $f_0(n) := S(n) = n + 1$
2. $f_{\alpha+1} := f_\alpha^n(n) := \underbrace{f_\alpha(f_\alpha(\ldots f_\alpha(n)))}_{f_\alpha \text{ applied } n \text{ times}}$
3. $f_\alpha(n) := f_{\alpha[n]}(n)$ if α is a limit ordinal.

$\alpha[n]$ in this case represents the nth term in the fixed fundamental sequence assigned to the ordinal α.

Remark 4.2. The fast-growing hierarchy is **ill-defined** unless a **specific choice of a system of fundamental sequences is explicitly fixed** in the context.

Wait, but what is a fundamental sequence? For now, we will ignore that warning as we will won't be dealing with limit ordinals.

4.1 The Fast-Growing Hierarchy for finite ordinals

Before we start plugging in infinite ordinals into the FGH, let's try figuring out the growth rate of the first few f_1, f_2, f_3, f_4:

$$f_1(n) = f_0^n(n) = \underbrace{f_0(f_0(\ldots f_0(n)))}_{\text{applied } n \text{ times}} = n + \underbrace{1 + 1 + \ldots + 1}_{n \text{ times}} = 2n$$

$$f_2(n) = f_1^n(n) = \underbrace{f_1(f_1(\ldots f_1(n)))}_{\text{applied } n \text{ times}} = \underbrace{2 \cdot 2 \cdot \ldots 2}_{n \text{ times}} \cdot n = 2^n \cdot n > 2^n$$

$$f_3(n) = f_2^n(n) = \underbrace{f_2(f_2(\ldots f_2(n)))}_{\text{applied } n \text{ times}} > \underbrace{2 \uparrow 2 \uparrow \ldots \uparrow 2}_{n \text{ times}} \uparrow n$$

$$> \underbrace{2 \uparrow 2 \uparrow \ldots \uparrow 2}_{n \text{ times}} = 2 \uparrow\uparrow n$$

$$f_4(n) = f_3^n(n) = \underbrace{f_3(f_3(\ldots f_3(n)))}_{\text{applied } n \text{ times}} > \underbrace{2 \uparrow\uparrow 2 \uparrow\uparrow \ldots \uparrow\uparrow 2}_{n \text{ times}} \uparrow\uparrow n$$

$$> \underbrace{2 \uparrow\uparrow 2 \uparrow\uparrow \ldots \uparrow\uparrow 2}_{n \text{ times}} = 2 \uparrow\uparrow\uparrow n$$

We see that for every new f_m, we get increasingly powerful hyperoperators:

FGH function	corresponding growth rate
$f_1(n)$	$2n$ (Multiplication-level)
$f_2(n)$	$> 2^n$ (Exponentiation-level)
$f_3(n)$	$> 2 \uparrow\uparrow n$ (Tetration-level)
$f_4(n)$	$> 2 \uparrow\uparrow\uparrow n$ (Pentation-level)
$f_m(n)$	$> 2 \uparrow^{m-1} n$ ($(m-1)$th hyperoperator-level)

4.2 Fast-Growing Hierarchy for ordinals $\leq \varepsilon_0$

Informal Treatment of the Fast-Growing Hierarchy for ω

With the functions $f_m(n)$ we have so far, we can easily make enormous numbers like $f_{69}(420)$. But can we grow faster than that? If we *diagonlize* across all $f_m(n)$, we get $f_n(n)$ in red:

f_0	$f_0(0)$	$f_0(1)$	$f_0(2)$	$f_0(3)$	$f_0(4)$
f_1	$f_1(0)$	$f_1(1)$	$f_1(2)$	$f_1(3)$	$f_1(4)$
f_2	$f_2(0)$	$f_2(1)$	$f_2(2)$	$f_2(3)$	$f_2(4)$
f_3	$f_3(0)$	$f_3(1)$	$f_3(2)$	$f_3(3)$	$f_3(4)$
f_4	$f_4(0)$	$f_4(1)$	$f_4(2)$	$f_4(3)$	$f_4(4)$

This $f_n(n)$ will eventually grow faster than any fixed $f_m(n)$ like $f_{1000}(n)$. We can formalize this as such:

Definition 4.3.

For 2 functions $f, g : \mathbb{N} \mapsto \mathbb{N}$, we say that f **dominates**, or **eventually dominates** g if there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $f(n) > g(n)$

We use $f_\omega(n)$ to denote $f_n(n)$. From this, we can continue counting up:

$f_{\omega+1}(n) = f_\omega^n(n)$

$f_{\omega+2}(n) = f_{\omega+1}^n(n)$

$f_{\omega 2}(n) = f_{\omega+n}(n)$

$f_{\omega 2+1}(n) = f_{\omega 2}^n(n)$

$f_{\omega^2}(n) = f_{\omega\omega}(n) = f_{\omega n}(n)$

One property of the fast-growing hierarchy is that it's *hard to go to the next tier*. For example, let's say we have a function that generates big numbers, $N(100) = f_{\omega^2}(100)$, and we want to try to "see" if we can reach $f_{\omega^3}(100)$ by using the $N()$ function. We can first try something like $N(100)^{N(100)}$, but that barely makes a dent googologoically. After all, exponentiation is at a measly f_2. What about nesting N, like $N(N(N(100)) = N^3(100)$? What if we go $N^{100}(100)$? Well, that is "just" $f_{\omega^2+1}(100) = f_{\omega^2}^{100}(100) = N^{100}(100)$. And to go to "just" $f_{\omega^2+2}(100)$, we have:

$$\begin{aligned}
f_{\omega^2+2}(100) &= f_{\omega^2+1}^{100}(100) \\
&= f_{\omega^2+1}^{99}(f_{\omega^2+1}(100)) \\
&= f_{\omega^2+1}^{99}(N^{100}(100)) \\
&= f_{\omega^2+1}^{98}(f_{\omega^2+1}(N^{100}(100))) \\
&= f_{\omega^2+1}^{98}(N^{N^{100}(100)}(N^{100}(100)) \\
&> f_{\omega^2+1}^{98}\left(N^{N^{100}(100)}(100)\right) \\
&> f_{\omega^2+1}^{97}\left(N^{N^{N^{100}(100)}(100)}(100)\right)
\end{aligned}$$

We can sort of see how $N^{N^{\cdot^{\cdot^{N^{100}(100)}\cdot}\cdot}(100)}(100)$ with 100 layers, is a lower bound of $f_{\omega^2+2}(100)$. Beyond that, it starts to get very tedious to even try to express $f_{\omega^2+3}(100)$, let alone $f_{\omega^2+\omega}(100)$. And ω^2 is honestly still quite a small ordinal! We can easily come up with a bigger ordinal to plug into the fast-growing hierarchy, like ω^ω. Our resulting number, $f_{\omega^\omega}(100)$, essentially cannot be expressed in terms of $N(100)$ at all (at least, not without using the fast-growing hierarchy).

Formal Treatment of the Fast-Growing Hierarchy for ordinals $\leq \varepsilon_0$

In the warning earlier, it specified that we need to specify a system of fundamental sequences for ordinals in order for FGH to be properly defined. Here, we will formally define fundamental sequences for ordinals $\leq \varepsilon_0$.

Before we define that, let's first have a standard way to express ordinals $\leq \varepsilon_0$:

Theorem 4.4 (Cantor Normal Form).

For every ordinal α there is a unique sequence of ordinals $\alpha_0 \geq \alpha_1 \geq \ldots \geq \alpha_n$ such that

$$\alpha = \omega^{\alpha_0} + \omega^{\alpha_1} + \ldots + \omega^{\alpha_n}$$

The basic rules of CNF is that you can only use addition and the "ω power" function $\omega^{\bullet}(\alpha) = \omega^{\alpha}$. Note that what exactly the α_n in ω^{α_n} is written as is not enforced, so $\omega^{1+\omega}$ is technically a valid way to write ω^{ω} in CNF. Another way to illustrate it is that everything in **red** is enforced, while those in **black** are not:

$$\begin{aligned}\omega^{\omega} + \omega^2 \cdot 2 + \omega \cdot 3 &= \omega^{\omega} + \omega^{2} + \omega^{2} + \omega^{1} + \omega^{1} + \omega^{1} \\ &= \omega^{1+\omega} + \omega^{1+1} + \omega^{\omega^0+\omega^0} + \omega^{\omega^0} + \omega^{S(0)} + \omega^{\text{the smallest ordinal greater than 0}}\end{aligned}$$

Now that we have a standardized way to uniquely write each ordinal, we can finally go into specifying fundamental sequences. The usual way to define these fundamental sequences based on Cantor Normal Form is the **Wainer Hierarchy**.

Definition 4.5.

The **Wainer Hierarchy** is a system of fundamental sequences for ordinals $\leq \varepsilon_0$ (expressed in Cantor Normal Form):

1. $0[n] := 0$
2. $(\omega^{\alpha_0} + \omega^{\alpha_1} + \ldots + \omega^{\alpha_n})[n] := \omega^{\alpha_0} + \omega^{\alpha_1} + \ldots + \omega^{\alpha_n}[n]$ (i.e. take the fundamental sequence of the last term, which is the term with the smallest α)
3. $\omega^0[n] = 1[n] := 0$
4. $\omega^{\alpha+1}[n] = \omega^{\alpha} \cdot n$
5. $\omega^{\alpha}[n] = \omega^{\alpha[n]}$ if α is a limit ordinal
6. $\varepsilon_0[0] := 0$, and $\varepsilon_0[n+1] = \omega^{\varepsilon_0[n]}$

Example 4.6.

If we want to find out what $f_{\omega^{\omega^2}+\omega^{\omega}}(3)$ is, we first evaluate $\left(\omega^{\omega^2} + \omega^{\omega}\right)[3]$

$$\begin{aligned}\left(\omega^{\omega^2} + \omega^{\omega}\right)[3] &= \omega^{\omega^2} + \omega^{\omega}[3] \\ &= \omega^{\omega^2} + \omega^{\omega[3]} \\ &= \omega^{\omega^2} + \omega^3\end{aligned}$$

So $f_{\omega^{\omega^2}+\omega^{\omega}}(3) = f_{\omega^{\omega^2}+\omega^3}(3)$, so we still need to evaluate $\left(\omega^{\omega^2} + \omega^3\right)[3]$:

$$\begin{aligned}\left(\omega^{\omega^2} + \omega^3\right)[3] &= \omega^{\omega^2} + \omega^2 \cdot \omega[3] \\ &= \omega^{\omega^2} + \omega^2 \cdot 3\end{aligned}$$

Then we find the fundamental sequence of *that*:

$$\begin{aligned}\omega^{\omega^2} + (\omega^2 \cdot 3)[3] &= \omega^{\omega^2} + \omega^2 \cdot 2 + \omega^2[3] \\ &= \omega^{\omega^2} + \omega^2 \cdot 2 + \omega \cdot \omega[3] \\ &= \omega^{\omega^2} + \omega^2 \cdot 2 + \omega \cdot 3\end{aligned}$$

And repeat again:

$$\begin{aligned}\omega^{\omega^2}+\omega^2\cdot 2+(\omega\cdot 3)[3] &= \omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+\omega[3]\\ &= \omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+3\end{aligned}$$

So $f_{\omega^{\omega^2}+\omega^\omega}(3) = f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+3}(3)$. Then

$$\begin{aligned}&f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+3}(3)\\ &= f^{3}_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}(3)\\ &= f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}\left(f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}\left(f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}(3)\right)\right)\\ &= f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}\left(f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}\left(f^{3}_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+1}(3)\right)\right)\\ &= f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+2}\left(f^{f^3_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+1}(3)}_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+1}\left(f^{3}_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2+1}(3)\right)\right)\end{aligned}$$

And we will stop there. It's quite clear that this grows wayyy out of control that even expressing it in terms of $f_{\omega^{\omega^2}+\omega^2\cdot 2+\omega\cdot 2}$ is basically impossible

Example 4.7.

$$\begin{aligned}\varepsilon_0[0] &= 0\\ \varepsilon_0[1] &= \omega^{\varepsilon_0[0]} = \omega^0 = 1\\ \varepsilon_0[2] &= \omega^{\varepsilon_0[1]} = \omega^1 = \omega\\ \varepsilon_0[3] &= \omega^{\varepsilon_0[2]} = \omega^\omega\\ \varepsilon_0[4] &= \omega^{\varepsilon_0[3]} = \omega^{\omega^\omega}\\ &\dots\end{aligned}$$

So:

$$\begin{aligned}f_{\varepsilon_0}(1) &= f_1(1)\\ f_{\varepsilon_0}(2) &= f_\omega(2)\\ f_{\varepsilon_0}(3) &= f_{\omega^\omega}(3)\\ f_{\varepsilon_0}(4) &= f_{\omega^{\omega^\omega}}(4)\\ &\dots\end{aligned}$$

4.3 Other growing hierarchies

Aside from the fast-growing hierarchy, we have other slower hierarchies, though at sufficiently large ordinals they "catch up" with the fast-growing hierarchy.

Definition 4.8.

Slow-growing hierarchy (SGH):

1. $g_0(n) := 0$
2. $g_{\alpha+1}(n) := g_\alpha(n) + 1$
3. $g_\alpha(n) = g_{\alpha[n]}$ when α is a limit ordinal

Definition 4.9.

Middle-growing hierarchy (MGH):

1. $m_0(n) := n + 1$
2. $m_{\alpha+1}(n) := m_\alpha^2(n) = m_\alpha(m_\alpha(n))$
3. $m_\alpha(n) := m_{\alpha[n]}$ when α is a limit ordinal

Definition 4.10.

Hardy hierarchy (HH):
1. $H_0(n) := n$
2. $H_{\alpha+1}(n) := H_\alpha(n + 1)$
3. $H_\alpha(n) := H_{\alpha[n]}$ when α is a limit ordinal

5 Formal treatment of orders and ordinals

Order types are the most general way to define ordinals. Under this definition, an ordinal is an equivalence class of *well-ordered sets* under the relation of *order isomorphism*. There's a lot to unpack there, so we'll introduce the concepts one at a time:

5.1 Orders and well-orders

Definition 5.1.

A **relation** R on a set X is a set of ordered pairs from X, i.e., $R \subseteq X \times X = \{(x, y) \mid x, y \in X\}$. We write xRy for $(x, y) \in R$

Specifically, such a relation is a *binary relation* or *2-ary relation* as it is a relation between 2 sets. However, we will mainly be using binary relations, so we will just call them relations.

Example 5.2.

The relation "less than" on a set $\{1, 2, 3\}$ can be defined as the following set of ordered pairs from X: $R = \{(x, y) \mid x, y \in X \text{ and } x < y\} = \{(1, 2), (1, 3), (2, 3)\}$.

We see that R is a subset of $X \times X$:

$$\{(1, 2), (1, 3), (2, 3)\} \subset \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

Definition 5.3.

A relation $<$ on X is called a **strict linear order** on X if and only if the following are true for all $x, y, z \in X$:

1. $<$ is transitive: If $x < y$ and $y < z$, then $x < z$
2. $<$ is asymmetric: If $x < y$, then $y \not< x$ (which is $\neg(y < x)$)
3. $<$ is total: If $x \neq y$, then either $x < y$ or $y < x$

For strict linear orders, we'll usually use symbol $<$ and its variants like $\prec$. The *partial linear order* $x \leq y$ means $x = y$ or $x < y$. The symbol $>$ can similarly be defined as $x > y$ if and only if $y < x$.

Definition 5.4.

A relation $<$ is a **well-ordering** on the set X is a relation with a **strict linear ordering** of X such that **every (non-empty) subset of X has a least element under $<$**.

A least element of Y is some $y \in Y$ such that $y \leq z$ for every $z \in Y$, and is unique for a linearly ordered set (if a least element exists).

Definition 5.5.

A **well-ordered set** $(X, <_X)$ is a set X and a relation $<_X$ that is a well-ordering of X

Proposition 5.6.

Suppose $(X, <)$ is a well-ordered set. For every $x \in X$, if it is not the maximum element, then there is a unique $x' > x$ such that there is no $z \in X$ such that $x < z < x'$. (x' here is called the **successor** of x in $(X, <)$)

Proof. Let $Y = \{y \in X \mid y > x\}$, the set of all elements greater than x. Since $<$ is a well-ordering of X, then $Y \subseteq X$ must have a least element x'. Since $x' \in Y$, $x < x'$.

Since x' is the least element in Y, there can't be any $z \in Y$ such that $x < z$ and $z < x'$

To prove x' is unique, if we assuming there exists another number x'' with the same property that there is no z such that $x < z < x''$, we have to show $x' = x''$. Since $x < x''$, $x'' \in Y$. Since x' is the least element in Y, $x' \leq x''$. We can't have $x' < x''$, as it leads to $x < x' < x''$ which violates the property mentioned. Therefore $x' = x''$, and thus x' is unique. ■

Another property of well-ordered set is that they do not contain an infinite descending sequence. This property is needed to do induction along well-ordered sets. It also serves as another way to prove that an ordered set is well-ordered.

> **Proposition 5.7.**
>
> Every well-ordered set does not contain an infinite descending sequence $x_1 > x_2 > \dots$.
>
> Similarly, every strictly linearly ordered set which does not contain an infinite descending sequence is a well-ordered set.

Proof. Suppose X is well-ordered by $<$ but yet there is an infinite descending sequence $x_1 > x_2 > \dots$. We create a subset $Y \subseteq X$ such that $Y = \{x_i \mid i \in \mathbb{N} \text{ and } i \geq 1\}$. Since Y is a nonempty subset of X, it has a least element. But if $y \in Y$ is the least element, then such an infinite sequence would not be able to exist.

Suppose $(X, <)$ is a strictly linearly ordered set which does not contain an infinite descending sequence, but is not a well-ordered set. As it is not a well-ordered set, there exists a subset $P \subseteq X$ such that there is no least element. Therefore, for any $x_i \in P$, we can find an $x_{i+1} \in P$ such that $x_i > x_{i+1}$. We can construct an infinite descending sequence $x_1 > x_2 > x_3 > \dots$, a contradiction. Therefore $(X, <)$ must be a well-ordered set. ■

We have a similar result for **non-increasing** sequences:

> **Proposition 5.8.**
>
> If $(X, <)$ is a well-ordered set, and $x_1 \geq x_2 \geq x_3 \geq \dots$ is a non-increasing infinite sequence, then the sequence eventually stabilizes, i.e. for some k, $x_k = x_i$ for all $i \geq k$.

Proof. Assume the converse, that for all k, $x_k \neq x_{i(k)}$ for some $i(k) > k$.

Since $x_k \geq \dots \geq x_{i(k)}$, by transitivity, $x_k \geq x_{i(k)}$. But since $x_k \neq x_{i(k)}$, $x_k > x_{i(k)}$. This leads to an infinite descending sequence, which cannot exist in a well-ordered set, a contradiction. ■

5.2 Order Isomorphisms and Order Types

> **Definition 5.9.**
>
> An **order isomorphism** between two *ordered* sets $(X, <_X), (Y, <_Y)$ is a bijection $f : S \mapsto T$ such that for all $x, y \in X$, $x <_X y \Leftrightarrow f(x) <_Y f(y)$.
>
> If that was too dense, to put it in another way, it is a mapping $f : X \mapsto Y$ such that:
> 1. f is total, i.e., defined for all $x \in X$
> 2. f is injective, i.e., if $x \neq y$ then $f(x) \neq f(y)$
> 3. f is surjective onto Y, i.e. for every $y \in Y$, there is an $x \in X$ such that $f(x) = y$
> 4. f is order-preserving, i.e., $x <_X y$ if and only if $f(x) <_Y f(y)$
>
> Note: the first three criteria are for the definition of a bijection.

If there is an order isomorphism between two well-ordered sets, we say they are **order isomorphic**, or that they have the same **order type**.

Note: an order isomorphism only requires that the sets be *ordered*. It does not require them to be *well-ordered*. Nonetheless we will usually be dealing with well-ordered sets, as the ordinals are well-ordered under $\in$:

Definition 5.10.

Every ordinal α has the following properties:

1. transitivity, i.e., for all $\beta \in \alpha$, $\beta \subseteq \alpha$, and
2. $\in$ is a well-ordering on α

1. For all $\beta \in \alpha$, $\beta < \alpha$ and therefore the elements inside β would be inside α, i.e. $\beta \subseteq \alpha$.
2. An infinite descending sequence can't exist in α as it has a least element 0.

This is another way we can define von-Neumann ordinals, and leads to the same structure, where the successor of an ordinal α is $\alpha \cup \{\alpha\}$, and an ordinal is the set of all ordinals smaller than it.

Proposition 5.11.

If $(\alpha_1, \alpha_2, ...)$ is a sequence of ordinals that is increasing (i.e., $\alpha_i < \alpha_{i+1}$) then:

$$\bigcup_i \alpha_i = \{x \mid x \in \alpha_i \text{ for some } i\}$$

is an ordinal, and is the *least ordinal* $\beta > \alpha_i$ for all $i = 1, 2, ...$.

The ordinal $\bigcup_i \alpha_i$ is known as the **limit**, **supremum**, or **strict supremum**[1] of a sequence α_i. We denote this with the the symbol $\overset{+}{\sup}$.

Proof. Suppose $\alpha \in \bigcup_i \alpha_i$. Then for some i, $\alpha \in \alpha_i$. Since α_i is an ordinal, so are all of its elements, therefore α is also an ordinal. This means that every element in $\bigcup_i \alpha_i$ is an ordinal.

Now we have to show that $\bigcup_i \alpha_i$ contains every ordinal less than any member of it. This might sound confusing, but let's take a look at ω. ω contains every ordinal less than any member of it. For example, 500 is an ordinal in ω, and all ordinals < 500 are also contained inside ω. This is true for any ordinal $n \in \omega$.

So for any ordinal $\alpha \in \bigcup_i \alpha_i$, for some i, $\alpha \in \alpha_i$. Now consider all ordinals $\beta < \alpha$. The ordinal β has to be $\in \alpha_i$ as α_i is an ordinal and therefore contains all ordinals below it ($\beta < \alpha < \alpha_i$).

Therefore for any ordinal $\alpha \in \bigcup_i \alpha_i$, for all $\beta < \alpha$, $\beta \in \bigcup_i \alpha_i$, and as such $\bigcup_i \alpha_i$ is an ordinal. ■

Proposition 5.12 (Order isomorphism transfers well-order).

If $(X, <_X)$ and $(Y, <_Y)$ are order isomorphic and $(X, <_X)$ is a well-ordered set, $(Y, <_Y)$ is also a well-ordered set.

Proof. Let f be an order isomorphism between X and Y. Suppose$(X, <_X)$ is a well-ordered set but $(Y, <_Y)$ isn't. Then there is an infinite descending sequence $y_1 >_Y y_2 >_Y ....$ Because f is onto, for each i there exists there is an $x_i \in X$ such that $f(x_i) = y_i$. Since $y_i >_Y y_{i+1}$, $f(x_i) >_Y f(x_{i+1})$ and

[1]In analysis, supremum usually uses partial $\geq$ rather than strict $>$: it usually refers to the least number s such that for all $x \in X$, $s \geq x$. Note that for limit ordinals, there is no difference between a strict supremum and the "typical" supremum.

by the order-preserving property, $x_i >_X x_{i+1}$. But then $x_1 >_X x_2 >_X \ldots$ is an infinite descending sequence in X, a contradiction. ■

Example 5.13.

Show that the set of natural numbers $\mathbb{N}$ and the set of even natural numbers $E = \{2n \mid n \in \mathbb{N}\}$, with their usual order, have the same order type.

$f(x) = 2x$ is an order isomorphism between $\mathbb{N}$ and E as:

1. f is defined for all $\mathbb{N}$
2. If $x \neq y, 2x \neq 2y$
3. For every $n \in E, n = 2 \cdot m$ where $m \in N$. m satisfies $f(m) = n$
4. If $x < y, 2x < 2y$ and vice versa

And in fact, since $\omega = \mathbb{N}$ is an ordinal, it is well-ordered. Therefore by Proposition 5.12, E which has an order isomorphism to ω is also well-ordered.

In the above example, we show that these two well-ordered sets are order isomorphic have the same **order type**. Since the set of all natural numbers is ω, we can generalize this to say that any set that has order isomorphism to $\omega = \mathbb{N}$ has order type ω. In general, **any set that has order isomorphism to an ordinal α has order type α**.

Example 5.14.

Show that the set $X = \left\{\frac{1}{n} \mid n \in \mathbb{N} \text{ and } n \neq 0\right\} \cup \{0\}$ with the ordering $>$:

1. has the same cardinality as ω
2. is **not** order isomorphic to ω
3. is order isomorphic to $\omega + 1$
4. is a well-ordered set

(note that the ordering of X is "greater than" rather than the typical "lesser than". As such we will color $>$ to represent the ordering of X)

1. For cardinality, we do not need to consider the order of the set. We can define a bijection $f : \omega \mapsto X$ as such:

$$f(n) = \begin{cases} 0 & \text{if } n = 0 \\ \frac{1}{n} & \text{if } n \neq 0 \end{cases}$$

   (The validity of this bijection ~~should be obvious~~ is left as an exercise to the reader)

2. The main reason why X is not isomorphic to ω is that when you order the set by $>$, you get $\left\{\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \ldots, 0\right\}$. There is an element "at the end", whereas for $\omega = \{0, 1, 2, \ldots\}$ there is no such element at the end.

   Let's prove by contradiction. Suppose an order isomorphism $f : \omega \mapsto X$ exists. Then there must be some $n \in \omega$ that maps to 0, i.e., there must exist some n such that $f(n) = 0$.

   However, there exists an $n + 1 \in \omega$ such that $n < n + 1$. The order-preserving property therefore requires that $f(n) > f(n + 1)$, i.e., there must exist a $f(n + 1) \in X$ such that $0 > f(n + 1)$. But there is no such element in X, as $\frac{1}{n} > 0$ and $0 = 0$, a contradiction.

3. $\omega + 1 = \{0, 1, 2, \ldots, \omega\}$, so intuitively (though you shouldn't trust intuition when dealing with infinities) we can see how we can map this to X while preserving order.

   The following bijection $f : \omega + 1 \mapsto X$ is order isomorphic:

$$f(\alpha) = \begin{cases} \frac{1}{\alpha+1} & \text{if } \alpha < \omega \\ 0 & \text{if } \alpha = \omega \end{cases}$$

4. Since $\omega + 1$ is a well-ordered set, and there exists an order isomorphism from $\omega + 1$ to X, by Proposition 5.12 X is a well-ordered set.

This is the crux of the difference between ordinals and cardinals. **Cardinality** is about comparing sizes via **bijections** between **unordered sets**, and assigning a **cardinal number**. **Order Types** is about comparing sizes via **order isomorphisms** between **(well-)ordered sets**, and assigning an **ordinal number**.

In our original definition, an ordinal is "an **equivalence class** of well-ordered sets under the relation of order isomorphism". We can interpret this as: "an ordinal represents the class of all well-ordered sets that are order isomorphic to each other". So in the equivalence class corresponding to $\omega + 1$, we have well-ordered sets like $(\{0, 1, 2, ..., \omega\}, <)$, and $\left(\left\{\frac{1}{n} \mid n \in \mathbb{N}, n \neq 0\right\} \cup \{0\}, >\right)$.

Now, let's look at how the concepts of well-orders and order types applies in other settings:

5.3 Lexicographical Ordering

Let us consider the ordering of the set of pairs of natural numbers $\mathbb{N}^2 = \{(n, m) \mid n, m \in \mathbb{N}\}$. The lexicographical ordering $\underset{\text{lex}}{\prec}$ is defined as:

$$(k, l) \underset{\text{lex}}{\prec} (n, m) \text{ if and only if (a) } k < n \text{ or (b) } k = n \text{ and } l < m$$

It's called a lexicographical ordering as it compares these pairs like how a dictionary would compare words - it compares the first element, and only moves onto the second element if the first element is the same.

$$(0,0) \underset{\text{lex}}{\prec} (0,1) \underset{\text{lex}}{\prec} (0,2) \underset{\text{lex}}{\prec} ... \underset{\text{lex}}{\prec} (1,0) \underset{\text{lex}}{\prec} (1,1) \underset{\text{lex}}{\prec} ... \underset{\text{lex}}{\prec} (2,0) \underset{\text{lex}}{\prec} ...$$

We can extend this definition of lexicographical ordering to longer sequences. Similarly to $\mathbb{N}^2$, we call the set of all sequences of length k $\mathbb{N}^k$.

Definition 5.15.

The **lexicographical ordering** $\underset{\text{lex}}{\prec}$ on $\mathbb{N}^k$ is defined as:

$(n_1, ..., n_k) \underset{\text{lex}}{\prec} (m_1, ..., m_k)$ if and only if, for some $j < k$, the following is true:

(a) $n_i = m_i$ for all i when $0 < i \leq j$

(b) $n_{j+1} < m_{j+1}$

Example 5.16.

$(1, 2, 3) \underset{\text{lex}}{\prec} (3, 2, 1)$ because for $j = 0$:

(a) for all $0 < i \leq j, n_i = m_i$ (since there is no $0 < i \leq 0$, this is considered vacuously true)

(b) $n_1 < m_1$

$(1, 2, 3) \underset{\text{lex}}{\prec} (1, 2, 5)$ because for $j = 2$:

(a) for all $0 < i \leq j, n_i = m_i$

(b) $n_3 < m_3$

Personally, I find that lexicographical ordering defined via math is quite clunky. I think easier if you think about it conceptually (sorting it like how a dictionary would) than trying to find what j value satisfies the criteria.

With the definition of $\underset{\text{lex}}{\prec}$ extended to any sequence of length k, we can now show that:

Proposition 5.17.
$\left(\mathbb{N}^k, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set.

Proof. Let $Y \subseteq \mathbb{N}^k$. We now have to prove there exists a least element in Y.

Let n_1 be the smallest n such that $(n, m_2, m_3, ..., m_k) \in Y$. Then there is no $(m_1, m_2, m_3, ..., m_k) \in Y$ such that $m_1 < n_1$. Now let n_2 be the smallest n such that $(n_1, n, m_3, ..., m_k) \in Y$. Again, there is no $(m_1, m_2, m_3, ..., m_k) \in Y$ such that $m_1 < n_1$ or $n_1 = m_1$ and $m_2 < n_2$. Continue this process until you finally define n_k as the least n such that $(n_1, n_2, n_3, ..., n) \in Y$. Then $(n_1, n_2, n_3, ..., n_k) \in Y$ is the least element in the $\underset{\text{lex}}{\prec}$ ordering. ■

Proposition 5.18.
$\left(\mathbb{N}^k, \underset{\text{lex}}{\prec}\right)$ has order type ω^k.

Proof. We use successor induction:

1. Base case: $\left(\mathbb{N}^1, \underset{\text{lex}}{\prec}\right)$ has order type ω^1
2. Successor case: Suppose $\left(\mathbb{N}^k, \underset{\text{lex}}{\prec}\right)$ has order type ω^k. This means there exists an order isomorphism $o : \mathbb{N}^k \mapsto \omega^k$. From this, we show that we can construct an order isomorphism $o' : \mathbb{N}^{k+1} \mapsto \omega^{k+1}$

   Firstly, it is apparent that there is an order isomorphism from $\mathbb{N}^{k+1}$ to ordered pairs (n, s) where $n \in \mathbb{N}$ and $s \in \mathbb{N}^k$, where n represents the first entry in a sequence in $\mathbb{N}^{k+1}$ and $s \in \mathbb{N}^k$ represents the rest.

   Given a sequence $s' \in \mathbb{N}^{k+1}$, we can then define o' as such:

$$\begin{aligned} o'(s') &= o'(n, s) \text{ for some } n \in \mathbb{N} \text{ and } s \in \mathbb{N}^k \\ &= \omega^k \cdot n + o(s) \end{aligned}$$

   We prove that this new map o' is order-preserving by contradiction. Suppose there exists two sequences $s_1', s_2' \in \mathbb{N}^{k+1}$, such that $s_1' \underset{\text{lex}}{\prec} s_2'$ but $o'(s_1') \nless s'(s_2')$. We can then express each sequence as ordered pairs as shown earlier, i.e., s_1' becomes (n_1, s_1) and s_2' becomes (n_2, s_2). If $n_1 = n_2$, then $s_1 \underset{\text{lex}}{\prec} s_2$ but $o(s_1) \nless o(s_2)$, which means that o is not order-preserving, which contradictions our assumption. And if $n_1 < n_2$, then

$$\omega^k \cdot n_1 + o(s_1) < \omega^k \cdot n_2 + o(s_2)$$

   since $o(s) < \omega^k$ for all $n \in \mathbb{N}^k$, and therefore $o'(s_1') < o'(s_2')$, a contradiction.

■

However, what if we want to compare sequences of arbitrary length? Let $\mathbb{N}^*$ be the sequences of natural number of arbitrary length:

$$\begin{gathered} (0), (1), (2), ... \\ (0, 0), (0, 1), (0, 2), ... \\ (0, 0, 0), (0, 0, 1), (0, 0, 2), ... \end{gathered}$$

We can extend our usual definition of lexicographic ordering as such:

Definition 5.19.

For 2 sequences s, t:

$$s = (n_1, n_2, ..., n_k)$$
$$t = (m_1, m_2, ..., m_l)$$

$s \underset{\text{lex}}{\prec} t$ if and only if either of the following is true:

1. $n_1 = m_1, n_2 = m_2, ..., n_k = m_k$, and $k < l$

   Example 5.20.

   $(1, 3, 5) \underset{\text{lex}}{\prec} (1, 3, 5, 7, 9)$

2. there is a $j < k$ such that $n_1 = m_1, ..., n_j = m_j$ and $n_{j+1} < m_{j+1}$

   Example 5.21.

   $(1, 3, 5, 7, 9) \underset{\text{lex}}{\prec} (1, 3, 7)$

   The first two elements are the same but in the third element they are different. The third element in the first sequence (5) < the third element in the second sequence (7).

   Put another way, $j = 2, n_1 = m_1, n_2 = m_2, n_3 < m_3$

However, $\left(\mathbb{N}^*, \underset{\text{lex}}{\prec}\right)$ is **not a well-ordered set**, as we can construct an infinite descending sequence of elements:

$$(1) \underset{\text{lex}}{\succ} (0, 1) \underset{\text{lex}}{\succ} (0, 0, 1) \underset{\text{lex}}{\succ} ...$$

However, if we only consider **decreasing sequences**, $\underset{\text{lex}}{\prec}$ is a well-order (Note that in the above example, none of them are decreasing.)

Proposition 5.22.

Let $\mathbb{N}^*_>$ be the set of all decreasing sequences $(n_1, n_2, ..., n_k)$ where $n_1 > n_2 > ... > n_k$. $\left(\mathbb{N}^*_>, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set.

Proof. Suppose for contradiction that there exists an infinite descending sequence of elements in $\mathbb{N}^*_>$: $s_1 \underset{\text{lex}}{\succ} s_2 \underset{\text{lex}}{\succ} ...$, where $s_1, s_2, ... \in \mathbb{N}^*_>$.

Let $s_i = \left(n^i_1, n^i_2, ..., n^i_{k_i}\right) \in \mathbb{N}^*_>$.

If we consider the first elements of $s_1, s_2, s_3, ...$ we get a non-increasing sequence $n^1_1 \geq n^2_1 \geq n^3_1 \geq ...$. For every k, $n^k_1 \geq n^{k+1}_1$, as if $n^k_1 < n^{k+1}_1$, then lexicographically $s_k \underset{\text{lex}}{\prec} s_{k+1}$ and it won't be in the infnite descending sequence.

Since it is a non-increasing sequence in the well-ordered set $\mathbb{N}$, it must stabilize at some index. Let's call the index j_1, and the value it stabilizes at m_1. Therefore for all $i \geq j_1$, the first element of s_i must be m_1: $s_i = \left(m_1, n^i_2, n^i_3, ..., n^i_{k_i}\right)$.

Now let's consider all sequences after the index j_1. We can similarly construct a non-increasing sequence from the second elements in those sequences: $n^{j_1}_2 \geq n^{j_1+1}_2 \geq n^{j_1+2}_2 \geq ...$. It similarly stabilizes at an index j_2 with a value of m_2. Therefore for all $i \geq j_2$, the second element of s_i must be m_2: $s_i = (m_1, m_2, n^i_3, ..., n^i_{k_i})$.

Now since $s_i \in \mathbb{N}^*_<$, s_i *is a decreasing sequence*, so $m_1 > m_2$. Continuing this, we get $m_1 > m_2 > m_3 > \ldots$, which is an infinite descending sequence in $\mathbb{N}$, a contradiction. Therefore $\left(\mathbb{N}^*_>, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set. ∎

Note that there is nothing special about $\mathbb{N}$ here, we can replace $\mathbb{N}$ with any well-ordered set:

> **Proposition 5.23 (Decreasing sequences of a well-ordered set are well-ordered).**
> Let $(X, <)$ be a well-ordered set. Let $X^*_>$ be the set of all decreasing sequences $(x_1, x_2, \ldots, x_k)$ where $x_1 > x_2 > \ldots > x_k$ and $x_i \in X$.
> $\left(X^*_>, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set.

In fact, the set of all **non-increasing** sequences of elements of a well-ordered set X is also a well-ordered by $\underset{\text{lex}}{\prec}$. A non-increasing sequence of elements in X is a sequence $(x_1, x_2, \ldots, x_k)$ where $x_i \geq x_{i+1}$ (rather than strictly decreasing $x_i > x_{i+1}$).

To show this, we can split non-increasing sequences into a sequence of constant sequences, e.g. $(4, 4, 4, 3, 2, 2, 1, 1, 1, 1) \to ((4, 4, 4), (3), (2, 2), (1, 1, 1, 1))$.

> **Lemma 5.24.**
> Let $(X, <)$ be a well-ordered set. Let $X^*_=$ be the set of all constant sequences $(x, \ldots, x) \in X$.
> e.g. if $X = \{1, 2, 3\}$, $X^*_= = \{(1), (1, 1), (1, 1, 1), \ldots, (2), (2, 2), \ldots, (3), (3, 3), \ldots\}$
> $\left(X^*_=, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set.

Proof. Let $Y \subseteq X^*_=$. Now we have to show there exists a least element in Y.

Lexicographically, we compare sequences by their first entry. As such, the least element must have the least first entry. Let x be the value of the least first entry.

Since all sequences in $X^*_=$ are constant sequences, all candidates to be the least element will be sequences like $(x, x), (x, x, x, x), (x, x, x)$. The least element (lexicographically) would be the *shortest* constant sequence of x. ∎

> **Proposition 5.25 (Non-increasing sequences well-ordered).**
> If $(X, <)$ is a well-ordered set, the set of non-increasing sequences of elements $\left(X^*_\geq, \underset{\text{lex}}{\prec}\right)$ is a well-ordered set.

Proof.

Since the set of constant sequences $X^*_=$ is well-ordered, by Proposition 5.23, the set of decreasing sequences of constant sequences $(X^*_=)^*_{\underset{\text{lex}}{\succ}}$ is also well-ordered. Now, by Proposition 5.12 all we need to do is construct an order ismorphism from $(X^*_=)^*_{\underset{\text{lex}}{\succ}}$ to $X^*_\geq$ to prove that $X^*_\geq$ is a well-ordered set.

Let y_i denote constant sequences in $X^*_=$. We can construct an isomorphism $f : (X^*_=)^*_{\underset{\text{lex}}{\succ}} \mapsto X^*_\geq$:

$$f((y_1, y_2, \ldots, y_{k'})) = \left(\underbrace{x_1, \ldots, x_1}_{\text{contents of } y_1}, \underbrace{x_2, \ldots, x_2}_{\text{contents of } y_2}, \ldots, \underbrace{x_{k'}, \ldots, x_{k'}}_{\text{contents of } y_{k'}}\right)$$

Both sets are ordered lexicographically. $X^*_\geq$ is ordered by $\underset{\text{lex}}{\prec}$, and $(X^*_=)^*_{\underset{\text{lex}}{\succ}}$ is ordered lexicographically based on $\underset{\text{lex}}{\prec}$, so this is an order-preserving isomorphism.

Therefore is $X^*_\geq$ is a well-ordered set. ∎

All this discussion on lexicographic orderings is quite abstract, but we will use the concepts of orders (especially on non-increasing sets) in the next chapter.

6 Ordinal Notations (up to ε_0)

An **ordinal notation**, or **ordinal notation system** is basically a way to express an ordinal as a **finite string**, i.e., finite sequences of finite symbols. Think of the decimal system, that allows you to express any number into a finite sequence of the symbols $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$. Of course, we also need to define some ordering of these strings too.

Definition 6.1.

An **Ordinal Notation** is a well-ordered set $(OT, \prec)$, where OT is a *recursive* set of finite formal strings with a finite alphabet, and $\prec$ is a *recursive* order.

(*recursive* here means that there exists an algorithm to compute them, i.e. computable)

6.1 Iterated Cantor Normal Form

Definition 6.2.

Iterated Cantor Normal Form (ICNF) is an ordinal notation for ordinals $< \varepsilon_0$.

The ordinal notation here is a string consisting of finitely many of the following symbols:

(We use a different font to highlight the fact that these are not true ordinals)

- $\mathsf{0}$
- $\mathsf{+}$
- The function $\mathsf{\omega}^{\bullet}(\alpha)$ which can also be written as $\mathsf{\omega}^{\alpha}$

The ICNF ordinal notation $o(\alpha)$ for an ordinal $\alpha < \varepsilon_0$ is:

$$o(\alpha) := \begin{cases} \mathsf{0} & \text{if } \alpha = 0 \\ \mathsf{\omega}^{o(\alpha_1)} \mathsf{+} \mathsf{\omega}^{o(\alpha_2)} \mathsf{+} \ldots \mathsf{+} \mathsf{\omega}^{o(\alpha_n)} & \text{if the CNF of } \alpha \text{ is } \omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n} \end{cases}$$

This works as a notation because Cantor normal form is unique, and for $\alpha < \varepsilon_0$, $\alpha_i < \alpha$.

We can define a well-ordering $\prec$ that borrows from the normal order of ordinals by saying $o(\alpha) \prec o(\beta)$ if and only if $\alpha < \beta$. However, such an ordering is not recursive as showing that $\alpha \in \beta$ requires using infinite sets. We can order the strings on our own without relying on the order of ordinals (we will define one later).

Example 6.3.

$$\begin{aligned} o(\omega^\omega + \omega^3 + \omega^2 + 1) &= o(\omega^\omega + \omega^3 + \omega^2 + \omega^0) \text{ (convert to CNF within } o()) \\ &= \mathsf{\omega}^{o(\omega)} \mathsf{+} \mathsf{\omega}^{3} \mathsf{+} \mathsf{\omega}^{2} \mathsf{+} \mathsf{\omega}^{0} \\ &= \mathsf{\omega}^{o(\omega^1)} \mathsf{+} \mathsf{\omega}^{o(\omega^0+\omega^0+\omega^0)} \mathsf{+} \mathsf{\omega}^{o(\omega^0+\omega^0)} \mathsf{+} \mathsf{\omega}^{o(\omega^0)} \\ &\text{(convert to CNF within } o()) \\ &= \mathsf{\omega}^{\mathsf{\omega}^{o(1)}} \mathsf{+} \mathsf{\omega}^{\mathsf{\omega}^0\mathsf{+}\mathsf{\omega}^0\mathsf{+}\mathsf{\omega}^0} \mathsf{+} \mathsf{\omega}^{\mathsf{\omega}^0\mathsf{+}\mathsf{\omega}^0} \mathsf{+} \mathsf{\omega}^{\mathsf{\omega}^0} \\ &= \mathsf{\omega}^{\mathsf{\omega}^{\mathsf{\omega}^0}} \mathsf{+} \mathsf{\omega}^{\mathsf{\omega}^0\mathsf{+}\mathsf{\omega}^0\mathsf{+}\mathsf{\omega}^0} \mathsf{+} \mathsf{\omega}^{\mathsf{\omega}^0\mathsf{+}\mathsf{\omega}^0} \mathsf{+} \mathsf{\omega}^{\mathsf{\omega}^0} \\ &\text{(convert to CNF within } o()) \end{aligned}$$

Definition 6.4.

Here we will define a recursive ordering for the ICNF ordinal notation. We define the following inductively:

1. the set $OT_{=k}$ of ordinal notations of height k,
2. the set $OT_{\leq k}$ of ordinal notations of height $\leq k$,
3. an ordering $\prec_k$ on $O_{\leq k}$

The sets $OT_{=k}$ and $OT_{\leq k}$ are defined as such:

1. $\mathtt{0}$ is the only ordinal notation of height 0, so $OT_{=0} = OT_{\leq 0} = \{\mathtt{0}\}$, and $\prec_0 = \{\}$, i.e., $\mathtt{0} \not\prec_0 \mathtt{0}$
2. If $OT_{=k}, OT_{\leq k}, \prec_k$ are defined, then the set $OT_{=k+1}$, the set of all ordinal notations of height $k+1$, is defined as:
$$\omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n}$$
where $\alpha_1 \in OT_{=k}$, and $\alpha_1 \succeq_k \alpha_2 \succeq_k \ldots \succeq_k \alpha_n$ $(\alpha_i \in OT_{\leq k})$.
We color greek symbols in **green** to represent ordinal notations, rather than ordinals. Note that those colored in **red** still represents the symbols themselves. To use a programming analogy, symbols in **green** are like variables containing strings (specifically, ordinal notation strings).

The order $\prec_{k+1}$ on $OT_{<k+1}$ is defined as such:

- Case 1: α and β are both of height $\leq k$ (α and $\beta \in OT_{\leq k}$).
  In this case, we just use $\prec_k$ to compare them, i.e. $\alpha \prec_{k+1} \beta$ if $\alpha \prec_k \beta$.
- Case 2: α is of height $\leq k$ but β is of height $k+1$ ($\alpha \in OT_{\leq k}$ and $\beta \in OT_{=k+1}$). In this case, $\alpha \prec_{k+1} \beta$
- Case 3: α and β are both of height $k+1$.
  In this case, we can express α and β as such they are of the form:
$$\alpha = \omega^{\alpha_1} + \ldots + \omega^{\alpha_n}$$
$$\beta = \omega^{\beta_1} + \ldots + \omega^{\beta_m}$$
  We order this lexicographically, comparing α_1 and β_1. If they are equal, move onto α_2 and β_2, and if they are still equal, move on until either you find a $\alpha_i \prec \beta_i$ or, if all exponents are equal, that α has less terms than β. A formal way to write this is:
  $\alpha \prec_{k+1} \beta$ if either of the following hold:
  - $\alpha_i = \beta_i$ for all $i \leq j$ and $j = n < m$
    e.g. $\omega^{\omega^0} + \omega^0 \prec_2 \omega^{\omega^0} + \omega^0 + \omega^0$, here $n = 2, m = 3, j = n = 2$.
  - $\alpha_i = \beta_i$ for all $i \leq j$ and $\alpha_{j+1} < \beta_{j+1}, j < n$
    e.g. $\omega^{\omega^0} + \omega^0 \prec_2 \omega^{\omega^0} + \omega^{\omega^0}$, here $j = 1$

Finally, we can define the set of all ordinal notations $OT = OT_{\leq 0} \cup OT_{\leq 1} \cup OT_{\leq 2} \cup \ldots$, and an ordering $\prec = \prec_0 \cup \prec_1 \cup \prec_2 \cup \ldots$

We still have to prove that ordering $\prec$ on the set OT is a well-ordering. We will assume that it is a strict linear ordering – the proof of that is left as an exercise to the reader (totally not because I'm lazy).

Proposition 6.5.
$(OT_{\leq k}, \prec)$ is a well-ordered set

Proof.

$OT_{\leq 0} = \{0\}$ is trivially well-ordered, any nonempty subset is just $\{0\}$, of which 0 is the smallest element.

Now for induction, assume $(OT_{\leq k}, \prec)$ is well-ordered. The set $(OT_{\leq k+1}, \prec_{k+1})$ is order isomorphic to the set of nonincreasing sequences of ordinal notations from $(OT_{\leq k}, \prec_k)$. Why? because we can make a bijection $f : OT_{\leq k+1} \mapsto OT^*_{\leq k}$ like this:

$f(\omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n}) = (\alpha_1, \alpha_2, \ldots, \alpha_n)$.

It should be apparent that the expression in the left hand side $\omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n} \in OT_{\leq k+1}$, and $\alpha_1 \succeq \alpha_2 \succeq \ldots \succeq \alpha_n$, and thus the right hand side is an non-increasing sequence of expressions $\alpha_i \in OT_{\leq k}$ of variable length.

Since $OT_{\leq k+1}$ is order isomorphic to the set of all non-increasing sequence of elements from $OT_{\leq k}$, and $OT_{\leq k}$ is well-ordered, by Proposition 5.25, $(OT_{\leq k+1}, \prec)$ must therefore be well-ordered too. ■

Proposition 6.6.

$(OT, \prec)$ is well-ordered

Proof. Let $\alpha_1 \succ \alpha_2 \succ \ldots$ be an infinite descending sequence of ordinal notations. The first α_1 is of some height k. Since any ordinal notation of height $> k$ would be greater than any ordinal notation of height k, the height of every α_i is $\leq k$. So $\alpha_1 \succ \alpha_2 \succ \ldots$ is an infinite descending sequence in $(OT_{\leq k}, \prec_k)$. However, $(OT_{\leq k}, \prec_k)$ is well-ordered, a contradiction. Therefore no infinite descending sequence can exist in $(OT, \prec)$, so it is a well-ordered set. ■

While ordinal notations look like actual ordinals with symbols like ω^0, they are really just complicated sequences of sequences of ... sequences of symbols. Ordinal notations of height 0 is just 0. Then the ordinal notations of height 1 are ω^0, $\omega^0 + \omega^0$, $\omega^0 + \omega^0 + \omega^0$, ... But we can instead represent them as sequences of ordinal notations of height 0, so $\omega^0 = (0)$, $\omega^0 + \omega^0 = (0, 0)$, and so on. We can then lexicographically order these sequences of 0 by length to get our order for $OT_{\leq 1}$, letting 0 be smaller than any sequence of 0s:

$$0 \prec_1 (0) \prec_1 (0, 0) \prec_1 (0, 0, 0) \prec_1 \ldots$$

With this order, we can make a set of non-decreasing sequences of terms from $OT_{\leq 1}$, which corresponds to $OT_{\leq 2}$. For example, $\omega^{\omega^0 + \omega^0} + \omega^0 = ((0, 0), 0)$. We can define a lexicographic order on these sequences based on the order $\prec_1$ to get $\prec_2$, and with this method we can keep building ordinal notations of greater and greater height.

Ordinal notations themselves are **not transfinite sets** (unlike actual ordinals) – this is why they are highlighted so distinctly in red. If you look closely, we do not need to know anything about transfinite ordinals to show that they are well-ordered (we just needed to know facts about lexicographical orderings of sequences).

6.2 Ordering of Trees and Goodstein sequences

We now have two order-isomorphic well-ordered sets of order type ε_0:

1. The ordinals $< \varepsilon_0$ ordered by $<$, or equivalently $\in$
2. Ordinal notations $< \varepsilon_0$ ordered by $\prec$ (This ordinal notation is known as Iterated Cantor Normal Form)

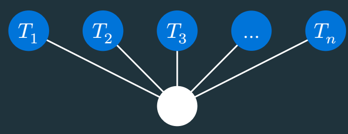
Another example of a well-ordered set of order type ε_0 is the set of all finite, finitely branching trees.

Definition 6.7.

The set T of finite, finitely branching tree can be inductively defined as such:

1. The tree containing a single node is a tree of height 1. We can represent this as $()$ or as a diagram:

2. If $T_1, \ldots, T_n$ are trees, and k is the maximum height among $T_1, \ldots, T_n$, a tree of height $k+1$ is $(T_1, T_2, \ldots, T_n)$

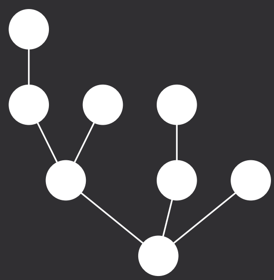


(nodes in blue represents trees)

3. Nothing else is a tree

Example 6.8.

The tree `((((),()),()),(()),())` is of height 4:

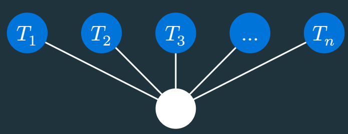


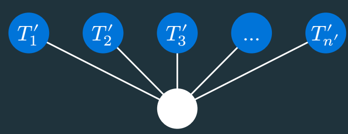
Definition 6.9.

Now let's inductively define an order $<$ on these trees:

1. The tree of height 1 is only equal to itself, and is $<$ all trees of height > 1
2. Suppose T and T' are trees of height > 1, and let h be the maximum of their heights. Assume that $<$ has been defined for trees of height $< h$.

   We can then write $T = (T_1, T_2, \ldots, T_n)$ and $T' = (T'_1, T'_2, \ldots, T'_{n'})$, where $T_i \geq T_{i+1}$, $T'_i \geq T'_{i+1}$ (i.e. arranged in a non-decreasing sequence)

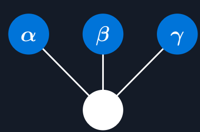
   $T =$ 

   $T' =$ 

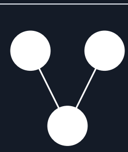
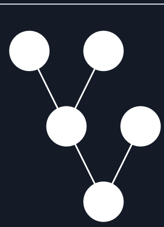
   We can then order these lexicographically, i.e. $T < T'$ if:

   1. $T_i = T'_i$ and $n < n'$
   2. There exists a k such that for all $i < k$, $T_i = T'_i$ and $T_k < T'_k$

This should already be sounding familiar to you – this is analogous to our ordinal notation system, where we can correspond $\omega^{\alpha} + \omega^{\beta} + \omega^{\gamma}$ to a tree like this:



Some examples are

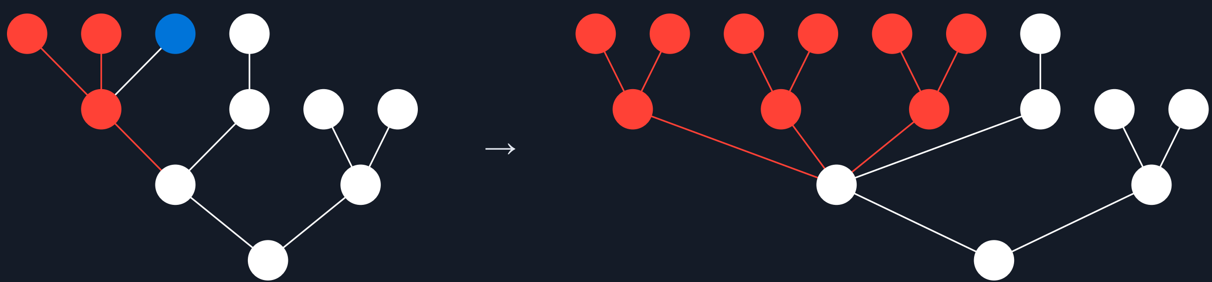
Ordinal notation	Associated tree
0	
ω^0	
$\omega^0 + \omega^0$	
$\omega^{\omega^0+\omega^0} + \omega^0$	

With this, we can construct an order isomorphism from T to ε_0, and show that T is of order type ε_0. We can basically show via induction that $OT_{\leq k}$, the set of all ordinal notations of height $\leq k$ is order isomorphic to $T_{\leq k}$, the set of all trees of height $\leq k$ since the way the ordering is defined (lexicographic ordering of elements of lower height) is the same. (informal proof)

Kirby-Paris Hydra

Imagine the previous tree, where every leaf node (i.e. not connected to any node above) is a head of a Hydra. Every time we cut off a head, the leaf's parent node and all of its remaining descendents will be multiplied n times.

Here's is an example: if we cut the blue leaf node, the part in red gets duplicated $n = 3$ times:



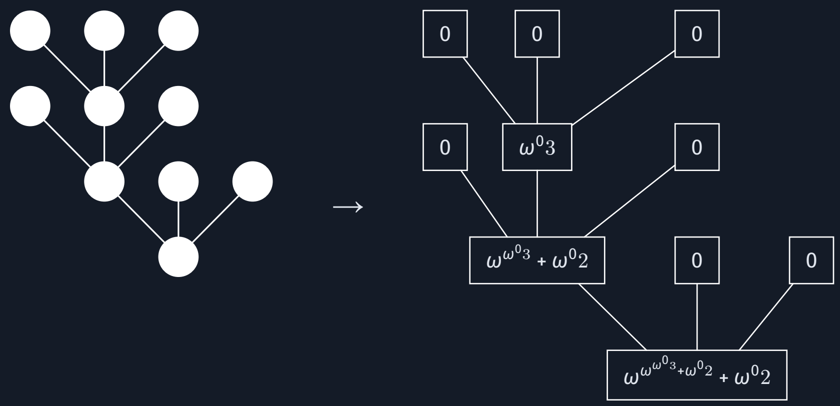
> **Theorem 6.10 (Kirby-Paris Hydra).**
>
> If you keep cutting off heads (in any manner), the Hydra eventually dies.

Proof.

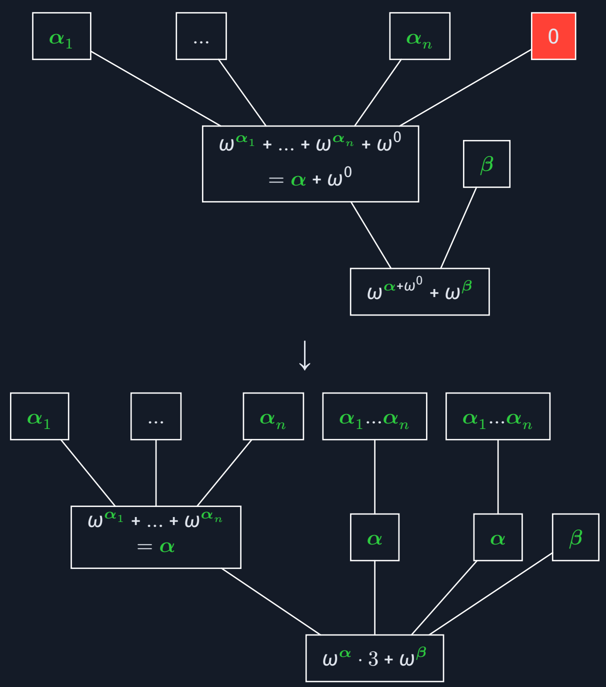
We assign ordinal notations $< \varepsilon_0$ to nodes in the tree:

1. Leaf nodes are assigned 0.
2. If the ordinal notations assigned to a node's descendants are $\alpha_1 \succeq \alpha_2 \succeq \ldots \succeq \alpha_n$, the node itself is assigned $\omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n}$

For example:



Now let's look at what happens when a leaf node (filled in black) is removed. We set $n = 3$ for this example. Here we denote $\alpha = \omega^{\alpha_1} + \ldots \omega^{\alpha_n}$



The ordinal notation at the bottom decreases whenever we remove a head:

$$\omega^{\alpha} \cdot n < \omega^{\alpha+\omega^0} \text{ since}$$

$$\omega^{\alpha} \cdot n < \omega^{\alpha+1}$$

Therefore, eventually, the root will reach 0 and the hydra will die.

■

Goodstein sequences

In base 10, we represent a number using powers of 10:

$$25739 = 2 \cdot 10^4 + 5 \cdot 10^3 + 7 \cdot 10^2 + 3 \cdot 10^1 + 9$$

In other bases, e.g. base 3, we can do the same:

$$25739 = 3^9 + 2 \cdot 3^7 + 2 \cdot 3^4 + 2 \cdot 3^3 + 2 \cdot 3 + 2$$

We can take a page out of our Iterated Cantor Normal Form and demand that the exponents of 3 must also be in terms of power of 3:

$$25739 = 3^{3^2} + 2 \cdot 3^{2 \cdot 3+1} + 2 \cdot 3^{3+1} + 2 \cdot 3^3 + 2 \cdot 3 + 2$$

We call this "hereditary" base-3 representation.

Now the next term in the **Goodstein sequence** for this number would be equivalent to taking every "3^" and replacing it with a "4^", then subtracting 1 from it.

$$4297067018 = 4^{4^2} + 2 \cdot 4^{2 \cdot 4+1} + 2 \cdot 4^{4+1} + 2 \cdot 4^4 + 2 \cdot 4 + 2 - 1$$

Definition 6.11.

The **Goodstein sequence** for some number n is the sequence $(n_2, n_3, n_4, ...)$ where the number n_{i+1} is obtained by writing n_i in hereditary base-i notation, changing the is into $(i+1)$s and subtracting 1.

Example 6.12.

Let's start with $n_2 = 10$. We first write 10 in hereditary base-2 notation:

$$n_2 = 2^{2+1} + 2$$

Then n_3 is:

$$n_3 = 3^{3+1} + 3 - 1 = 3^{3+1} + 2 = 83$$

We can repeat this for $n_4, n_5, ...$:

$$\begin{aligned}
n_4 &= 4^{4+1} + 2 - 1 = 4^{4+1} + 1 = 1025 \\
n_5 &= 5^{5+1} + 1 - 1 = 5^{5+1} = 15625 \\
n_6 &= 6^{6+1} - 1 = 279935 \\
&= 5 \cdot 6^6 + 5 \cdot 6^5 + 5 \cdot 6^4 + 5 \cdot 6^3 + 5 \cdot 6^2 + 5 \cdot 6 + 5 \\
n_7 &= 5 \cdot 7^7 + 5 \cdot 7^5 + 5 \cdot 7^4 + 5 \cdot 7^3 + 5 \cdot 7^2 + 5 \cdot 7 + 5 - 1 \\
&= 5 \cdot 7^7 + 5 \cdot 7^5 + 5 \cdot 7^4 + 5 \cdot 7^3 + 5 \cdot 7^2 + 5 \cdot 7 + 4 \\
&= 4215754
\end{aligned}$$

The goodstein sequence seems to grow forever, but this sequence eventually terminates at 0!

Theorem 6.13.

The Goodstein sequence for n eventually reaches 0, for all n

Proof. let $\alpha(n, b)$ be the ordinal notation resulting from it by replacing every b by ω.

With this, for each n_i in the Goodstein sequence, we can assign an ordinal notation $\alpha(n_i, i)$ to it. We now show that $\alpha(n_{i+1}, i+1) \prec \alpha(n_i, i)$.

Suppose $n_i = a_k i^k + ... + a_j i^j + a_0$. Then $\alpha(n_i, i) = \omega^k \cdot a_k + ... + \omega^j \cdot a_j + a_0$.

If $a_0 > 0$, then

$$\begin{aligned}
n_{i+1} &= a_k (i+1)^k + ... + a_j (i+1)^j + a_0 - 1 \\
\alpha(n_{i+1}, i+1) &= \omega^k \cdot a_i + ... + \omega^j \cdot a_j + (a_0 - 1) \\
&\prec \omega^k \cdot a_k + ... + \omega^j \cdot a_j + a_0 = \alpha(n_i, i)
\end{aligned}$$

If $a_0 = 0$, then:

$$
\begin{aligned}
n_{i+1} &= a_k(i+1)^k + \ldots + a_j(i+1)^j - 1 \\
&= a_k(i+1)^k + \ldots + (a_j - 1)(i+1)^j + (i+1)^j - 1 \\
&= a_k(i+1)^k + \ldots + (a_j - 1)(i+1)^j + \sum_{s=0}^{j-1} i(i+1)^s \\
\alpha(n_{i+1}, i+1) &= \omega^k \cdot a_k + \ldots + \omega^j \cdot (a_j - 1) + \sum_{s=0}^{j-1} \omega^s \cdot i \\
&\prec \omega^k \cdot a_k + \ldots + \omega^j \cdot a_j + a_0 = \alpha(n_i, i)
\end{aligned}
$$

In the last line, we compare the factor on ω^j (the largest power of ω in which they differed), and find that $\alpha(n_i, i) \succ \alpha(n_{i+1}, i+1)$ no matter the value of a_0, and therefore the sequence reaches 0. ∎

6.3 The Primitive Sequence System (PrSS)

Definition 6.14 (Primitive Sequence System (PrSS)).

Let the set T_{PrSS} be the set of sequences $s = (s_1, s_2, \ldots, s_l)$ such that:

1. s is either empty $()$ or starts with 0: $(0, \ldots)$.
2. For any s_i in s, let it's **parent** s_j be the *latest element* before s_i such that $s_j < s_i$. s_j must be equal to $s_i - 1$.

   Put another way, let the *parent index* $p(i) \in \mathbb{N}$ be the largest possible $p(i) < i$ such that $s_{p(i)} < s_i$. For all s_i in the sequence, $s_{p(i)} = s_i - 1$.

   Example 6.15.

   $(0, 1, 3)$ is not a valid sequence as the latest element before 3 that is less than 3 is 1, which is not $3 - 1 = 2$.

   $(0, 1, 2, 1, 2)$ is a valid sequence:
   - It starts with 0
   - $s_2 = 1$. The latest element less than 1 is $s_1 = 0$, which is 1 less than $s_2 = 1$
   - $s_3 = 2$. Similarly, the latest element less than 2 is $s_2 = 1$ which is 1 less than $s_3 = 2$
   - $s_4 = 1$. The latest element less than $s_4 = 1$ is $s_1 = 0$, so no issues
   - $s_5 = 2$. The latest element less than $s_5 = 2$ is $s_4 = 1$, so no issues
3. The ordering $<$ is simply the lexicographic ordering of these sequences.

The set T_{PrSS} is the set of all valid PrSS sequences. We can use this notation to define either *a fast-growing function*, or a *system of fundamental sequences*:

Definition 6.16.

For a sequence $s = (s_1, s_2, \ldots, s_l) \in T_{\text{PrSS}}$, we define $s[n]$ as the n^{th} member in its *fundamental sequence*.

- $()[n] = n$. Otherwise, for nonempty s:
- The **good root** g and the **bad root** b are defined as such:
  - $g = (s_1, \ldots, s_{p(l)-1})$
  - $b = (s_{p(l)}, \ldots, s_{l-1})$
  - if $p(l)$ does not exist (i.e. there is no $s_i < s_l$, for example if $s_l = 0$), then: $g = (s_1, \ldots, s_{l-1})$ and b is empty

- $s[n] = (g, b, b, ..., b)$ with n copies of b, though this can be replaced with any increasing function $f(n)$. The original definition used $f(n) = n^2$.

The action of finding the fundamental sequence of a PrSS sequence is often called **expanding**.

As for the *fast-growing function* also confusingly labelled $s[n]$:

- We split the sequence into good root g and bad root b again.
- $s[n] = (g, b, b, ..., b)[f(n)]$ with $f(n)$ copies of b. The original definition used $f(n) = n^2$. We will instead use a different definition that makes analysis easier:

$$f(n) = \begin{cases} n & \text{if } s_l \neq 0 \\ n+1 & \text{if } s_l = 0 \end{cases}$$

  i.e. n only increases by 1 if the sequence ends with 0.

The final result is the value in the [] when the sequence is eventually empty, i.e. $()[n] = n$

Example 6.17.

Expand the following:

1. $(0, 0)[3]$
2. $(0, 1)[3]$
3. $(0, 1, 2, 3, 2)[3]$
4. $(0, 1, 2, 3, 4, 3, 4, 2, 3)[3]$

Seriously, try it yourself. I'll provide the answers in case you get stuck:

1. 0 has no parent, so good root is 0 and bad root is empty. We get (0). Note that n doesn't matter when the last element is 0.
2. The parent of 1 is 0, so we have an empty good root and a bad root of 0. We replicate the bad root $n = 3$ times, so we get $(0, 0, 0)$
3. The parent of 2 is 1, so we have a good root of 0 and a bad root of $1, 2, 3$, so we get $(0, 1, 2, 3, 1, 2, 3, 1, 2, 3)$
4. $(0, 1, 2, 3, 4, 2, 3, 2, 2, 2)$

Example 6.18.

Evaluate $(0, 1, 1)[3]$ as a fast-growing function with our definition of $f(n)$

$$
\begin{aligned}
&(0,1,1)[3]\\
&=(0,1,0,1)[3]\\
&=(0,1,0,0,0)[3]\\
&=(0,1,0,0)[4]\\
&=(0,1,0)[5]\\
&=(0,1)[6]\\
&=(0,0,0,0,0,0)[6]\\
&=(0,0,0,0,0)[7]\\
&=(0,0,0,0)[8]\\
&=(0,0,0)[9]\\
&=(0,0)[10]\\
&=(0)[11]\\
&=()[12]\\
&=12
\end{aligned}
$$

Now imagine if we used a bigger value of n, making many more copies of the bad root. Or image if we used $f(n) = n^2$ instead of using n or $n+1$ like we did. $s[n]$ is a very powerful function. But we can constrain it's growth rate in the Hardy hierarchy as we'll show later.

Order Isomorphism to ε_0

We now show that PrSS is order isomorphic to ε_0.

Now, just like in ICNF where we use a non-increasing sequence of ordinal notations of height $\leq k$ to define a new height $k+1$:

$$\alpha_1 \geq \alpha_2 \geq \ldots \geq \alpha_n$$

$$\omega^{\alpha_1} + \omega^{\alpha_2} + \ldots + \omega^{\alpha_n}$$

In PrSS we can make a new notation of new height $k+1$ from sequences of height $\leq k$ as such:

Given a non-increasing *sequence of PrSS sequences* of height $\leq k$, e.g.

$$(0,1) \underset{\text{lex}}{\succeq} (0,0,0) \underset{\text{lex}}{\succeq} (0,0) \underset{\text{lex}}{\succeq} (0,0)$$

, we add 1 to each term in the sequence and join them up with 0s:

$$(0,1) \underset{\text{lex}}{\succeq} (0,0,0) \underset{\text{lex}}{\succeq} (0,0) \underset{\text{lex}}{\succeq} (0,0) \to (1,2) \underset{\text{lex}}{\succeq} (1,1,1) \underset{\text{lex}}{\succeq} (1,1) \underset{\text{lex}}{\succeq} (1,1)$$

$$(0,(1,2),0,(1,1,1),0,(1,1),0,(1,1))$$

$$(0,1,2,0,1,1,1,0,1,1,0,1,1)$$

Since the lexicographic ordering of *sequences of PrSS* is also lexicographic (i.e. to compare 2 *sequences of PrSS*, we first compare the first sequence in both, then move onto the second if they are equal), for any two *sequences of PrSS* S_1 and S_2, $S_1 \underset{\text{lex}}{\prec} S_2$ if and only if $f(S_1) \underset{\text{lex}}{\prec} f(S_2)$.

In this was, we can define the set of standard PrSS sequences OT_{PrSS} as the set of all notations of height k created by this process, in a very similar method we used to define ordinal notation of Iterative Cantor Normal Form.

Here are some examples of the PrSS correspondence with ordinals. I added a third column for the PrSS "hydra" to show how elements in the sequence relate to nodes of the ordinal notation tree.

Ordinal	Ordinal Notation	PrSS tree	PrSS
0	0	`•`	$()$
1	ω^0	`  0` `•─┘`	(0)
2	$\omega^0 + \omega^0$	`  0 0` `•─┴─┘`	$(0, 0)$
3	$\omega^0 + \omega^0 + \omega^0$	`  0 0 0` `•─┴─┴─┘`	$(0, 0, 0)$

ω	ω^{ω^0}	`    1`
		`  0─┘`
		`•─┘`
			$(0, 1)$

$\omega + 1$	$\omega^{\omega^0} + \omega^0$	`    1`
		`  0─┘ 0`
		`•─┴───┘`
			$(0, 1, 0)$

$\omega + 2$	$\omega^{\omega^0} + \omega^0 + \omega^0$	`    1`
		`  0─┘ 0 0`
		`•─┴───┴─┘`
			$(0, 1, 0, 0)$

$\omega \cdot 2$	$\omega^{\omega^0} + \omega^{\omega^0}$	`    1   1`
		`  0─┘ 0─┘`
		`•─┴───┘`
			$(0, 1, 0, 1)$

$\omega \cdot 2 + 1$	$\omega^{\omega^0} + \omega^{\omega^0} + \omega^0$	`    1   1`
		`  0─┘ 0─┘ 0`
		`•─┴───┴───┘`
			$(0, 1, 0, 1, 0)$

ω^2	$\omega^{\omega^0 + \omega^0}$	`    1 1`
		`  0─┴─┘`
		`•─┘`
			$(0, 1, 1)$

ω^3	$\omega^{\omega^0 + \omega^0 + \omega^0}$	`    1 1 1`
		`  0─┴─┴─┘`
		`•─┘`
			$(0, 1, 1, 1)$

ω^ω	$\omega^{\omega^{\omega^0}}$	`      2`
		`    1─┘`
		`  0─┘`
		`•─┘`
			$(0, 1, 2)$

$\omega^{\omega^3+2} + 2$	$\omega^{\omega^{\omega^0 3} + \omega^0 2} + \omega^0 2$	`      2 2 2`
		`    1─┴─┴─┘ 1 1`
		`  0─┴───────┴─┘ 0 0`
		`•─┴─────────────┴─┘`
			$(0, 1, 2, 2, 2, 1, 1, 0, 0)$

Ordinal	Ordinal Notation	PrSS tree	PrSS

ω^{ω^ω}	$\omega^{\omega^{\omega^{\omega^0}}}$	`      3`
		`    2─┘`
		`  1─┘`
		`0─┘`
		`•─┘`
			$(0, 1, 2, 3)$

We see that each number in the sequence represents the literal height of the term on this hydra. When taking funamental sequences, this "hydra" behaves quite similarly to the Kirby-Paris Hydra we saw earlier:

$(0, 1, 2, 3, 3, 3)[3] = (0, 1, 2, 3, 3, 2, 3, 3, 2, 3, 3)$

```
         3 3 3                  3 3   3 3   3 3
       2─┴─┴─┘                2─┴─┘ 2─┴─┘ 2─┴─┘
     1─┘                    1─┴─────┴─────┘
   0─┘                    0─┘
 •─┘                    •─┘
```

With green representing the good root and red representing the bad root.

Growth rate of $s[n]$

Since we have assigned a mapping from each PrSS sequence to an ordinal, let's compare PrSS expansion rules with the Hardy Hierarchy, using the Wainer Hierarchy for fundamental sequences.

PrSS	Hardy Hierarchy
$()[n] = n$	$H_0(n) = n$
$(0)[n] = ()[n+1]$	$H_1(n) = H_0(n+1)$
$(0,0)[n] = (0)[n+1]$	$H_2(n) = H_1(n+1)$
$(0,1)[n] = \underbrace{(0,0,0,...,0)}_{n \text{ copies of } 0}[n]$	$H_\omega(n) = H_n(n)$
$(0,1,0)[n] = (0,1)[n]$	$H_{\omega+1} = H_\omega(n)$
$(0,1,0,1)[n] = (0,1,\underbrace{0,0,0,...,0}_{n \text{ copies of } 0})[n]$	$H_{\omega \cdot 2}(n) = H_{\omega+n}(n)$
$(0,1,1)[n] = \underbrace{(0,1,0,1,...,0,1)}_{n \text{ copies of } (0,1)}[n]$	$H_{\omega^2}(n) = H_{\omega n}(n)$
$(0,1,2)[n] = (0,1,\underbrace{1,1,1,...,1}_{n \text{ copies of } 1})[n]$	$H_{\omega^\omega}(n) = H_{\omega^n}(n)$

We see the correspondence between each PrSS sequence and its associated ordinal in our modified Hardy Hierarchy. Therefore the maximum growth rate of PrSS is $(0, 1, 2, 3, ...)[n]$ which corresponds to $H_{\varepsilon_0}(n)$.

But what about the fast-growing hierarchy? How does that relate to the Hardy Hierarchy? While the Hardy Hierarchy appears to grow slower than the fast-growing hierarchy, we can show that $f_{\varepsilon_0}(n) \approx H_{\varepsilon_0}(n)$, i.e., the Hardy Hierarchy "catches up" at ε_0.

Lemma 6.19.

$$f_\alpha(n) = H_{\omega^\alpha}(n)$$

Proof.

1. Base case: $f_0(n) = H_{\omega^0}(n) = H_1(n) = n + 1$
2. Successor case: If $f_\alpha(n) = H_{\omega^\alpha}(n)$, then:

$$\begin{aligned} H_{\omega^\alpha + \omega^\alpha}(n) &= H_{\omega^\alpha}(H_{\omega^\alpha}(n)) \\ &= f_\alpha^2(n) \end{aligned}$$

$$\begin{aligned} H_{\omega^{\alpha+1}}(n) &= H_{\omega^\alpha \cdot n}(n) \\ &= f_\alpha^n(n) \\ &= f_{\alpha+1}(n) \end{aligned}$$

1. Limit case: If for a limit ordinal α, then we presume for all $\beta < \alpha$, $f_\beta(n) = H_{\omega^\beta}(n)$, then:

$$\begin{aligned} H_{\omega^\alpha}(n) &= H_{\omega^{\alpha[n]}}(n) \\ &= f_{\alpha[n]}(n) \text{ (Since } \alpha[n] < \alpha) \\ &= f_\alpha(n) \end{aligned}$$

■

Proposition 6.20.

$$f_{\varepsilon_0}(n) \approx H_{\varepsilon_0}(n+1) \approx H_{\varepsilon_0}(n)$$

Proof.

$$\begin{aligned} f_{\varepsilon_0}(n) &= f_{\varepsilon_0[n]}(n) \\ &= f_{\omega \uparrow\uparrow n}(n) \\ &= H_{\omega \uparrow\uparrow n+1}(n) \\ &= H_{\varepsilon_0[n+1]}(n) \\ &\approx H_{\varepsilon_0}(n+1) \\ &\approx H_{\varepsilon_0}(n) \end{aligned}$$

■

Formal Order Isomorphism to ε_0

However, we have another way to define the set of standard PrSS sequences OT_{PrSS}:

Definition 6.21.

For two sequences s, t, we define the relation $s \sqsubset t$ if and only if there exists $n_1, n_2, ..., n_k \in \mathbb{N}$ such that

$$s = t[n_1][n_2]...[n_k]$$

That is, $s \sqsubset t$ if we can derive s from t using expansions. Then naturally for $\sqsupset$ we define $s \sqsupset t$ if and only if $t \sqsubset s$.

Definition 6.22.

The set of standard PrSS sequences OT_{PrSS} are defined as:

$$OT_{\text{PrSS}} = \{s \in T \mid \text{There exists an } n \in \mathbb{N} \text{ such that } s \sqsubset (0, 1, ..., n)\}$$

This might feel like a strange way to define the set of standard sequences OT_{PrSS}, but when we go over more powerful extensions of this notation in later chapters, this "top-down" approach will be more useful.

Lemma 6.23.

For all $s \in T$, $n \in \mathbb{N}$, $s[n] \underset{\text{lex}}{\prec} s$

Proof. Let $s = (s_1, ..., s_m)$. Let the parent of s_m be $s_{p(m)}$. As such s can be re-written as:

$$s = (\underbrace{s_1, ..., s_{p(m)-1}}_{G}, \underbrace{s_{p(m)}, ..., s_{m-1}}_{B}, s_m)$$

Then $s[n] \underset{\text{lex}}{\prec} s$, as the parent has to be smaller than s_m (i.e. $s_{p(m)} < s_m$)

$$\begin{aligned} s[n] &= (\underbrace{s_1, ..., s_{p(m)-1}}_{G}, \underbrace{s_{p(m)}, ..., s_{m-1}}_{B}, \underbrace{s_{p(m)}, ..., s_{m-1}}_{B}, ...) \\ &\underset{\text{lex}}{\prec} \left(s_1, ..., s_{p(m)-1}, s_{p(m)}, ..., s_{m-1}, s_m\right) \\ &= (s_1, ..., s_m) = s \end{aligned}$$

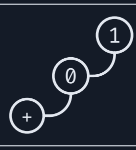
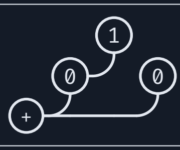
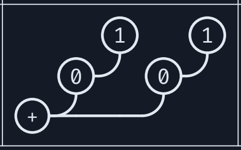
■

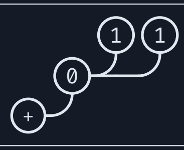
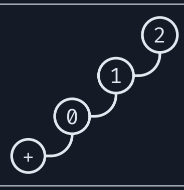
Lemma 6.24.

Expansion (i.e. taking fundamental sequences) does not lead to an infinite descending chain

$$s \underset{\text{lex}}{\succ} s[n_1] \underset{\text{lex}}{\succ} s[n_1][n_2] \underset{\text{lex}}{\succ} ...$$

Proof. We previously saw how we can assign each PrSS sequence a corresponding tree (by taking each number in the PrSS sequence literally as its height), and that we can assign each tree to a corresponding ordinal notation. In this manner we assigned each PrSS sequence to an ordinal:

PrSS	Tree	Ordinal Notation	Ordinal
$()$		0	0
(0)		ω^0	1
$(0, 1)$		ω^{ω^0}	ω
$(0, 1, 0)$		$\omega^{\omega^0} + \omega^0$	$\omega + 1$
$(0, 1, 0, 1)$		$\omega^{\omega^0} + \omega^{\omega^0}$	$\omega 2$

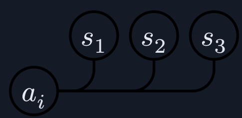
PrSS	Tree	Ordinal Notation	Ordinal
$(0, 1, 1)$		$\omega^{\omega^0+\omega^0}$	ω^2
$(0, 1, 2)$		$\omega^{\omega^{\omega^0}}$	ω^ω

Earlier, we also saw how expansion behaves identically to Kirby-Paris Hydra, which we have already proven its termination by associating each tree with an ordinal. So we can essentially just piggyback off on our termination of Kirby-Paris Hydra proof to prove this. ■

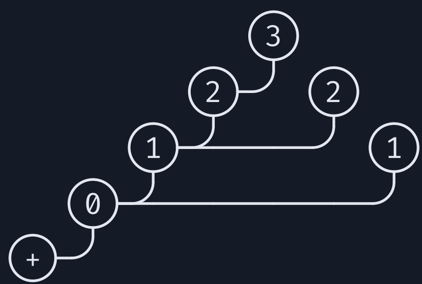
To connect this to our previous definition of OT_{PrSS}, we will show the following:

> **Theorem 6.25.**
>
> For every standard PrSS sequence in OT_{PrSS}, when interpreted as a tree, each node's children are arranged in non-increasing lexicographic order.
>
> Let's call this property *"normality"*. (This is not a formal name, just a name I made up for this property).
>
> This property is analogous to the property of cantor normal form, where the exponents of ω are in non-increasing order.

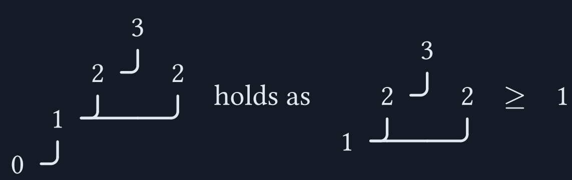
To put this theorem in another way, for any element a_i in a PrSS sequence s, with children subtrees $s_1, s_2, s_3, \ldots$:



If $s \in OT_{\text{PrSS}}$, then $s_1 \underset{\text{lex}}{\succeq} s_2 \underset{\text{lex}}{\succeq} \ldots$. For example, for the sequence $(0, 1, 2, 3, 2, 1)$, we have:



We see that for each of the nodes with multiple children, this property holds:



This property also applies for the root node, for when a sequence has multiple 0s.

Proof. Since $(0, 1, \ldots, n)$ satisfies this property, we just need to show that expansion preserves this property to show that all OT_{PrSS} sequences satisfy this property.

1. If a sequence ends with 0, its fundamental sequence is just the same sequence with the last 0 chopped off, so the normality property is preserved.

   Let $s \in OT_{\text{PrSS}}$ be:

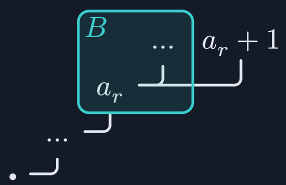
   then we know $s_1 \geq \ldots \geq s_k \geq 0$. So $s[n]$ is just chopping of the last 0:

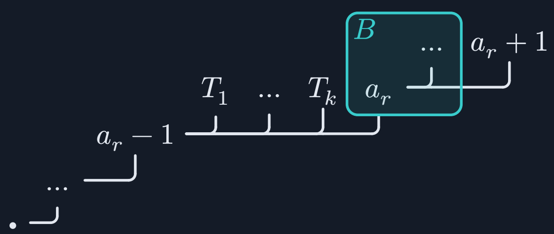
   and we still have $s_1 \geq \ldots \geq s_k$.

2. If a sequence ends with a number > 0, then we proceed with the normal Good Root/Bad Root expansion rules:

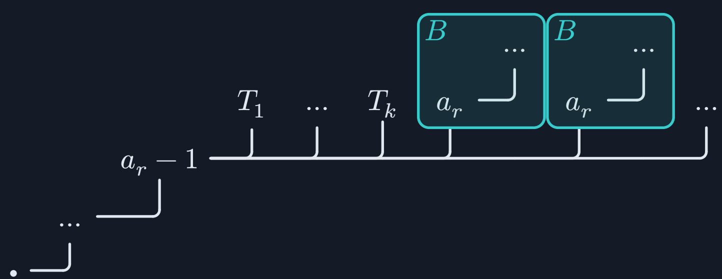
   Let $s \in OT_{\text{PrSS}}$ have its last number be $a_r + 1$, and whose bad root starts with a_r:

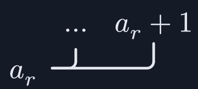
   

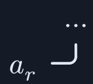
   Zooming out, we see that expansion occurs by copying the bad root subtree onto the rightmost element of value $a_r - 1$ (If $a_r = 0$, it instead adds bad roots onto the root node)

   $\downarrow$

   and since $T_1 \geq \ldots \geq T_k \geq$ 

   $\geq$ 

   our new expanded sequence also satisfies the "normality" property.

As such all sequences in OT_{PrSS} satisfies this "normality" property. ∎

Note that this "normality" property corresponds to the restriction that for a standard form term $\omega^{\alpha_1} + \ldots + \omega^{\alpha_n}$ in ICNF, $\alpha_1 \geq \ldots \geq \alpha_n$.

Theorem 6.26.

$s \sqsubset t \Leftrightarrow s \underset{\text{lex}}{\prec} t$ when $s, t \in OT_{\text{PrSS}}$.

Proof. Since $s[n] \underset{\text{lex}}{\prec} s$, $s \sqsubset t \Rightarrow s \underset{\text{lex}}{\prec} t$ is already shown. We just need show that $s \underset{\text{lex}}{\prec} t \Rightarrow s \sqsubset t$.

Let's define a new function $E(u)$, where $E : OT_{\text{PrSS}} \to OT_{\text{PrSS}}$.

If there exists a k such that $u[k] \geq s$, then $E(u) = \min\{u[k] \mid u[k] \geq s\}$, (i.e. E returns the smallest element of u's fundamental sequence that is greater than or equal to s). Otherwise if no such k exists, $E(u) = u$. Note that this means if $E(u) \neq u$, $E(u) \sqsubset u$

Then we create a sequence of sequences $t_0, t_1, t_2, \ldots$ where $t_0 = t$, and $t_{n+1} = E(t_n)$. Since there are no infinite descending sequences that arise from expansion in OT_{PrSS}, there exists an n such that

$$t_0 > \ldots > t_{n-1} > t_n = t_{n+1} = \ldots$$

So $t_n \geq S$. If $t_n = s$, then we are done, as $s = t_n \sqsubset t_{n-1} \sqsubset \ldots \sqsubset t_0 \Rightarrow s \sqsubset t$. If $t_n > s$, then this means that for all k, $t_n[k] < s$, but yet, $s < t_n$. If we divide t_n into its good root, bad root, and the last element, we realize that s cannot satisfy the "normality" property, a contradiction:

$$t_n = (\underbrace{a_1, \ldots,}_{G} \underbrace{a_r, \ldots,}_{B} a_r + 1)$$

$$t_n[k] = (\underbrace{a_1, \ldots,}_{G} \underbrace{a_r, \ldots,}_{B} \underbrace{a_r, \ldots,}_{B} \underbrace{a_r, \ldots,}_{B} \ldots)$$

$$s = (\underbrace{a_1, \ldots,}_{G} \underbrace{a_r, \ldots,}_{B} \underbrace{a_r, ???}_{X})$$

In this case, s needs to be of the form (G, B, X), and X needs to start with a_r (since $t_n[k] < s$ for all k, X cannot start with a number smaller than a_r), but it cannot start with $a_r + 1$ either (since $s < t_n$). X also needs to be greater than $(B, B, B, \ldots)$, however, this means that among $a_r - 1$'s parents, we have $B < X$, which violates "normality". However $s \in OT_{\text{PrSS}}$, a contradiction.

As such, we have $t_n = s$, and $s \sqsubset t$. ■

We can then define a map $\text{Seq} : \varepsilon_0 \mapsto OT_{\text{PrSS}}$ to show that $(OT, <)$ is order isomorphic to $(\varepsilon_0, \in)$.

For an ordinal $\alpha < \varepsilon_0$, $\text{Seq}(\alpha)$ is defined as such:

1. $\text{Seq}(0) = ()$
2. $\text{Seq}(\omega^{\alpha_1} + \ldots + \omega^{\alpha_n}) = S'(\alpha_1) \oplus \ldots \oplus S'(\alpha_n)$, where $\oplus$ represents concatenation and $S'(\alpha)$ is defined as:

$$S'(\alpha) = S'((a_1, a_2, \ldots, a_l)) = (0, a_1 + 1, a_2 + 1, \ldots, a_l + 1)$$

Lemma 6.27.

Seq is order-preserving, i.e., if $\alpha < \beta$ then $\text{Seq}(\alpha) < \text{Seq}(\beta)$ (for $\alpha < \beta < \varepsilon_0$)

Proof. Suppose:

$$\alpha = \omega^{\alpha_1} + \ldots + \omega^{\alpha_n}$$
$$\beta = \omega^{\beta_1} + \ldots + \omega^{\beta_n}$$

if $\alpha < \beta$, then when we lexicographically compare the sequences of their ω-exponents, $(\alpha_1, \ldots, \alpha_n)$ and $(\beta_1, \ldots, \beta_n)$, and we must have $(\alpha_1, \ldots, \alpha_n) \underset{\text{lex}}{\prec} (\beta_1, \ldots, \beta_n)$. Then:

$$\text{Seq}(\alpha) = S'(\text{Seq}(\alpha_1)) \oplus \ldots \oplus S'(\text{Seq}(\alpha_n))$$
$$\text{Seq}(\beta) = S'(\text{Seq}(\beta_1)) \oplus \ldots \oplus S'(\text{Seq}(\beta_n))$$

, and since $(\alpha_1, ..., \alpha_n) \underset{\text{lex}}{\prec} (\beta_1, ..., \beta_n)$, $\mathrm{Seq}(\alpha) < \mathrm{Seq}(\beta)$ ∎

Lemma 6.28.

Proof.

∎

Lemma 6.29.

7 Veblen Hierarchy

7.1 What lies beyond ε_0?

We previously spent a lot of time on ordinals $< \varepsilon_0$. In this chapter, we finally see what lies beyond ε_0. We first start with $\varepsilon_0 + 1$, then $\varepsilon_0 + 2, \varepsilon_0 + \omega, \varepsilon_0 + \omega^2, \varepsilon_0 + \omega^\omega$, until we get $\varepsilon_0 + \varepsilon_0 = \varepsilon_0 \cdot 2$. We then have the following ordinals, and their expression in Cantor Normal Form (CNF):

(Note that CNF applies to **all** ordinals, not just ordinals $< \varepsilon_0$. It is the *Ordinal Notations* discussed last chapter that have this restriction.)

$$\begin{aligned}
\varepsilon_0 \cdot 3 &= \omega^{\varepsilon_0} + \omega^{\varepsilon_0} + \omega^{\varepsilon_0} \\
\varepsilon_0 \cdot \omega &= \omega^{\varepsilon_0 + 1} \\
\varepsilon_0^2 &= \omega^{\varepsilon_0 \cdot 2} \\
\varepsilon_0^\omega &= \omega^{\omega^{\varepsilon_0 + 1}} \\
\varepsilon_0^{\varepsilon_0} &= \omega^{\omega^{\varepsilon_0 \cdot 2}} \\
\varepsilon_0^{\varepsilon_0^\omega} &= \omega^{\omega^{\omega^{\varepsilon_0 + 1}}} \\
\varepsilon_0^{\varepsilon_0^{\varepsilon_0}} &= \omega^{\omega^{\omega^{\varepsilon_0 \cdot 2}}} \\
\varepsilon_0^{\varepsilon_0^{\varepsilon_0^\omega}} &= \omega^{\omega^{\omega^{\omega^{\varepsilon_0 + 1}}}}
\end{aligned}$$

Note that while $\omega^{\varepsilon_0} = \varepsilon_0, \varepsilon_0^\omega \neq \varepsilon_0$.

$$\begin{aligned}
\omega^{\varepsilon_0} &= \bigcup_{\gamma < \varepsilon_0} (\omega^\gamma) = \overset{+}{\sup}\{\omega, \omega^\omega, \omega^{\omega^\omega}, \omega^{\omega^{\omega^\omega}}, \ldots\} = \varepsilon_0 \\
\varepsilon_0^\omega &= \bigcup_{\gamma < \omega} (\varepsilon_0^\gamma) > \varepsilon_0
\end{aligned}$$

The limit of stacking $\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0+1}}}}$ is ε_1, or more formally:

$$\varepsilon_1 := \overset{+}{\sup}\left\{\varepsilon_0 + 1, \omega^{\varepsilon_0+1}, \omega^{\omega^{\varepsilon_0+1}}, \ldots\right\}$$

ε_1 can also be defined as:

$$\varepsilon_1 = \overset{+}{\sup}\left\{\varepsilon_0, \varepsilon_0^{\varepsilon_0}, \varepsilon_0^{\varepsilon_0^{\varepsilon_0}}, \ldots\right\}$$

However, you may be noticing that these are not the same sequence: $\varepsilon_0^{\varepsilon_0} = \omega^{\omega^{\varepsilon_0 \cdot 2}}$, which is not of the form $\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0+1}}}}$. However, we can show that these two definitions lead to the same supremum of ε_1.

Proof. Let's fix $\varepsilon_1 = \overset{+}{\sup}\left\{\varepsilon_0 + 1, \omega^{\varepsilon_0+1}, \omega^{\omega^{\varepsilon_0+1}}, \ldots\right\}$. Suppose for contradiction that

$$\overset{+}{\sup}\left\{\varepsilon_0, \varepsilon_0^{\varepsilon_0}, \varepsilon_0^{\varepsilon_0^{\varepsilon_0}}, \ldots\right\} > \overset{+}{\sup}\left\{\varepsilon_0 + 1, \omega^{\varepsilon_0+1}, \omega^{\omega^{\varepsilon_0+1}}, \ldots\right\} = \varepsilon_1$$

That means there must exist a term $\varepsilon_0^{\cdot^{\cdot^{\varepsilon_0}}}$ greater than all $\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0+1}}}}$. As seen earlier, we can express $\varepsilon_0^{\cdot^{\cdot^{\varepsilon_0}}}$ as some $\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0 \cdot 2}}}}$. However, since $\varepsilon_0 + 1 \leq \varepsilon_0 \cdot 2 \leq \omega^{\varepsilon_0+1}$,

$$\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0+1}}}} \leq \omega^{\cdot^{\cdot^{\omega^{\varepsilon_0 \cdot 2}}}} \leq \omega^{\cdot^{\cdot^{\omega^{\omega^{\varepsilon_0+1}}}}}$$

So there cannot exist a term $\varepsilon_0^{\cdot^{\cdot^{\varepsilon_0}}}$ greater than all $\omega^{\cdot^{\cdot^{\omega^{\varepsilon_0+1}}}}$, and as such:

$$\overset{+}{\sup}\left\{\varepsilon_0, \varepsilon_0^{\varepsilon_0}, \varepsilon_0^{\varepsilon_0^{\varepsilon_0}}, \ldots\right\} = \overset{+}{\sup}\left\{\varepsilon_0 + 1, \omega^{\varepsilon_0+1}, \omega^{\omega^{\varepsilon_0+1}}, \ldots\right\} = \varepsilon_1$$

In fact, we can see that for any ordinal $\varepsilon_0 < \alpha < \varepsilon_1$:

$$\varepsilon_1 = \overset{+}{\sup}\left\{\omega^{\alpha}, \omega^{\omega^{\alpha}}, \omega^{\omega^{\omega^{\alpha}}}, \ldots\right\}$$

■

The fact that there are a wide variety of sequences with the same supremum means that beyond this point, fundamental sequences for these ordinals do not have an obvious "standard".

ε_1 is the second fixed point of $\omega^{\alpha} = \alpha$. Note that it is also a fixed point of $\varepsilon_0^{\alpha} = \alpha$.

We can repeat this process to get $\varepsilon_2 = \overset{+}{\sup}\left\{\varepsilon_1 + 1, \omega^{\varepsilon+1}, \omega^{\omega^{\varepsilon_1+1}}, \ldots\right\}$ and so on. ε_n is the $(1+n)$th fixed point of $\omega^{\alpha} = \alpha$, and ε_{ω} is the ω^{th} fixed point of $\omega^{\alpha} = \alpha$. Formally:

$$\varepsilon_{\omega} = \overset{+}{\sup}\{\varepsilon_0, \varepsilon_1, \varepsilon_2, \ldots\} = \overset{+}{\sup}\{\varepsilon_n \mid n \in \mathbb{N}\}$$

and in general for a limit ordinal λ, $\varepsilon_{\lambda} = \overset{+}{\sup}\{\varepsilon_{\alpha} \mid \alpha < \lambda\}$

We can keep repeating until we get something like $\varepsilon_{\varepsilon_0}$ and $\varepsilon_{\varepsilon_{\varepsilon_0}}$. Eventually we reach:

$$\zeta_0 = \varepsilon_{\varepsilon_{\varepsilon_{\ddots}}} = \overset{+}{\sup}\left\{\varepsilon_0, \varepsilon_{\varepsilon_0}, \varepsilon_{\varepsilon_{\varepsilon_0}}, \ldots\right\}$$

ζ_0 is the first fixed point of $\varepsilon_{\alpha} = \alpha$, since $\varepsilon_{\zeta_0} = \varepsilon_{\left(\varepsilon_{\varepsilon_{\ddots}}\right)} = \varepsilon_{\varepsilon_{\varepsilon_{\ddots}}} = \zeta_0$. Note that it is also a fixed point of $\omega^{\alpha} = \alpha$, as $\omega^{\zeta_0} = \omega^{\varepsilon_{\zeta_0}} = \varepsilon_{\zeta_0} = \zeta_0$.

What lies beyond ζ_0?

We have the usual $\zeta_0 + 1$, all the way up to $\zeta_0 + \zeta_0 = \zeta_0 \cdot 2$. Then we have $\zeta_0 \cdot n$, all the way up to $\zeta_0 \cdot \omega = \omega^{\zeta_0 + 1}$. We can similarly construct:

$$\varepsilon_{\zeta_0+1} = \overset{+}{\sup}\left\{\zeta_0 + 1, \omega^{\zeta_0+1}, \omega^{\omega^{\zeta_0+1}}, \ldots\right\}$$

In order to climb to the next ζ, we need to nested deeper and deeper εs:

$$\varepsilon_{\varepsilon_{\zeta_0+1}} = \overset{+}{\sup}\left\{\varepsilon_{\zeta_0+1}, \omega^{\varepsilon_{\zeta_0+1}}, \omega^{\omega^{\varepsilon_{\zeta_0+1}}}, \ldots\right\}$$

$$\zeta_1 = \overset{+}{\sup}\left\{\zeta_0 + 1, \varepsilon_{\zeta_0+1}, \varepsilon_{\varepsilon_{\zeta_0+1}}, \varepsilon_{\varepsilon_{\varepsilon_{\zeta_0+1}}}, \ldots\right\}$$

When we finally reach $\zeta_{\zeta_{\zeta_{\ddots}}}$ we can define a new ordinal η_0 the first fixed point of $\zeta_{\alpha} = \alpha$. We can keep going, defining an even larger ordinal such that $\eta_{\alpha} = \alpha$ but eventually we will run our of symbols in this process. but at this point it's clear we need another way to express iteratively using fixed points.

Is there always a fixed point?

We have seen that $\omega^{\alpha} = \alpha$ has a fixed point, and we can convince ourselves it exists by informally saying $\omega^{\varepsilon_0} = \omega^{\omega^{\omega^{\iddots}}} = \omega^{\omega^{\iddots}} = \varepsilon_0$. We can similarly convince ourselves that the fixed point $\varepsilon_{\alpha} = \alpha$ also exists by saying $\varepsilon_{\zeta_0} = \varepsilon_{\left(\varepsilon_{\varepsilon_{\ddots}}\right)} = \varepsilon_{\varepsilon_{\varepsilon_{\ddots}}} = \zeta_0$. But does a fixed point exist for every function? The answer is yes, if the function meets certain criteria:

> **Theorem 7.1 (Veblen's Fixed Point Theorem).**
>
> > **Definition 7.2.**
> >
> > A **normal function** is a function $f : \mathrm{Ord} \mapsto \mathrm{Ord}$ such that:
> >
> > 1. f is **strictly increasing**: $f(\alpha) < f(\beta)$ whenever $\alpha < \beta$
> > 2. f is **continuous**: if β is a limit ordinal, then $f(\beta) = \overset{+}{\sup}\{f(\alpha) \mid \alpha < \beta\}$
>
> If a function is normal, then it has a fixed point $f(\beta) = \beta$. In fact there are infinitely many fixed points $f(\beta) = \beta$ that they make a proper class.

Proof. Let α be an arbitrary ordinal, and left f be a normal function. Now let's define β as:

$$\begin{aligned}\beta &= \overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\} \\ &= \overset{+}{\sup}\{f^0(\alpha) = \alpha, f(\alpha), f(f(\alpha)), f^3(\alpha), ...\}\end{aligned}$$

Where $f^n(\alpha)$ denotes function iteration. We can now show that β is a fixed point:

$$\begin{aligned}f(\beta) &= f\left(\overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\}\right) \\ &= \overset{+}{\sup}\{f^{n+1}(\alpha) \mid n \in \mathbb{N}\} \\ &= \beta\end{aligned}$$

However, this assumes $\alpha \leq f(\alpha)$, which must hold, as otherwise if $\alpha > f(\alpha)$, we can make an infinite descending sequence:

$$\alpha > f(\alpha) > f(f(\alpha)) > f^3(\alpha) > ...$$

However, the ordinals are well-ordered, so such a sequence can't exist.

Since for any ordinal α, $\alpha \leq f(\alpha)$, we can conclude that $\alpha \leq \beta$. We can in fact show that $\beta = \overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\}$ is the **least** fixed point f with the property $\alpha \leq \beta$:

Proof. Let γ be any fixed point of f, so $f(\gamma) = \gamma$, with the property $\alpha \leq \gamma$.

$$\begin{aligned}f(\alpha) &\leq f(\gamma) = \gamma \\ f(f(\alpha)) &\leq f(f(\gamma)) = \gamma \\ f^n(\alpha) &\leq \gamma \\ \beta = \overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\} &\leq \gamma\end{aligned}$$

■

Therefore there is a proper class of fixed points of f, no matter what f is. ■

We can define a function to find these fixed points:

Definition 7.3.

The **derivative** of normal functions f, denoted $f'(\alpha)$, is a function that enumerates over the fixed points of f.

$$f'(\alpha) = \overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\} \leq \gamma$$

Example 7.4.

The derivative of $f(\alpha) = \omega^\alpha$ is $f'(\alpha) = \varepsilon_\alpha$, e.g. ε_0 is the 1st fixed point of $f(\alpha) = \omega^\alpha$

We can generalize this to define a fixed point of multiple functions:

Definition 7.5.

For a family of the function $f_1, f_2, ..., f_k$, the *joint fixed point* of *all* $f_1, f_2, ..., f_k$ is defined as the fixed point of

$$f(\alpha) := f_1(f_2(...f_k(\alpha)))$$

So the $(1 + \alpha)^{\text{th}}$ fixed point of $f_1, f_2, ..., f_k$ is:

$$f'(\alpha) = \overset{+}{\sup}\{f^n(\alpha) \mid n \in \mathbb{N}\}$$

7.2 2-argument Veblen function

Definition 7.6.

The 2-argument/binary Veblen function $\boldsymbol{\varphi(\alpha, \beta)}$ is defined as such:

- $\varphi(0, \gamma) := \omega^\gamma$
- $\varphi(\beta, \gamma)$ for $\alpha \neq 0$ is the $(1+\gamma)^{\text{th}}$ fixed point of the functions $\varphi(\alpha, \bullet) = \bullet$ for $\alpha < \beta$

Since we will be dealing with functions with multiple arguments, it's hard to tell which argument is the one being fixed, so I denote the "dummy variable" of the fixed point to be $\bullet$.

Example 7.7.

For an ordinal α:

$$\begin{aligned} \omega^\alpha &= \varphi(0, \alpha) \\ \varepsilon_\alpha &= \varphi(1, \alpha) \\ \zeta_\alpha &= \varphi(2, \alpha) \\ \eta_\alpha &= \varphi(3, \alpha) \end{aligned}$$

Show that $\varepsilon_0 = \varphi(1, 0)$, and $\zeta_2 = \varphi(2, 2)$

1. $\varphi(1, 0)$ is the $(1+0)^{\text{th}}$ fixed point of the function $\varphi(\gamma, \bullet) = \bullet$ for $\gamma < 1$ (i.e. $\gamma = 0$)
   $\varphi(1, 0)$ is the 1^{st} fixed point of the function $\varphi(0, \bullet) = \omega^\bullet = \bullet$, and the first fixed point of $\omega^\bullet = \bullet$ is ε_0.
2. $\varphi(2, 2)$ is the $(1+2)^{\text{th}}$ fixed point of the function $\varphi(\gamma, \bullet) = \bullet$ for $\gamma < 2$
   $\varphi(2, 2)$ is the 3^{rd} fixed point of the functions $\varphi(0, \bullet) = \omega^\bullet = \bullet$ and $\varphi(1, \bullet) = \varepsilon_\bullet = \bullet$, and ζ_2 is the third ordinal that is a fixed point for *both* functions. (ζ_0, ζ_1 are the first and second).

Let's go through some basic properties of the Veblen function:

Proposition 7.8.

1. If $\alpha < \beta$, then $\varphi(\alpha, \varphi(\beta, \gamma)) = \varphi(\beta, \gamma)$
2. If $\alpha < \beta$, then $\varphi(\gamma, \alpha) < \varphi(\gamma, \beta)$

Both properties are just consequences of the definition provided, so convince yourself that they are true.

Example 7.9.

Which is bigger? $\varphi(2, \varphi(1, 0))$ or $\varphi(3, 0)$?

Of course we can convert them into ordinals that we know (ζ_ω and η_0) but we don't have notation that keeps up.

$\varphi(3, 0)$ can be written as $\varphi(1, \varphi(3, 0))$, which can then be written as $\varphi(2, \varphi(1, \varphi(3, 0)))$ (Property 1)

Then we have:

$$\begin{aligned} 0 &< \varphi(3, 0) \\ \varphi(1, 0) &< \varphi(1, \varphi(3, 0)) \\ \varphi(2, \varphi(1, 0)) &< \varphi(2, \varphi(1, \varphi(3, 0))) = \varphi(3, 0) \end{aligned}$$

> Therefore $\varphi(3,0)$ is bigger.

In the previous section, we defined ε_1 as:

$$\begin{aligned}\varepsilon_1 &= \overset{+}{\sup}\left\{\varepsilon_0+1, \omega^{\varepsilon_0+1}, \omega^{\omega^{\varepsilon_0+1}}, \ldots\right\} \\ &= \overset{+}{\sup}\{\varphi(1,0)+1, \varphi(0,\varphi(1,0)+1), \varphi(0,\varphi(0,\varphi(1,0)+1)), \ldots\} \\ &= \overset{+}{\sup}\{\varphi^n(0,\varphi(1,0)+1) \mid n \in \mathbb{N}\} \\ &= \varphi(1,1)\end{aligned}$$

We also mentioned that instead of $\varphi(1,0)+1$ (i.e. ε_0+1), we can replace that with any ordinal α such that $\varphi(1,0) < \alpha < \varphi(1,1)$ (i.e. $\varepsilon_0 < \alpha < \varepsilon_1$). However, it is a bit self-referential to use $\varphi(1,1)$ in a condition that defines $\varphi(1,1)$. We can instead prove a similar statement for all $\varphi(\alpha+1,\beta+1)$:

> **Proposition 7.10.**
> $\varphi(\alpha+1,\beta+1)$ is the first fixed point of the function $f(\bullet) = \varphi(\alpha,\varphi(\alpha+1,\beta)+\bullet) = \bullet$

Proof. Let ξ be the first (i.e. *least*) fixed point of f, i.e., $f(\xi) = \varphi(\alpha,\varphi(\alpha+1,\beta)+\xi) = \xi$. We now show that $\xi = \varphi(\alpha+1,\beta+1)$

Since $\varphi(\alpha,\gamma) \geq \gamma$ for all γ, we can express ξ as:

$$\xi = \varphi(\alpha, \varphi(\alpha+1,\beta)+\xi) \geq \varphi(\alpha+1,\beta)+\xi$$

This gives us $\xi \geq \varphi(\alpha+1,\beta)+\xi$. Since obviously $\xi \not> \varphi(\alpha+1,\beta)+\xi$, we have:

$$\xi = \varphi(\alpha+1,\beta)+\xi$$

, and for that to be true, $\xi > \varphi(\alpha+1,\beta)$. Applying $\varphi(\alpha,\bullet)$ on both sides we get

$$\begin{aligned}\varphi(\alpha,\xi) &= \varphi(\alpha,\varphi(\alpha+1,\beta)+\xi) \\ &= \xi\end{aligned}$$

Therefore ξ is a fixed point of $\varphi(\alpha,\bullet)$, i.e., $\xi = \varphi(\alpha+1,\gamma)$ for some ordinal γ. And since $\xi > \varphi(\alpha+1,\beta)$, minimally $\xi \geq \varphi(\alpha+1,\beta+1)$.

We also notice that $\varphi(\alpha+1,\beta+1)$ is a fixed point of f:

$$\begin{aligned}f(\varphi(\alpha+1,\beta+1)) &= \varphi(\alpha,\varphi(\alpha+1,\beta)+\varphi(\alpha+1,\beta+1)) \\ &= \varphi(\alpha,\varphi(\alpha+1,\beta+1)) \\ &= \varphi(\alpha+1,\beta+1)\end{aligned}$$

Since ξ is the *least* fixed point of f, $\xi \leq \varphi(\alpha+1,\beta)$. Since we have just shown earlier that $\xi \geq \varphi(\alpha+1,\beta+1)$, we get:

$$\xi = \varphi(\alpha+1,\beta+1)$$

■

Note that with this definition our ε_1 sequence would be translated as:

$$\begin{aligned}\varphi(1,1) &= \overset{+}{\sup}\{f^{n(1)} \mid n \in \mathbb{N}\} \\ &= \overset{+}{\sup}\{\varphi(0, \varphi(1,0)+1), \varphi(0, \varphi(1,0)+\varphi(0,\varphi(1,0)+1)), ...\} \\ &= \overset{+}{\sup}\left\{\omega^{\varepsilon_0+1}, \omega^{\varepsilon_0+\omega^{\varepsilon_0+1}}, \omega^{\varepsilon_0+\omega^{\varepsilon_0+\omega^{\varepsilon_0+1}}}, ...\right\} \\ &= \overset{+}{\sup}\left\{\omega^{\varepsilon_0+1}, \omega^{\omega^{\varepsilon_0+1}}, \omega^{\omega^{\omega^{\varepsilon_0+1}}}, ...\right\}\end{aligned}$$

Now that we have defined the 2-argument Veblen function, what is the largest ordinal we can make? After η_0 we have $\varphi(4,0)$, $\varphi(10,0)$, or even $\varphi(10000,0)$. What about $\varphi(\omega,0)$? We can think of it as the first fixed point for *all* $\varphi(n, \bullet) = \bullet$ for every $n \in \mathbb{N}$, or as the supremum of all $\varphi(n,0)$:

$$\varphi(\omega,0) = \overset{+}{\sup}\{\varphi(n,0) \mid n \in \mathbb{N}\}$$

It should be obvious to deduce that it is also the supremum of all $\varphi(n,1)$ too, i.e., the value of the second argument does not matter. In general, for a limit ordinal λ,

$$\varphi(\lambda,0) = \overset{+}{\sup}\{\varphi(\alpha,0) \mid \alpha < \lambda\}$$

Continuing where we left off, we have $\varphi(\omega,0) = \varphi(\varphi(1,0),0)$, and we can keep nesting $\varphi(\varphi(\varphi(1,0),0),0)$, $\varphi(\varphi(\varphi(\varphi(1,0),0),0),0)$, until we reach the (first) fixed point of the Veblen function: $\Gamma_0 = \varphi(\Gamma_0, 0)$. Γ_0 is known as the **Feferman–Schütte ordinal**, which can also be expressed as:

$$\Gamma_0 = \overset{+}{\sup}\{\varphi(1,0), \varphi(\varphi(1,0),0), \varphi(\varphi(\varphi(1,0),0),0), ...\}$$

This is the largest ordinal we can reach with the two-argument veblen function.

7.3 Veblen Normal Form and Fundamental Sequences

Much like how all ordinals $< \varepsilon_0$ can be expressed in Cantor Normal Form, all ordinals $< \Gamma_0$ can be expressed in:

Theorem 7.11 (Veblen Normal Form).

Every ordinal $\alpha < \Gamma_0$, can be uniquely written in the following form:

$$\varphi(\beta_1, \gamma_1) + \varphi(\beta_2, \gamma_2) + ... + \varphi(\beta_k, \gamma_k)$$

where $\varphi(\beta_1, \gamma_1) \geq \varphi(\beta_2, \gamma_2) \geq ... \geq \varphi(\beta_k, \gamma_k)$ and $\gamma_i < \varphi(\beta_i, \gamma_i)$ for $i \in \{1, ..., k\}$

Definition 7.12.

The **Veblen Hierarchy** is a system of fundamental sequences for **limit** ordinals $\alpha < \Gamma_0$ when expressed in Veblen Normal Form:

1. $(\varphi(\beta_1, \gamma_1) + \varphi(\beta_2, \gamma_2) + ... + \varphi(\beta_k, \gamma_k))[n] := \varphi(\beta_1, \gamma_1) + \varphi(\beta_2, \gamma_2) + ... + \varphi(\beta_k, \gamma_k)[n]$
2. $\varphi(0, \gamma+1) := \omega^\gamma \cdot n$
3. $\varphi(\beta+1, 0)[n] := \varphi^n(\beta, 0)$
4. $\varphi(\beta+1, \gamma+1)[n] := \varphi^n(\beta, \varphi(\beta+1, \gamma)+1)$
5. $\varphi(\beta, \gamma)[n] := \varphi(\beta, \gamma[n])$ when γ is a limit ordinal
6. $\varphi(\beta, 0)[n] := \varphi(\beta[n], 0)$ when β is a limit ordinal
7. $\varphi(\beta, \gamma+1)[n] := \varphi(\beta[n], \varphi(\beta, \gamma)+1)$ when β is a limit ordinal

In rule 3 and 4, $\varphi^n()$ refers to function iteration, so $\varphi^n(\beta, \gamma) = \varphi(\beta, \varphi^{n-1}(\beta, \gamma))$ and $\varphi^1(\beta, \gamma) = \varphi(\beta, \gamma)$

Example 7.13.

- $\varphi(2,0)[n]$

$$\varphi(2,0)[1] = \varphi^1(1,0) = \varphi(1,0) = \varepsilon_0$$
$$\varphi(2,0)[2] = \varphi^2(1,0) = \varphi(1,\varphi(1,0)) = \varepsilon_{\varepsilon_0}$$
$$\varphi(2,0)[3] = \varphi^3(1,0) = \varphi(1,\varphi(1,\varphi(1,0))) = \varepsilon_{\varepsilon_{\varepsilon_0}}$$

- $\varphi(2,1)[n]$

$$\varphi(2,1)[1] = \varphi^1(1,\varphi(2,0)+1) = \varepsilon_{\zeta_0+1}$$
$$\varphi(2,1)[2] = \varphi^2(1,\varphi(2,0)+1) = \varphi(1,\varphi(1,\varphi(2,0)+1)) = \varepsilon_{\varepsilon_{\zeta_0+1}}$$
$$\varphi(2,1)[3] = \varphi^3(1,\varphi(2,0)+1) = \varphi(1,\varphi(1,\varphi(1,\varphi(2,0)+1))) = \varepsilon_{\varepsilon_{\varepsilon_{\zeta_0+1}}}$$

- $\varphi(\omega,0)[n]$

$$\varphi(\omega,0)[n] = \varphi(\omega[n],0) = \varphi(n,0)$$

- $\varphi(\omega,1)[n]$

$$\varphi(\omega,1)[n] = \varphi(\omega[n],\varphi(\omega,0)+1) = \varphi(n,\varphi(\omega,0)+1)$$

7.4 Multivariate Veblen function

3-argument veblen function

So how do we go beyond Γ_0? Γ_0 is the 1^{st} fixed point of $\varphi(\bullet,0) = \bullet$. Γ_1 is therefore the 2^{nd} fixed point, and Γ_2 is the 3^{rd}, and so on. This "Gamma function" can be thought of as $\Gamma_\beta = (1+\beta)^{\text{th}}$ fixed point of $\varphi(\bullet,0) = \bullet$. So we can make Γ_ω, Γ_{ε_0}, or even Γ_{Γ_0}. We have seen this pattern before – eventually we reach the *Gamma fixed point* of $\Gamma_\bullet = \bullet$, which can be though of as an infinite nesting of Gammas: $\Gamma_{\Gamma_{\ddots}}$.
We can then iterate over the fixed points of $\Gamma_\bullet = \bullet$, but we can systematize this with the *3-argument veblen function*. We let $\Gamma_0 = \varphi(1,0,0)$, and then $\Gamma_1 = \varphi(1,0,1)$. In general $\Gamma_\beta = \varphi(1,0,\beta)$, and first Gamma fixed point $\Gamma_{\Gamma_{\ddots}} = \varphi(1,1,0)$ can be denoted as the first fixed point of $\varphi(1,0,\bullet) = \bullet$.
We can create a definition of the 3-argument veblen function as such:

Definition 7.14.

- $\varphi(0,\beta,\gamma) = \varphi(\beta,\gamma)$, converting from the 3-argument veblen to the 2-argument veblen
- $\varphi(\alpha,0,\gamma)$ is the $(1+\gamma)^{\text{th}}$ fixed point of $\varphi(\alpha',\bullet,0) = \bullet$ for all $\alpha' < \alpha$
- $\varphi(\alpha,\beta,\gamma), \beta \neq 0$ is the $(1+\gamma)^{\text{th}}$ fixed point of $\varphi(\alpha,\beta',\bullet)$ for all $\beta' < \beta$

Example 7.15.

$\varphi(2,0,0)$ is the 1^{st} fixed point of $\varphi(\alpha',\bullet,0) = \bullet$ for all $\alpha' < \alpha = 2$, so $\alpha' = \{0,1\}$
- $\varphi(0,\bullet,0) = \bullet$
- $\varphi(1,\bullet,0) = \bullet$

Extension to finitely many arguments

Now what about a 4-argument veblen function? or an n-argument veblen function? We can extend it to any number of arguments as such:

Definition 7.16.

Let S be an arbitrary string of ordinals, and O be a string of 0s.

- $\varphi(\gamma) = \gamma$
- $\varphi(O, S) = \varphi(S)$, i.e., you can remove zeroes from the left
- $\varphi(S, \beta, O, \gamma)$ is the $(1+\gamma)^{\text{th}}$ fixed point of $\varphi(S, \alpha, \bullet, O) = \bullet$ for all $\alpha < \beta$

Example 7.17.

1. $\varphi(1, 0, 0, 0)$

   We have $S = \text{empty}, \beta = 1, O = (0, 0), \gamma = 1$:

   $$\varphi\left(\underbrace{1}_{\beta}, \underbrace{0, 0}_{O}, \underbrace{0}_{\gamma}\right)$$

   As such $\varphi(1, 0, 0, 0)$ is the $(1+0)^{\text{th}}$ fixed point of $\varphi(\alpha, \bullet, 0, 0) = \bullet$ for all $\alpha < 1$, i.e. $\alpha = 0$. Rephrasing, we have:

   $\varphi(1, 0, 0, 0)$ is the 1^{st} fixed point of $\varphi(\bullet, 0, 0) = \bullet$.
2. $\varphi(1, 2, 3, 4)$ We have $S = (1, 2), \beta = 3, O = \text{empty}, \gamma = 4$

   As such $\varphi(1, 2, 3, 4)$ is the $(1+4)^{\text{th}}$ fixed point of $\varphi(1, 2, \alpha, \bullet) = \bullet$ for all $\alpha < 3$. Rephrasing, we have:

   $\varphi(1, 2, 3, 4)$ is the 5^{th} fixed point of $\varphi(1, 2, \alpha, \bullet) = \bullet$ for $\alpha \in \{0, 1, 2\}$.

With this, we can define ordinals like $\varphi(1, 0, 0, 0)$, $\varphi(1, 0, 0, 0, 0)$, and so on. Taking this to its limit yields the **Small Veblen Ordinal**:

$$\text{SVO} = \sup^{+}\{\varphi(1), \varphi(1, 0), \varphi(1, 0, 0), \varphi(1, 0, 0, 0), \varphi(1, 0, 0, 0, 0), \ldots\}$$

We also have an "Extended Veblen Normal Form":

Theorem 7.18.

Every ordinal $\alpha < \text{SVO}$, can be uniquely written in the following form:

$$\varphi(s_1) + \varphi(s_2) + \ldots + \varphi(s_k)$$

where:

- $\varphi(s_1) \geq \varphi(s_2) \geq \ldots \geq \varphi(s_k)$
- s_i is an arbitrary string of ordinals $\alpha_{i,1}, \alpha_{i,2}, \ldots, \alpha_{i,k_i}$ where $i \in \{1, \ldots, k\}$
- $\alpha_{i,1} > 0$ and $\alpha_{i,j} < \varphi(s_i)$ for all $i \in \{1, \ldots, k\}$ and $j \in \{1, \ldots, k_i\}$

However, I'm going to skip straight to the veblen function with *transfinitely* many arguments, as that generalizes to finitely many arguments too.

7.5 Transfinitary Veblen function

Finitely Supported functions

Definition 7.19.

A function $f : X \mapsto Y$ if a **finitely supported** if there are only *finitely many* values of $x \in X$ such that $f(x) \neq 0$ (assuming $0 \in Y$).

We call the set of values $x \in X$ such that $f(x) \neq 0$ the *support* of the function. $\{x | x \in X \text{ and } f(x) \neq 0\})$

If $f : X \mapsto Y$ is finitely supported, we write it as $f : X \mapsto_0 Y$.

Example 7.20.

$$f(\beta) = \begin{cases} 2 \text{ if } \beta = 0 \\ 3 \text{ if } \beta = 1 \\ 0 \text{ else} \end{cases}$$

is an example of a finitely supported function from $\mathrm{Ord} \mapsto_0 \mathrm{Ord}$

We use finitely supported functions for this as our veblen function will now *take in* a finitely supported function $f : \mathrm{Ord} \mapsto_0 \mathrm{Ord}$, so our veblen function can be thought of as $\varphi : (\mathrm{Ord} \mapsto_0 \mathrm{Ord}) \mapsto \mathrm{Ord}$.

Example 7.21.

In our previous example, we defined an f where $f(0) = 2$, $f(1) = 3$, $f(\text{anything else}) = 0$. Then

$$\varphi(f) = \varphi(f(1), f(0)) = \varphi(3, 2)$$

and in general, the arguments are arranged "right to left":

$$\varphi(f) = \varphi(..., f(2), f(1), f(0))$$

The easiest way to create a new finitely supported function from another is to "update" one of their values:

Definition 7.22.

For a finitely supported function f, $f[\alpha@\beta]$ is defined as such:

$$f[\alpha@\beta](\xi) := \begin{cases} f(\xi) & \text{if } \xi \neq \beta \\ \alpha & \text{if } \xi = \beta \end{cases}$$

You can think of $f[\alpha@\beta]$ as "f with α at β".

As shorthands:

1. If $f(\xi) = 0$ for all ξ, i.e., f is the "zero function", then $f[\alpha@\beta]$ can be written as just $[\alpha@\beta]$
2. $f[\alpha_1@\beta_1][\alpha_2@\beta_2]...[\alpha_n@\beta_n]$ can be written as $f[\alpha_1@\beta_1, \alpha_2@\beta_2, ..., \alpha_n@\beta_n]$

Proposition 7.23.

The set of all finitely supported functions $\mathrm{Ord} \mapsto_0 \mathrm{Ord}$ is well-ordered as such:

For two finitely supported functions $f, g : \mathrm{Ord} \mapsto_0 \mathrm{Ord}$, $f < g$ if for some ordinal α:

1. $f(\beta) = g(\beta)$ for all $\beta < \alpha$, and
2. $f(\alpha) < g(\alpha)$

This should be fairly easy to show, it is quite similar to how $\mathbb{N}^2$ is well-ordered.

Definition 7.24.

The **degree** $\deg(f)$ of a finitely supported function $f : X \mapsto_0 Y$ is the smallest $d \in X$ such that $f(d) \neq 0$ (i.e. the smallest element of X that is in the support of f)

Definition

Definition 7.25.

The multivariate Veblen function is the function $\varphi : (\mathrm{Ord} \mapsto_0 \mathrm{Ord}) \mapsto \mathrm{Ord}$ defined as follows: Let $f : \mathrm{Ord} \mapsto_0 \mathrm{Ord}$, and let $g = f[0@0]$. If g is the zero function, we set $\varphi(f) = \omega^{f(0)}$. Otherwise, let $d = \deg(g)$. We define $\varphi(f)$ as the $(1 + f(0))^{\text{th}}$ common fixed point of the functions $\varphi(g[\alpha@d, \bullet@d']) = \bullet$ for all $\alpha < f(d)$ and $d' < d$.

Example 7.26.
$\varphi(1, 1, 0) = \varphi(f)$ where f is:

$$f(\xi) = \begin{cases} 0 \text{ if } \xi = 0 \\ 1 \text{ if } \xi = 1 \\ 1 \text{ if } \xi = 2 \\ 0 \text{ else} \end{cases}$$

$g = f[0@0]$ basically sets $g(0) = 0$, which doesnt change anything since $f(0)$ is already 0. So g is not the zero function.

$d = \deg(f) = 1$. So $\varphi(f)$ is the $(1 + f(0))^{\text{th}} = 1^{\text{st}}$ common fixed point of: $\varphi(g[\alpha@d, \bullet@d']) = \bullet$ for all $\alpha < f(d) = 1$ and $d' < d = 1$. So we are basically stuck with $\alpha = 0$ and $d' = 0$, so $\varphi(1, 1, 0)$ is the first fixed point of $\varphi(f[0@1, \bullet@0]) = \varphi(1, 0, \bullet) = \bullet$

Example 7.27.
$\varphi([1@\omega]) = \varphi(f)$ where f is:

$$f(\xi) = \begin{cases} 1 \text{ if } \xi = \omega \\ 0 \text{ else} \end{cases}$$

$g = f[0@0] = f$ which doesn't change anything since $f(0) = 0$. $\deg(f) = \omega$, so from the definition, $\varphi(f)$ is the 1^{st} common fixed point of all functions $\varphi(g[\alpha@d, \bullet@d']) = \bullet$ for all $\alpha < f(d) = 1$ and $d' < d = \omega$. Plugging this in yields $\varphi(g[0@\omega, \bullet@d']) = \bullet$ for all $d' < \omega$, or put another way it is the common fixed points of all $\varphi(1@n)$ for $n \in \mathbb{N}$.

This is equivalent to the *Small Veblen Ordinal*.

But how do we even know that these fixed points exist? We need to first show that the multivariate veblen are all normal functions.

Proposition 7.28.
If $f : \mathrm{Ord} \mapsto_0 \mathrm{Ord}$, and $\deg(f) \geq d$, then $g(\alpha) = \varphi(f[\alpha@d])$ is a non-negative normal function.

Proof. [WIP] ∎

Now that we have extended the veblen function yet again, what is the new limit? We can construct $\mathrm{SVO} = \varphi([1@\omega])$, but why stop at ω? We can have $\varphi([1@\Gamma_0])$, $\varphi([1@\mathrm{SVO}]) = \varphi([1@\varphi([1@\omega])])$, or even $\varphi([1@\varphi([1@\varphi([1@\omega])])])$. I think we have seen this pattern enough times by now – we can make a fixed point $\varphi([1@\bullet]) = \bullet$, and the (first) ordinal that satisfies is known as the **Large Veblen Ordinal** LVO. If we let $f(\alpha) = \varphi([1@\alpha])$, it can also be written as:

$$\begin{aligned} \mathrm{LVO} &= \overset{+}{\sup}\{f^n(0) \mid n \in \mathbb{N}\} \\ &= \overset{+}{\sup}\{f(0), f(f(0)), f(f(f(0))), ...\} \\ &= \overset{+}{\sup}\{\varphi([1@0]) = \omega, \varphi([1@\varphi[1@0]]) = \mathrm{SVO}, \varphi([1@\varphi[1@\varphi([1@0])]]), ...\} \end{aligned}$$

We do have to first show that $\varphi([1@\alpha])$ is normal before we know that such a fixed point exists:

Proposition 7.29.

The function $f(\alpha) = \varphi([1@\alpha])$ is normal.

Proof. We need to show that it is:

1. strictly increasing

   $\varphi([1@y]) = \varphi([\varphi([1@y])@x]) > \varphi([1@x])$ (why? [WIP])
2. continuous [WIP]

■

Theorem 7.30 (Tranfinitary Veblen Normal Form).

All ordinals $\alpha < \mathrm{LVO}$ can be expressed as

Definition 7.31.

The fundamental sequences for the multivariate veblen function is: [WIP]

Schьtte's Klammersymbolen

This is another way to write the veblen function with transfinitely many arguments. If we have a finitely covered function f like so:

$$f(\xi) = \begin{cases} \alpha_0 & \text{if } \xi = 0 \\ \alpha_1 & \text{if } \xi = 1 \\ & \vdots \\ \alpha_\beta & \text{if } \xi = \beta \\ 0 & \text{else} \end{cases}$$

We can "decompose" it to $[\alpha_0@0, \alpha_1@1, ..., \alpha_\beta@\beta]$. In Schütte's Klammersymbolen, we write this as:

$$\begin{pmatrix} \alpha_\beta & \dots & \alpha_1 & \alpha_0 \\ \beta & \dots & 1 & 0 \end{pmatrix}$$

So for example, the Small Veblen Ordinal would be $\varphi\binom{1}{\omega}$, and the Large Veblen Ordinal would be represented as the following fixed point:

$$\mathrm{LVO} = \varphi\begin{pmatrix} 1 \\ \mathrm{LVO} \end{pmatrix}$$

7.6 Dimensional Veblen

Some fellas made another extension to the Veblen function. I won't cover it here but you can check it out here: https://arxiv.org/pdf/2310.12832

8 Buchholz's Ordinal Collapsing Function

Buchholz's **Ordinal Collapsing Function** (BOCF) is a way for us to create even larger (countable) ordinals. Ironically, to make these larger ordinals, we will be using even larger *uncountable* ordinals to get there.

But first, how exactly are ordinals of higher cardinality (ω_1, $\omega_2, \ldots$) defined? Let's first define the *successor cardinal* of $\aleph_0$, $\aleph_1$, and its assigned ordinal ω_1 as **the set of all ordinals whose cardinality $\leq \aleph_0$**:

$$\aleph_1 = \omega_1 = \{\alpha \in \mathrm{Ord} : |\alpha| \leq \aleph_0\}$$

where $|\alpha|$ represents the cardinality of an ordinal α. Similarly, we have the cardinal $\aleph_2$ and its assigned ordinal ω_2 to be the set of all ordinals whose cardinality is $\leq \aleph_1$. We can repeat this to make $\aleph_3, \aleph_4, \ldots$. For a limit ordinal λ, we define $\aleph_\lambda$ as such:

$$\aleph_\lambda = \bigcup_{\beta<\lambda} \aleph_\beta$$

Definition 8.1 (Buchholz's ψ function).

Let Ω_ν be defined as such:

$$\Omega_\nu := \begin{cases} 1 & \text{if } \nu = 0 \\ \omega_\nu & \text{if } \nu > 0 \end{cases}$$

Buchholz's ordinal collapsing function ψ and the set $C_\nu(\alpha)$ is defined as such:

1. $\Omega_\nu \subseteq C_\nu(\alpha)$
2. For any two ordinals $\xi, \eta \in C_\nu(\alpha)$, their sum $\xi + \eta \in C_\nu(\alpha)$
3. For any ordinal $\xi \in C_\nu(\alpha)$, as long as $\xi < \alpha$, then $\psi_\mu(\xi) \in C_\nu(\alpha)$ for all $\mu \leq \omega$
4. $\psi_\nu(\alpha) = \min\{\gamma \in \mathrm{Ord} \mid \gamma \notin C_\nu(\alpha)\}$, i.e., the smallest ordinal not inside $C_\nu(\alpha)$

This is a lot to unpack, so let's first start with $\nu = 0$:

Example 8.2.

Let's first find what $\psi_0(0)$ is.

In order to figure out what $\psi_0(0)$, we first need to know what's inside $C_0(0)$. Going down the list, we have:

1. $\Omega_0 = 1 = \{0\} \subseteq C_0(0)$

   so currently $C_0(0) = \{0\}$
2. For any two ordinals in $C_0(0)$, their sum is also in $C_0(0)$

   Another way to interpret this statement is to say that you can take a collection of ordinals in $C_0(0)$ and apply addition a finite number of times on them. In this case, all we have is 0, so nothing new gets added here.
3. For any ordinal $\xi \in C_0(0)$, as long as $\xi < 0$, then $\psi_\mu(\xi) \in C_0(0)$ for all $\mu \leq \omega$

   In this case, there are no ordinals $\xi < 0$ so, nothing new get added here either.
4. $\psi_0(0)$ is the smallest ordinal *not* contained inside $C_0(0) = \{0\}$, which is just 1.

In fact, we can generalize this and show that:

Lemma 8.3.

$\psi_\nu(0) = \Omega_\nu$

Proof. From the definition, $\Omega_\nu \subseteq C_\nu(0)$, i.e. for all $\alpha \in \Omega_\nu, \alpha \in C_\nu(0)$.

For all $\alpha, \beta \in \Omega_\nu, \alpha + \beta \in \Omega_\nu$, so adding doesn't generate us anything new.

Since there are no ordinals $\xi < 0$, we can't use ψ to generate us anything new.

So $C_\nu(0) = \{\gamma \in \mathrm{Ord} \mid \gamma < \Omega_\nu\}$. Then $\psi_\nu(0)$ is the smallest ordinal not in $C_\nu(0)$ which is Ω_ν. ■

Example 8.4.

Let's proceed to find the value of $\psi_0(1)$:

1. $\Omega_0 \subseteq C_0(0)$, so we once again have $0 \in C_0(0)$.
2. Adding doesn't do anything

The difference this time is that we can use $\psi_0(0)$:

3. For any ordinal $\xi \in C_0(0)$, as long as $\xi < 1$, then $\psi_\mu(\xi) \in C_0(0)$ for all $\mu \leq \omega$

   The only valid values of ξ is just 0, so we have $\psi_0(0), \psi_1(0), ..., \psi_\omega(0) \in C_0(0)$. $\psi_0(0) = 1 \in C_0(0)$, so we finally have something new. We haven't gotten to $\psi_1(0), \psi_2(0), ..., \psi_\omega(0)$ yet, and we'll ignore them for now.
2. We can now use addition to show that $1 + 1 = 2 \in C_0(0), 2 + 1 = 3 \in C_0(0)$, and so on to show that all natural numbers $\mathbb{N} \subseteq C_0(0)$. In fact, $\mathbb{N} = C_0(0)$.
4. $\psi_0(1)$ is the smallest ordinal not in $C_0(0)$, which is ω

8.1 Properties

Before we continue further, let's look at some properties about this function. Let's first introduce the concept of an **additively principal** ordinal:

Definition 8.5.

An **additively principal** ordinal α is any non-zero ordinal such that for any $\beta, \gamma < \alpha, \beta + \gamma < \alpha$.

We denote the *class* of all additively principal numbers as P. P is precisely all ordinals of the form ω^ξ, where ξ is some ordinal.

Formally, we can define P as such:

$$\begin{aligned} P &:= \{\alpha \in \mathrm{Ord} \mid 0 < \alpha \text{ and for all } \beta, \gamma < \alpha, \beta + \gamma < \alpha\} \\ &= \{\omega^\xi \mid \xi \in \mathrm{Ord}\} \end{aligned}$$

We can also define a function $P(\alpha)$ that "decomposes" an ordinal α into its additively principal "components":

Definition 8.6.

We define $P(\alpha)$ as:

1. $P(0) := \emptyset$
2. For $\alpha > 0$, there are uniquely determined $\alpha_0, \alpha_1, ..., \alpha_n \in P$ with $\alpha = \alpha_0 + \alpha_1 + ... + \alpha_n$.

   We set $P(\alpha) := \{\alpha_0, \alpha_1, ..., \alpha_n\}$

This notion is useful as the Buchholz function ψ, as each $C_\nu(\alpha)$ represents all ordinals that can be derived from addition and the ψ function itself:

Lemma 8.7.

1. If $\zeta, \eta \in C_\nu(\alpha)$, then $\zeta + \eta \in C_\nu(\alpha)$
2. An ordinal $\gamma \in C_\nu(\alpha)$ if and only if $P(\gamma) \subseteq C_\nu(\alpha)$
3. If $\zeta + \eta \in C_\nu(\alpha)$, then $\eta \in C_\nu(\alpha)$

Proof.

1. In the definition
2. Since $\gamma = \gamma_0 + \gamma_1 + ... + \gamma_n$ where $\gamma_i \in P(\gamma)$, and $P(\gamma) \subseteq C_\nu(\alpha)$, then $\gamma_i \in C_\nu(\alpha)$ for all $i \leq n$, and as such, we can "reconstruct" γ from elements of $C_\nu(\alpha)$:
$$\gamma = \gamma_0 + ... + \gamma_n \in C_\nu(\gamma)$$
3. Since $\xi + \eta \in C_\nu(\alpha)$, we have $P(\eta) \subseteq P(\eta + \xi) \subseteq C_\nu(\alpha)$, and thus, $\eta \in C_\nu(\alpha)$

■

We can additionally show that the ψ function only outputs additively principal ordinals:

Lemma 8.8.

$\psi_\nu(\alpha) \in P$ for all ordinals ν, α.

Proof. Suppose $\psi_\nu(\alpha) \notin P$. Then $P(\psi_\nu(\alpha)) \subseteq \psi_\nu(\alpha)$, which is to say, for every element $p \in P(\psi_\nu(\alpha))$, $p < \psi_\nu(\alpha)$, which also means $p \in C_\nu(\alpha)$, thus $P(\psi_\nu(\alpha)) \subseteq C_\nu(\alpha)$. However, this implies $\psi_\nu(\alpha) \in C_\nu(\alpha)$, a contradiction. ■

Another interesting property of this function is that subscript of ψ represents the cardinality of it's result:

Proposition 8.9.

$\Omega_\nu \leq \psi_\nu(\alpha) < \Omega_{\nu+1}$ for all ordinals α.

Proof. Since $\Omega_\nu \subseteq C_\nu(\alpha)$, we get $\Omega_\nu \leq \psi_\nu(\alpha)$. Since the cardinality of $C_\nu(\alpha)$ is less than $\Omega_{\nu+1}$, we have $\psi_\nu(\alpha) < \Omega_{\nu+1}$. ■

So ψ_0 will only spit out countable ordinals $1 \leq \psi_0(\alpha) < \Omega$, ψ_1 will only be in the range $\Omega \leq \psi_1 < \Omega_2$, etc....

8.2 Bigger ordinals

Now let's look at $\psi_0(2)$. Since $C_0(1) \subseteq C_0(2)$, we will mainly focus on what *additional* elements are contained with $C_0(2)$. For starters, since $1 < 2$, we can access $\psi_0(1) = \omega, \psi_1(1), ...$. Since we can also use addition, ordinals like $\omega + 1, \omega + \omega$, or even $\omega \cdot 1000$ are within $C_0(2)$. However, as we cannot add ω to itself transfinitely many times, the least ordinal not contained within $C_0(2)$ is $\psi_0(2) = \omega^2$.

So far, as we have increased α, $\psi_0(\alpha)$ has increased. In fact, this is true for any ψ_ν:

Lemma 8.10.

1. If $\alpha \leq \beta$, then $C_\nu(\alpha) \subseteq C_\nu(\beta)$ and $\psi_\nu(\alpha) \leq \psi_\nu(\beta)$
2. If $\alpha < \beta$ and $\alpha \in C_\nu(\alpha)$, then $\psi_\nu(\alpha) < \psi_\nu(\beta)$

Proof.

1. Trivial
2. From the previous statement, we can conclude that $C_\nu(\alpha) \subseteq C_\nu(\beta)$ and $\psi_\nu(\alpha) \leq \psi_\nu(\beta)$. Since $C_\nu(\alpha) \subseteq C_\nu(\beta)$ and $\alpha \in C_\nu(\alpha)$, $\alpha \in C_\nu(\beta)$, and since $\alpha < \beta$, $\psi_\nu(\alpha) \in C_\nu(\beta)$

■

Lemma 8.11.

If $\gamma = \psi_{\nu_1}(\alpha_1) = \psi_{\nu_2}(\alpha_2)$, then $\nu_1 = \nu_2$ and $\alpha_1 = \alpha_2$, i.e., all outputs of ψ are uniquely represented.

Proof. Left as an exercise to the reader ■

Continuing from where we left off, we can repeat this for $C_0(3)$ to get $\psi_0(3) = \omega^3$, $\psi_0(4) = \omega^4$ and so on. It seems like $\psi_0(\alpha) = \omega^\alpha$, and we can prove that fact: We first prove a property about how $\psi_\nu(\alpha + 1)$ relates to $\psi_\nu(\alpha)$:

Lemma 8.12.

1. $\psi_\nu(\alpha + 1) = \min\{\gamma \in P \mid \psi_\nu(\alpha) < \gamma\}$ if $\alpha \in C_\nu(\alpha)$
2. $\psi_\nu(\alpha + 1) = \psi_\nu(\alpha)$ if $\alpha \notin C_\nu(\alpha)$
3. If α is a limit ordinal, $\psi_\nu(\alpha) = \overset{+}{\sup}\{\psi_\nu(\xi) \mid \xi < \alpha \text{ and } \xi \in C_\nu(\xi)\}$

Proof. [WIP]

1. Since $\alpha \in C_\nu(\alpha)$, and $\alpha \in C_\nu(\alpha)$, $\psi_\nu(\alpha) < \psi_\nu(\alpha + 1)$. Suppose for contradiction there exists an ordinal γ such that $\psi_\nu(\alpha) \leq \gamma < \psi_\nu(\alpha + 1)$, and $\gamma \in P$.

■

Lemma 8.13.

1. For all ordinals $\alpha < \varepsilon_0$, $\alpha \in C_0(\alpha)$ and $\psi_0(\alpha) = \omega^\alpha$
2. For all ordinals $\alpha < \varepsilon_{\Omega_\nu + 1}$ where $\nu \neq 0$, $\alpha \in C_\nu(\alpha)$ and $\psi_\nu(\alpha) = \omega^{\Omega_\nu + \alpha}$

(The expression $\varepsilon_{\Omega_\nu + 1}$ really just means the supremum of $\left\{\Omega_\nu, \Omega_\nu^{\Omega_\nu}, \Omega_\nu^{\Omega_\nu^{\Omega_\nu}}, \ldots\right\}$, i.e. an infinite power tower of Ω_ν, similar to how ε_0 is an infinite power tower of ω)

Proof. We set:

$$\varepsilon(\nu) := \begin{cases} \varepsilon_0 & \nu = 0 \\ \varepsilon_{\Omega_\nu + 1} & \nu > 0 \end{cases}$$

and

$$\alpha * \nu := \begin{cases} \alpha & \nu = 0 \\ \Omega_\nu + \alpha & \nu > 0 \end{cases}$$

1. Base Case: We have $0 \in C_\nu(0)$ and we know that $\psi_\nu(0) = \Omega_\nu$.
2. Successor Case: Suppose $\alpha \in C_0(\alpha)$ and $\psi_0(\alpha) = \omega^{\alpha * \nu}$. Then $\alpha + 1 \in C_0(\alpha + 1)$ and $\psi_0(\alpha + 1)$, and $\psi_\nu(\alpha + 1) = \omega^{\alpha * \nu + 1}$ from part (a) of the earlier lemma.
3. Limit Case: Suppose $\alpha < \varepsilon(\nu), \alpha \in \mathrm{Lim}$ and that for all ordinals $\xi < \alpha$, $\xi \in C_\nu(\xi)$ and $\psi_\nu(\xi) = \omega^{\alpha * \nu}$. Then by part (b) of the earlier lemma,

$$\psi_\nu(\xi) = \overset{+}{\sup}\{\omega^{\xi * \nu} \mid \xi < \alpha\} = \omega^{\alpha * \nu}$$

Now we still have to prove $\alpha \in C_\nu(\alpha)$.

- For $\alpha < \Omega_\nu$, α is trivially included.
- If $\alpha = \Omega_\nu$, $\alpha = \psi_\nu(0)$ and $0 < \alpha$, so $\Omega_\nu \in C_\nu(\alpha)$.
- For $\Omega_\nu < \alpha < \varepsilon(\nu)$, we have $P(\alpha) \subseteq \alpha$. Then by induction hypothesis we have $\xi \in C_\nu(\xi) \subseteq C_\nu(\alpha)$ for all $\xi \in P(\alpha)$, so $\alpha \in C_\nu(\alpha)$

[WIP: what about if $P(\alpha) = \alpha$? its not in Buchholz paper... I suppose you can just say if $\alpha \in P$, $\alpha = \psi_\nu(\beta)$ for some $\beta < \alpha$, and by induction hypothesis $\beta \in C_\nu(\beta)$ so its fine]

$\blacksquare$

Using the lemma above, we have:

$$
\begin{aligned}
\psi_0(1) &= \omega \\
\psi_0(2) &= \omega^2 \\
\psi_0(\omega) &= \omega^\omega \\
\psi_0(\omega + 1) &= \omega^{\omega+1} \\
\psi_0(\omega^\omega) &= \omega^{\omega^\omega}
\end{aligned}
$$

Then what about $\psi_0(\varepsilon_0)$?

$$
\begin{aligned}
\psi_0(\varepsilon_0) &= \overset{+}{\sup}\{\psi_0(\xi) \mid \xi < \varepsilon_0 \text{ and } \xi \in C_0(\xi)\} \\
&= \{\omega^\xi \mid \xi < \varepsilon_0\} \\
&= \varepsilon_0
\end{aligned}
$$

Therefore we have $\psi_0(\varepsilon_0) = \varepsilon_0$, and $\varepsilon_0 \notin C_0(\varepsilon_0)$. What about $C_0(\varepsilon_0 + 1)$? Let's think about at what increasing the α in $C_0(\alpha)$ does. It loosens the restriction on what elements in $C_0(\alpha)$ that ψ_0 can be applied to. For example, $C_0(\omega)$ means that as long as $\xi \in C_0(\omega)$ and $\xi < \omega$, $\psi_0(\xi) \in C_0(\omega)$. In this case, we can produce ordinals like $\psi_0(2) = \omega^2$ within $C_0(\omega)$ (since $2 < \omega$). The only thing stopping us from feeding ω^2 into ψ_0, and letting $\psi_0(\omega^2) \in C_0(\omega)$ is the rule that $\xi < \omega$. However, for ε_0 and larger ordinals, no matter what (finite) combination of addition and ψ_0 we throw at it, we can't reach ε_0:

$$
\begin{aligned}
\psi_0(0) &= 1 \\
\psi_0(\psi_0(0)) = \psi_0(1) &= \omega \\
\psi_0(\psi_0(\psi_0(0))) = \psi_0(\omega) &= \omega^\omega \\
\psi_0(\psi_0(\psi_0(\psi_0(0)))) = \psi_0(\omega^2) &= \omega^{\omega^\omega}
\end{aligned}
$$

We can see that $\psi_0(\psi_0(...)) = \omega^{\omega^{\cdot^{\cdot^{\cdot}}}}$ "approaches" ε_0. The "limiting factor" is no longer the $\xi < \omega$ condition, but rather the "innate strength" of ψ_0. So no matter how large we make α, we will not be able to generate ε_0 within $C_0(\alpha)$. We can throw the largest ordinals we have come up with: $\psi_0(\Gamma_0) = \psi_0(\mathrm{SVO}) = \psi_0(\mathrm{LVO}) = \varepsilon_0$. This is where ψ_1 comes in.

We know (from an earlier lemma) that $\psi_1(0) = \Omega_1$. If you recall from the start, our $C_0(1)$ contains not only $\psi_0(0) = 1$, but also $\psi_1(0) = \Omega_1, \psi_2(0) = \Omega_2, \ldots$. From now on, we will use Ω as a shorthand for Ω_1. $\varepsilon_0 \notin C_0(\Omega)$, so $\psi_0(\Omega) = \varepsilon_0$. But since $\Omega \in C_0(\Omega + 1)$ as $\Omega < \Omega + 1$, we **can** use $\psi_0(\Omega)$ in $C_0(\Omega + 1)$, i.e., $\varepsilon_0 \in C_0(\Omega + 1)$, so we have:

$$
\begin{aligned}
\psi_0(\Omega) &= \varepsilon_0 \\
\psi_0(\Omega + 1) &= \varepsilon_0 \cdot \omega = \omega^{\varepsilon_0 + 1} \\
\psi_0(\Omega + 2) &= \varepsilon_0 \cdot \omega^2 = \omega^{\varepsilon_0 + 2} \\
\psi_0(\Omega + \omega) &= \varepsilon_0 \cdot \omega^\omega = \omega^{\varepsilon_0 + \omega} \\
\psi_0(\Omega + \varepsilon_0) &= \varepsilon_0 \cdot \omega^{\varepsilon_0} = \varepsilon_0^2 = \omega^{\varepsilon_0 + \varepsilon_0}
\end{aligned}
$$

We can actually prove that $\psi_0(\Omega + \alpha) = \omega^{\varepsilon_0 + \alpha}$:

Proof. We use transfinite induction:

1. Base case: $\psi_0(\Omega + 0) = \omega^{\varepsilon_0 + 0} = \varepsilon_0$ and $0 \in C_0(\Omega + 0)$
2. Successor ordinal case: If $\psi_0(\Omega + \alpha) = \omega^{\varepsilon_0 + \alpha}$ and $\alpha \in C_0(\Omega + \alpha)$, then

$$
\begin{aligned}
\psi_0(\Omega + \alpha + 1) &= \min\{\gamma \in P \mid \psi_0(\alpha) < \gamma\} \\
&= \min\{\omega^\xi \mid \omega^{\varepsilon_0 + \alpha} < \omega^\xi\} \\
&= \omega^{\varepsilon_0 + \alpha + 1}
\end{aligned}
$$

Since $\alpha, 1 \in C_0(\alpha) \subseteq C_0(\alpha + 1)$, $\alpha + 1 \in C_0(\alpha + 1)$

3. Limit ordinal case: If for a limit ordinal λ, for all ordinals $\alpha < \lambda$, $\psi_0(\Omega + \alpha) = \omega^{\varepsilon_0 + \alpha}$ and $\alpha \in C_0(\Omega + \alpha)$,

$$
\begin{aligned}
\psi_0(\Omega + \lambda) &= \overset{+}{\sup}\{\psi_0(\Omega + \alpha) \mid \alpha < \lambda \text{ and } \alpha \in C_0(\alpha)\} \\
&= \overset{+}{\sup}\{\omega^{\varepsilon_0 + \alpha} \mid \alpha < \lambda\} \\
&= \omega^{\varepsilon_0 + \lambda}
\end{aligned}
$$

However, we need to show that $\Omega + \lambda \in C_0(\Omega + \lambda)$ in order to *continue* the induction to higher ordinals. If $\Omega + \lambda \in C_0(\Omega + \lambda)$, then $\lambda \in C_0(\Omega + \lambda)$ and therefore we must have $\lambda < \psi_0(\Omega + \lambda)$. Since $\psi_0(\Omega + \lambda) = \omega^{\varepsilon_0 + \lambda}$, $\lambda < \omega^{\varepsilon_0 + \lambda}$. The first ordinal that violates this property is the *second* fixed point of $\omega^\bullet = \bullet$, i.e., ε_1. Though note that for ε_1, $\psi_0(\Omega + \varepsilon_1) = \omega^{\varepsilon_0 + \varepsilon_1} = \varepsilon_1$ still holds, just that $\varepsilon_1 \notin C_0(\Omega + \varepsilon_1)$

So therefore, for all $\alpha \leq \varepsilon_1$, $\psi_0(\Omega + \alpha) = \omega^{\varepsilon_0 + \alpha}$ ■

We see that our function gets "stuck" once again on $\psi_0(\Omega + \varepsilon_1)$:

$$
\begin{aligned}
\psi_0(\Omega) &= \varepsilon_0 \\
\psi_0(\Omega + \psi_0(\Omega)) &= \omega^{\varepsilon_0 + \varepsilon_0} = \varepsilon_0^2 \\
\psi_0(\Omega + \psi_0(\Omega)) &= \omega^{\varepsilon_0 + \varepsilon_0^2} = \omega^{\varepsilon_0^2} = \varepsilon_0^{\varepsilon_0} \\
\psi_0(\Omega + \psi_0(\Omega + \psi_0(\Omega))) &= \omega^{\varepsilon_0 + \varepsilon_0^{\varepsilon_0}} = \omega^{\varepsilon_0^{\varepsilon_0}} = \varepsilon_0^{\varepsilon_0^{\varepsilon_0}} \\
\psi_0(\Omega + \psi_0(\Omega + \psi_0(\Omega + \ldots))) &= \varepsilon_0^{\varepsilon_0^{\varepsilon_0^{\cdot^{\cdot^{\cdot}}}}}
\end{aligned}
$$

It only becomes "unstuck" once again when we have an ordinal $\alpha > \Omega + \varepsilon_1$ such that $\alpha \in C_0(\alpha)$. The next such ordinal is $\Omega + \Omega$, or $\Omega \cdot 2$, and we have $\psi_0(\Omega \cdot 2) = \varepsilon_1$.

We similarly have:

$$
\begin{aligned}
\psi_0(\Omega \cdot 3) &= \varepsilon_2 \\
\psi_0(\Omega \cdot \omega) &= \varepsilon_\omega \\
\psi_0(\Omega \cdot \varepsilon_0) &= \varepsilon_{\varepsilon_0}
\end{aligned}
$$

You might be wondering how we can get terms like $\Omega \cdot \omega$ from our ψ functions. We use the property that for $\alpha < \varepsilon_{\Omega_\nu + 1}$, $\psi_\nu(\alpha) = \omega^{\Omega_\nu + \alpha}$;

$$
\psi_1(1) = \omega^{\Omega + 1} = \omega^\Omega \cdot \omega^1 = \Omega \cdot \omega
$$

$$
\psi_1(\varepsilon_0) = \omega^{\Omega + \varepsilon_0} = \Omega \cdot \varepsilon_0
$$

and in general, $\psi_1(\alpha) = \Omega \cdot \omega^\alpha$, or just $\Omega \cdot \alpha$ if $\omega^\alpha = \alpha$.

So we can create another series of expressions "approaching" ζ_0:

$$
\begin{aligned}
\psi_0(\Omega) &= \varepsilon_0 \\
\psi_0(\psi_1(\psi_0(\Omega))) &= \psi_0(\Omega \cdot \varepsilon_0) \\
&= \varepsilon_{\varepsilon_0} \\
\psi_0(\psi_1(\psi_0(\psi_1(\psi_0(\Omega))))) &= \psi_0\left(\Omega \cdot \varepsilon_{\varepsilon_0}\right) \\
&= \varepsilon_{\varepsilon_{\varepsilon_0}} \\
\psi_0(\psi_1(\psi_0(\psi_1(\psi_0(\psi_1(...)))))) &= \psi_0(\Omega \cdot \psi_0(\Omega \cdot \psi_0(...))) \\
&= \psi_0\left(\Omega \cdot \varepsilon_{\varepsilon_{\ddots}}\right) \\
&= \varepsilon_{\varepsilon_{\varepsilon_{\ddots}}}
\end{aligned}
$$

This suggests that

$$\psi_0(\psi_1(\zeta_0)) = \psi_0(\Omega \cdot \zeta_0) = \zeta_0$$

Which checks out since ζ_0 is a fixed point of $\varepsilon_\bullet$:

$$\psi_0(\psi_1(\bullet)) = \psi_0(\Omega \cdot \bullet) = \varepsilon_\bullet$$

However, $\zeta_0 \notin C_0(\Omega \cdot \zeta_0)$, just like how $\varepsilon_0 \notin C_0(\varepsilon_0)$. The function once agains becomes "stuck" where $\psi_0(\Omega \cdot \zeta_0) = \psi_0(\Omega \cdot \Gamma_0) = \psi_0(\Omega \cdot \mathrm{LVO}) = \zeta_0$. We need to find the next ordinal α larger than ζ_0 such that $\alpha \in C_0(\Omega \cdot \alpha)$ (Spoiler alert: its Ω again).

Just like how $\psi_0(\Omega)$ is still ε_0, $\psi_0(\Omega \cdot \Omega) = \psi_0(\Omega^2)$ is still ζ_0, but since $\Omega^2 = \psi_1(\Omega) \in C_0(\Omega^2)$, $\psi_0(\Omega^2 + 1) = \zeta_0 \cdot \omega$ and we can produce larger and larger ordinals. In general, we can relate Buchholz's function to the Veblen function as such:

Proposition 8.14.

$$\psi_0(\Omega^\alpha \cdot (1 + \beta)) = \varphi(\alpha, \beta)$$

Proof. [WIP] probably shift to some appendix cause its quite long I think. ■

We can generalize this to the extended veblen function as such:

Proposition 8.15.

$$\psi_0\left(\Omega^{(\Omega^n \alpha_n + ... + \Omega \alpha_1 + \alpha_0)}(1 + \beta)\right) = \varphi(\alpha_n, ..., \alpha_1, \alpha_0, \beta)$$

You can verify that these expressions containing Ω satisfies $\alpha \in C_0(\alpha)$ by expressing them in terms of ψ_1:

$$
\begin{aligned}
\psi_1(0) &= \Omega \\
\psi_1(\alpha) &= \omega^{\Omega + \alpha} = \Omega \cdot \omega^\alpha \\
\psi_1(\psi_1(0)) &= \psi_1(\Omega) = \omega^{\Omega + \Omega} = \Omega^2 \\
\psi_1(\psi_1(\alpha)) &= \psi_1(\Omega \cdot \omega^\alpha) \\
\psi_1(\Omega \cdot \alpha) &= \omega^{\Omega + \Omega \cdot \alpha} = \omega^{\Omega \cdot \alpha} = \Omega^\alpha \\
\psi_1(\Omega^2) &= \Omega^\Omega \\
\psi_1(\Omega^\Omega) &= \Omega^{\Omega^\Omega}
\end{aligned}
$$

So for example:

$$
\begin{aligned}
\zeta_0 &= \psi_0(\Omega^2) \\
&= \psi_0(\psi_1(\Omega)) \\
&= \psi_0(\psi_1(\psi_1(0)))
\end{aligned}
$$

The limit of the 2-argument veblen function is Γ_0, which can be written as:

$$
\begin{aligned}
\Gamma_0 &= \psi_0(\Omega^\Omega) \\
&= \psi_0(\psi_1(\Omega^2)) \\
&= \psi_0(\psi_1(\psi_1(\Omega))) \\
&= \psi_0(\psi_1(\psi_1(\psi_1(0))))
\end{aligned}
$$

Continuing, we have

$$
\mathrm{SVO} = \psi_0(\Omega^{\Omega^\omega}), \mathrm{LVO} = \psi_0\left(\Omega^{\Omega^\Omega}\right)
$$

Now that we have "caught up" with the veblen function, we can go even further, with $\psi_0\left(\Omega^{\Omega^{\Omega^\Omega}}\right)$. Eventually we reach $\psi_0(\Omega^{\Omega^{\cdot^{\cdot}}}) = \psi_0(\varepsilon_{\Omega+1})$, the **Bachmann-Howard Ordinal (BHO)**. .

So far we've been using ψ_1 to construct all our terms containing Ω. However, just like how we previously got "stuck" at $\psi_0(\varepsilon_0) = \varepsilon_0$, we run into the issue where $\psi_1(\varepsilon_{\Omega+1}) = \varepsilon_{\Omega+1}$:

$$
\begin{aligned}
\psi_1(\varepsilon_{\Omega+1}) &= \sup^+\left(\psi_1(\Omega), \psi_1(\Omega^\Omega), \psi_1\left(\Omega^{\Omega^\Omega}\right), \ldots\right) \\
&= \sup^+\left\{\omega^{\Omega+\Omega}, \omega^{\Omega+\Omega^\Omega}, \omega^{\Omega+\Omega^{\Omega^\Omega}}, \ldots\right\} \\
&= \sup^+\left\{\Omega^2, \Omega^\Omega, \Omega^{\Omega^\Omega}, \ldots\right\} \\
&= \varepsilon_{\Omega+1}
\end{aligned}
$$

Since $\varepsilon_{\Omega+1} \notin C_1(\varepsilon_{\Omega+1})$ and $C_0(\varepsilon_{\Omega+1}) \subset C_1(\varepsilon_{\Omega+1})$, $\varepsilon_{\Omega+1} \notin C_0(\varepsilon_{\Omega+1})$. To go further than this, we will need to invoke higher Ωs, like Ω_2, Ω_3, and so on. The next ordinal larger than $\varepsilon_{\Omega+1}$ that we can access is $\psi_2(0) = \Omega_2$, so we can re-express the Bachmann-Howard Ordinal as $\psi_0(\Omega_2)$ instead.

But wait, if we have $\mathrm{BHO} = \psi_0(\Omega_2)$ and $\mathrm{BHO} = \psi_0(\varepsilon_{\Omega+1}) = \psi_0(\psi_1(\Omega_2))$, which is the standard way to write it? As we have seen earlier, having the property $\alpha \in C_\nu(\alpha)$ saves us a lot of headaches by avoiding places where the function "gets stuck". So $\psi_0(\Omega_2)$ is "nicer" as $\Omega_2 \in C_0(\Omega_2)$, whereas $\psi_0(\varepsilon_{\Omega+1}) = \psi_0(\psi_1(\Omega_2))$ is "less nice" as $\varepsilon_{\Omega+1} \notin C_0(\varepsilon_{\Omega+1})$. We'll elaborate more on this when we go over the Normal Form for Buchholz's ψ. From now on, in this section, we'll color terms where $\alpha \notin C_\nu(\alpha)$ and the values that they get "stuck" at in purple.

We then have:

$$
\begin{aligned}
\psi_0(\Omega_2 + 1) &= \mathrm{BHO} \cdot \omega \\
\psi_0(\Omega_2 + \omega) &= \mathrm{BHO} \cdot \omega^\omega \\
\psi_0(\Omega_2 + \mathrm{BHO}) &= \mathrm{BHO} \cdot \omega^{\mathrm{BHO}} = \mathrm{BHO}^2 \\
\psi_0(\Omega_2 + \mathrm{BHO}^2) &= \mathrm{BHO} \cdot \omega^{\mathrm{BHO}^2} = \mathrm{BHO}^{\mathrm{BHO}} \\
\psi_0\left(\Omega_2 + \mathrm{BHO}^{\mathrm{BHO}}\right) &= \mathrm{BHO} \cdot \omega^{\mathrm{BHO}^{\mathrm{BHO}}} = \mathrm{BHO}^{\mathrm{BHO}^{\mathrm{BHO}}} \\
\psi_0\left(\Omega_2 + \varepsilon_{\mathrm{BHO}+1}\right) &= \varepsilon_{\mathrm{BHO}+1}
\end{aligned}
$$

Just like how $\psi_0(\Omega + \varepsilon_1) = \varepsilon_1$, we have $\psi_0(\Omega_2 + \varepsilon_{\mathrm{BHO}+1}) = \varepsilon_{\mathrm{BHO}+1}$, so we replace it with an Ω and get $\psi_0(\Omega_2 + \Omega) = \varepsilon_{\mathrm{BHO}+1}$. With this we can find the next "fixed point":

$$\psi_0(\Omega_2 + \Omega) = \varepsilon_{\text{BHO}+1}$$
$$\psi_0(\Omega_2 + \Omega + \varepsilon_{\text{BHO}+1}) = \varepsilon_{\text{BHO}+1}^2$$
$$\psi_0(\Omega_2 + \Omega + \varepsilon_{\text{BHO}+1}^2) = \varepsilon_{\text{BHO}+1}^{\varepsilon_{\text{BHO}+1}}$$
$$\psi_0\left(\Omega_2 + \Omega + \varepsilon_{\text{BHO}+1}^{\varepsilon_{\text{BHO}+1}}\right) = \varepsilon_{\text{BHO}+1}^{\varepsilon_{\text{BHO}+1}^{\varepsilon_{\text{BHO}+1}}}$$
$$\psi_0(\Omega_2 + \Omega + \varepsilon_{\text{BHO}+2}) = \varepsilon_{\text{BHO}+2}$$
$$\psi_0(\Omega_2 + \Omega + \Omega) = \varepsilon_{\text{BHO}+2}$$

This is starting to look like the veblen correspondence all over again, where

$$\psi_0(\Omega_2 + \Omega^\alpha \cdot \beta) = \varphi(\alpha, \text{BHO} + \beta)$$

Now let's look at ψ_1:

$$\psi_1(\Omega_2) = \varepsilon_{\Omega+1}$$
$$\psi_1(\Omega_2 + 1) = \varepsilon_{\Omega+1} \cdot \omega$$
$$\psi_1(\Omega_2 + \varepsilon_{\Omega+1}) = \varepsilon_{\Omega+1} \cdot \omega^{\varepsilon_{\Omega+1}} = \varepsilon_{\Omega+1}^2$$
$$\psi_1(\Omega_2 + \varepsilon_{\Omega+1}^2) = \varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}}$$
$$\psi_1\left(\Omega_2 + \varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}}\right) = \varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}}}$$
$$\psi_1(\Omega_2 + \varepsilon_{\Omega+2}) = \varepsilon_{\Omega+2}$$

So we have to yet again find the next ordinal $\alpha > \Omega_2 + \varepsilon_{\Omega+2}$ such that $\alpha \in C_1(\alpha)$. The first ordinal to satisfy that criteria is $\alpha = \Omega_2 \cdot 2$, so we get $\psi_1(\Omega_2 \cdot 2) = \varepsilon_{\Omega+2}$. We again have a similar correspondence going on:

$$\psi_1(\Omega_2^\alpha \cdot \beta) = \varphi(\alpha, \Omega + \beta)$$

But we don't need the veblen function anymore. The limit of what we can do with Ω_2 is:

$$\psi_1\left(\Omega_2^{\Omega_2^{\Omega_2^{\cdot^{\cdot^{\cdot}}}}}\right) = \psi_1\left(\varepsilon_{\Omega_2+1}\right)$$

This is analogous the BHO earlier, where instead of $\psi_1(\varepsilon_{\Omega+1}) = \varepsilon_{\Omega+1}$, we have $\psi_2\left(\varepsilon_{\Omega_2+1}\right) = \varepsilon_{\Omega_2+1}$ (Convince yourself that this is true). So we get $\psi_1(\Omega_3) = \varepsilon_{\Omega_2+1}$, and $\psi_0\left(\varepsilon_{\Omega_2+1}\right) = \psi_0(\Omega_3)$.

We can keep going $\psi_0(\Omega_4), \psi_0(\Omega_5), \ldots$ up to $\psi_0(\Omega_\omega)$, which is known as **Buchholz's Ordinal (BO)**. This is the limit of all $\psi_0(\Omega_n)$ for finite n, and while we haven't gotten to the ordinal notation associated with Buchholz's ψ yet, there are many ordinal notations that correspond to ordinals below $\psi_0(\Omega_\omega)$.

Of course, the true limit of Buchhol'z OCF is $\psi_0\left(\Omega_\omega^{\Omega_\omega^{\Omega_\omega^{\cdot^{\cdot^{\cdot}}}}}\right) = \psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$, the **Takeuti–Feferman–Buchholz ordinal (TFBO)**, but it is not as commonly used in googology.

8.3 Normal Form and Fundamental Sequences

Theorem 8.16.

Every ordinal $\alpha < \psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$ can be uniquely expressed in the form:

$$\alpha = \psi_{k_1}(\alpha_1) + \ldots + \psi_{k_n}(\alpha_n)$$

Where:

- $\alpha_1, \ldots, \alpha_n \in C_0\left(\varepsilon_{\Omega_\omega+1}\right)$

- $k_1, ..., k_n \leq \omega$
- $\psi_{k_1}(\alpha_1) \geq ... \geq \psi_{k_n}(\alpha_n)$
- $\alpha_1 \in C_{k_1}(\alpha_1), ..., \alpha_n \in C_{k_n}(\alpha_n)$

Like in Cantor Normal Form, each α_i does not necessarily have to be expressed in normal form. For example, all of the following are in normal form:
- $\psi_0(\Omega^2)$
- $\psi_0(1 + \Omega^2)$
- $\psi_0(\psi_1(\Omega))$

because $\Omega^2 = 1 + \Omega^2 = \psi_1(\Omega) \in C_0(\Omega^2)$

Example 8.17.

Which of the following are in normal form:
1. $\psi_0(0)$
2. $\psi_0(\Omega)$
3. $\psi_0(\psi_0(\Omega))$
4. $\psi_0(\psi_1(\psi_2(\Omega_3)))$

Answers:
1. $0 \in C_0(0)$, normal form
2. $\Omega = \psi_1(0) \in C_0(\Omega)$, normal form
3. $\psi_0(\Omega) = \varepsilon_0 \notin C_0(\varepsilon_0)$, not normal form.

   Note that just because an ordinal α is expressed in normal form, that does not mean $\psi_\nu(\alpha)$ is in normal form. (e.g. $\psi_0(\Omega)$ is normal form but $\psi_0(\psi_0(\Omega))$ isn't)
4. Firstly we have $\psi_2(\Omega_3) = \varepsilon_{\Omega_2+1}$. Then we have $\psi_1(\psi_2(\Omega_3)) = \psi_1\left(\varepsilon_{\Omega_2+1}\right)$, which in normal form is $\psi_1(\Omega_3)$. We then construct the following inequality to solve for $\psi_0(\psi_1(\Omega_3))$:

$$\psi_1(\Omega_2) \leq \psi_1(\Omega_3) < \Omega_2$$
$$\psi_0(\psi_1(\Omega_2)) \leq \psi_0(\psi_1(\Omega_3)) \leq \psi_0(\Omega_2)$$
$$\text{BHO} \leq \psi_0(\psi_1(\Omega_3)) \leq \text{BHO}$$

   Therefore $\psi_0(\psi_1(\Omega_3)) = \text{BHO}$. However the BHO has only one unique normal form expression: $\psi_0(\Omega_2)$. So $\psi_0(\psi_1(\psi_2(\Omega_3))) = \psi_0(\psi_1(\Omega_3))$ are both not normal form.

Before we go into the fundamental sequences of ψ, we first introduce a concept known as **cofinality**:

Definition 8.18.

The **cofinality** of an ordinal α, denoted cof α, is the **least cardinal number** κ such that there exists a subset $S \subseteq \alpha$ of cardinality κ where α is smallest ordinal strictly greater than every element in S.

Example 8.19.

Find the cofinality of
1. $\omega 2$
2. $\omega 2 + 3$
3. $\omega_1 = \Omega$

Answers:

1. Let $S = \{\omega + n \mid n \in \mathbb{N}\} \subset \omega 2$. S has a cardinality of ω.

   However, we still need to rule out the possibility that there might be a subset S of finite cardinality (i.e. cardinality $< \omega$) that still satisfies the criteria that $\omega 2$ is the smallest ordinal greater than every element in S

   In a subset S which has a nonzero finite cardinality, there is a largest element β, and its successor $\beta + 1$ is the smallest ordinal greater than all elements of the subset S while being less than $\omega 2$. Therefore S can't have nonzero finite cardinality.

   If S has a cardinality of 0 (i.e. $S = \emptyset = \{\}$), then 0 is the smallest ordinal greater than all elements in S. Therefore S can't have a cardinality of 0.

   Therefore the smallest cardinality of S where $\omega 2$ is the least ordinal greater than all elements in S is ω.
2. $S = \{\omega 2 + 2\} \subset \omega 2 + 3$, and S has a cardinality of 1. S can't have a cardinality of 0 since the smallest ordinal larger than nothing is 0.
3. ω_1 (and its associated cardinal $\aleph_1$) is defined as the set of all ordinals whose cardinality is $\leq \aleph_0$. So the smallest subset where ω_1 is the

Now onto fundamental sequences. A fundamental sequence for an ordinal number α with cofinality $\operatorname{cof}(\alpha) = \beta$ is a strictly increasing sequence $(\alpha[0], \alpha[1], ...)$ with length β and limit α (i.e. $\alpha = \overset{+}{\sup}\{\alpha[\eta] \mid \eta < \beta\}$)

Definition 8.20 (System of fundamental sequences for ψ).

For ordinals $\alpha \leq \psi_0\left(\varepsilon_{\Omega_\omega + 1}\right)$ (expressed in normal form), the η^{th} term of the fundamental sequence of α, $\alpha[\eta]$ is defined as such:

1. If $\alpha = \psi_{k_1}(\alpha_1) + ... + \psi_{k_n}(\alpha_n)$, then $\operatorname{cof}(\alpha) = \operatorname{cof}\left(\psi_{k_n}(\alpha_n)\right)$, and $\alpha[\eta] = \psi_{k_1}(\alpha_1) + ... + \left(\psi_{k_n}(\alpha_n)[\eta]\right)$
2. If $\alpha = \psi_0(0) = 1$ then $\operatorname{cof}(\alpha) = 1$. We need a sequence of length 1, so we set $\alpha[0] = 0$
3. If $\alpha = \psi_{\nu+1}(0) = \Omega_{\nu+1}$, then $\operatorname{cof}(\alpha) = \Omega_{\nu+1}$. We need a sequence of length $\Omega_{\nu+1}$, so we set $\alpha[\eta] = \Omega_{\nu+1}[\eta] = \eta$
4. If $\alpha = \psi_\omega(0)$, then $\operatorname{cof}(\alpha) = \operatorname{cof}(\omega) = \omega$ and $\alpha[\eta] = \psi_{\omega[\eta]}(0) = \Omega_{\omega[\eta]} = \Omega_\eta$
5. If $\alpha = \psi_\nu(\beta + 1)$ then $\operatorname{cof}(\alpha) = \omega$ and $\alpha[\eta] = \psi_\nu(\beta) \cdot \eta$
6. If $\alpha = \psi_\nu(\beta)$ where β is a limit ordinal,
   (a) If $\operatorname{cof}(\beta) = \omega$ or $\operatorname{cof}(\beta) \leq \Omega_\nu$, then we have $\operatorname{cof}(\alpha) = \operatorname{cof}(\beta)$, so we need a sequence of length $\operatorname{cof}(\beta)$, so $\alpha[\eta] = \psi_\nu(\beta[\eta])$
   (b) Otherwise if $\operatorname{cof}(\beta) > \Omega_\nu$ and $\operatorname{cof}(\beta) \neq \omega$ then $\operatorname{cof}(\alpha) = \omega$. This is where the definitions diverge.

   We define μ as $\operatorname{cof}(\beta) = \Omega_{\mu+1}$.

   $\alpha[n] = \psi_\nu(\beta[\gamma[n]])$, where $\gamma[0] = 0$, and $\gamma[\eta + 1] = \psi_\mu(\beta[\gamma[\eta]])$. To illustrate this nesting process:

$$\alpha[0] = \psi_\nu(\beta[\gamma[0]]) = \psi_\nu(\beta[0])$$
$$\alpha[1] = \psi_\nu(\beta[\gamma[1]]) = \psi_\nu\left(\beta\left[\psi_\mu(\beta[0])\right]\right)$$
$$\alpha[2] = \psi_\nu(\beta[\gamma[2]]) = \psi_\nu\left(\beta\left[\psi_\mu\left(\beta\left[\psi_\mu(\beta[0])\right]\right)\right]\right)$$

In the case of $\mu = \nu$ however, we can use a shortcut: $\alpha[0] = \psi_\nu(\beta[0])$, and $\alpha[\eta+1] = \psi_\nu(\beta[\alpha[\eta]])$

However there are many other systems out there:

- Dennis Maksudov:

  Same definition, except $\gamma[0] = \Omega_\mu$.
- David_Exmachina:

  $\alpha[0] = 0$, and $\gamma[1] = 0$, otherwise $\gamma[\eta+1]$ unchanged.
- Buchholz's Original Paper

  Then $\alpha[\eta] = \psi_\nu(\beta[\psi_\mu(\beta[1])])$.

  Note that this is not as commonly used, as this "fundamental sequence" is independent of η (i.e. $\alpha[0] = \alpha[1] = \ldots$).

There's a lot to go through here, so let's go through some examples:

Example 8.21.

Calculate the fundamental sequences of the following

1. $\psi_1(0) = \Omega$
2. $\psi_0(\omega) = \omega^\omega$
3. $\psi_0(\Omega) = \varepsilon_0$
4. $\psi_0(\Omega+1) = \varepsilon_0 \cdot \omega = \omega^{\varepsilon_0+1}$
5. $\psi_0(\Omega 2) = \varepsilon_1$
6. $\psi_0(\Omega^2) = \zeta_0$
7. $\psi_0(\Omega_2) = \text{BHO}$

Before you peek into the answers, do try your hand at them!

Answers:

1. For $\psi_1(0) = \Omega$, this falls under Rule 3, $\operatorname{cof}(\psi_1(0)) = \operatorname{cof}(\Omega) = \Omega$, so we need a sequence of length Ω. As such $\psi_1(0)[\eta] = \Omega[\eta] = \eta$.

   In fact, for all $\Omega_\nu, \nu \in \mathbb{N} \setminus \{0\}, \Omega_\nu[\eta] = \eta$.
2. For $\psi_0(\omega)$, $\operatorname{cof}(\omega) = \omega$ (Rule 6(a)).

   So $\operatorname{cof}(\alpha) = \operatorname{cof}(\beta)$, and $\psi_0(\omega)[n] = \psi_0(\omega[n]) = \omega^n$. Since $\psi_0(\omega) = \omega^\omega$, $\omega^\omega[n] = \omega^n$ seems reasonable.
3. For $\psi_0(\Omega)$, $\operatorname{cof}(\Omega) > \Omega_0 = 1$, and $\operatorname{cof}(\Omega) \neq \omega$ (Rule 6(b)). So $\operatorname{cof}(\alpha) = \operatorname{cof}(\omega)$, and

$$\begin{aligned}
\psi_0(\Omega)[0] &= \psi_0(\Omega[0]) \\
&= \psi_0(0) \\
&= 1 \\
\psi_0(\Omega)[1] &= \psi_0(\Omega[\psi_0(\Omega)[0]]) \\
&= \psi_0(\Omega[1]) \\
&= \psi_0(1) \\
&= \omega
\end{aligned}$$

   So we have the following fundamental sequence, which makes sense:

$$\varepsilon_0[0] = 0$$
$$\varepsilon_0[1] = \omega$$
$$\varepsilon_0[2] = \omega^\omega$$
$$\varepsilon_0[3] = \omega^{\omega^\omega}$$
$$\vdots$$

4. For $\psi_0(\Omega + 1)$, this falls under Rule 5 when insides $\beta + 1$ is a successor ordinal. In this case we just have $\psi_0(\Omega + 1)[\eta] = \psi_0(\Omega) \cdot \eta$, i.e. $(\varepsilon_0 \cdot \omega)[n] = \varepsilon_0 \cdot n$, which seems reasonable.
5. For $\psi_0(\Omega 2)$, we have $\operatorname{cof}(\Omega 2) = \Omega$ which falls under Rule 6(b), and the specical shortcut where $\mu = \nu$. We first have

$$\psi_0(\Omega)[0] = \psi_0(\Omega 2[0])$$

But what is $\Omega 2[\eta]$? $\Omega 2[\eta] = \Omega + \Omega[\eta] = \Omega + \eta$. Continuing:

$$\begin{aligned}
\psi_0(\Omega 2)[0] &= \psi_0(\Omega 2[0]) \\
&= \psi_0(\Omega + 0) \\
&= \varepsilon_0 \\
\psi_0(\Omega 2)[1] &= \psi_0(\Omega 2[\varepsilon_0]) \\
&= \psi_0(\Omega + \varepsilon_0) \\
&= \varepsilon_0 \cdot \omega^{\varepsilon_0} \\
&= \varepsilon_0^2 = \omega^{\varepsilon_0 \cdot 2} \\
\psi_0(\Omega 2)[2] &= \psi_0(\Omega 2[\varepsilon_0^2]) \\
&= \psi_0(\Omega + \omega^{\varepsilon_0 \cdot 2}) \\
&= \varepsilon_0 \cdot \omega^{\omega^{\varepsilon_0 \cdot 2}} \\
&= \varepsilon_0^{\varepsilon_0} = \omega^{\omega^{\varepsilon_0 \cdot 2}}
\end{aligned}$$

So we have the fundamental sequence of $\varepsilon_1 = \psi_0(\Omega 2)$:

$$\varepsilon_1[0] = \varepsilon_0$$
$$\varepsilon_1[1] = \varepsilon_0^2 = \omega^{\varepsilon_0 \cdot 2}$$
$$\varepsilon_1[2] = \varepsilon_0^{\varepsilon_0} = \omega^{\omega^{\varepsilon_0 \cdot 2}}$$
$$\varepsilon_1[3] = \varepsilon_0^{\varepsilon_0^{\varepsilon_0}} = \omega^{\omega^{\omega^{\varepsilon_0 \cdot 2}}}$$
$$\vdots$$

Note that this is a **different** fundamental sequence than the one we defined with veblen's φ.
6. For $\psi_0(\Omega^2)$, we have $\operatorname{cof}(\Omega^2) = \Omega$ which falls under Rule 6(b), along with the special shortcut. So we have

$$\psi_0(\Omega)[0] = \psi_0(\Omega^2[0])$$

But what is $\Omega^2[\eta]$? Ω^2 itself is $\psi_1(\Omega)$, and $\psi_1(\Omega)[\eta] = \psi_1(\Omega[\eta]) = \psi_1(\eta) = \omega^{\Omega + \eta} = \Omega \cdot \omega^\eta$ (by Rule 6(a)). So $\Omega^2[\eta] = \Omega \cdot \eta$. Continuing:

$$
\begin{aligned}
\psi_0(\Omega)[0] &= \psi_0(\Omega^2[0]) \\
&= \psi_0(\Omega \cdot \omega^0) \\
&= \varepsilon_0 \\
\psi_0(\Omega)[1] &= \psi_0(\Omega^2[\psi_0(\Omega^2)[0]]) \\
&= \psi_0(\Omega^2[\varepsilon_0]) \\
&= \psi_0(\Omega \cdot \omega^{\varepsilon_0}) \\
&= \psi_0(\Omega \cdot \varepsilon_0) \\
&= \varepsilon_{\varepsilon_0} \text{ (Recall } \psi_0(\Omega^\alpha \cdot (1+\beta)) = \varphi(\alpha, \beta))
\end{aligned}
$$

So we have the fundamental sequence of $\zeta_0 = \psi_0(\Omega^2)$:

$$
\begin{aligned}
\zeta_0[0] &= \varepsilon_0 \\
\zeta_0[1] &= \varepsilon_{\varepsilon_0} \\
\zeta_0[2] &= \varepsilon_{\varepsilon_{\varepsilon_0}} \\
&\vdots
\end{aligned}
$$

This time, this is consistent with the one for veblen.

7. For $\psi_0(\Omega_2)$, $\mathrm{cof}(\Omega_2) = \Omega_2 > \Omega_0$ and $\mathrm{cof}(\Omega_2) \neq \omega$ so once again it falls under Rule 6(b), but we no longer have the special shortcut.

   Let's first evaluate $\gamma[\eta]$. Since $\mathrm{cof}(\Omega_2) = \Omega_2$, we have $\mu = 1$:

$$
\begin{aligned}
\gamma[0] &= 0 \\
\gamma[1] &= \psi_1(\Omega_2[\gamma[0]]) \\
&= \psi_1(\Omega_2[0]) \\
&= \Omega \\
\gamma[2] &= \psi_1(\Omega_2[\gamma[1]]) \\
&= \psi_1(\Omega_2[\Omega]) \\
&= \psi_1(\Omega) \\
&= \Omega^2 \\
\gamma[3] &= \psi_1(\Omega_2[\gamma[2]]) \\
&= \psi_1(\Omega^2) \\
&= \Omega^\Omega
\end{aligned}
$$

   This is a familiar pattern: $0, \Omega, \Omega^2, \Omega^\Omega, \Omega^{\Omega^\Omega}, \ldots$. Since $\psi_0(\Omega_2)[n] = \psi_0(\Omega_2[\gamma[n]]) = \psi_0(\gamma[n])$, we have the following fundamental sequence:

$$
\begin{aligned}
\psi_0(\Omega_2)[0] &= \psi_0(0) = 1 \\
\psi_0(\Omega_2)[1] &= \psi_0(\Omega) = \varepsilon_0 \\
\psi_0(\Omega_2)[2] &= \psi_0(\Omega^2) = \zeta_0 \\
\psi_0(\Omega_2)[3] &= \psi_0(\Omega^\Omega) = \Gamma_0 \\
\psi_0(\Omega_2)[4] &= \psi_0\left(\Omega^{\Omega^\Omega}\right) = \mathrm{LVO} \\
&\vdots
\end{aligned}
$$

9 Ordinal Notations up to $\psi_0(\Omega_\omega)$

Technically Buchholz's OCF goes up to $\psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$ but as we'll see later, $\psi_0(\Omega_\omega)$ is a more "natural" stopping point.

9.1 Ordinal Notation associated with BOCF

We introduce a set of ordinal notations OT corresponding to Buchholz's OCF, along with a recursive way to order them $\prec$, such that $(OT, \prec)$ is order isomorphic to $\left(C_0\left(\varepsilon_{\Omega_\omega+1}\right), \in\right)$

Definition 9.1.

Let T_B be the set of terms and PT be the set of all *prinicpal terms* (the terms in PT are associated to additively principal ordinals).

- $\mathsf{0} \in T$
- Given a term $t \in T$, $\mu \leq \omega$, $D_\mu(t) \in T$ and PT
- For principal terms $t_1, ..., t_k \in PT$, $t_1 + ... + t_k \in T_B$

Let $\prec$ be a binary relation on T.

For terms $s, t \in T$, $s \prec t$ is defined as:

- If $s = \mathsf{0}$ then $s \prec t$ if $t \neq \mathsf{0}$
- If $s = D_u(a)$ and $t = D_v(b)$, then $s \prec t$ if:
  - $u < v$, or
  - $u = v$ and $a \prec b$
- If $s = s_0 + ... + s_n, t = t_0 + ... + t_m$, then we do it lexicographically, i.e., $s \prec t$ if:
  - $s_i = t_i$ for all $i \leq n$ and $n < m$, i.e. s is a proper prefix of t
  - $s_i = t_i$ for all $i < k$ and $s_k \prec t_k$

For terms s, t, we have a family of the binary relations $s \in \mathcal{C}_\mu(t)$ for $\mu \leq \omega$ if and only if:

- $s \prec D_\mu(\mathsf{0})$
- $s = a + b$ for some $a \in PT$ and $b \in T \setminus \{\mathsf{0}\}$ and $a \in \mathcal{C}_\mu(t)$ and $b \in \mathcal{C}_\mu(t)$
- $s = D_\nu(a)$ for some $a \in T, \nu \leq \omega$ and $a \in \mathcal{C}_\mu(t)$ and $a \prec t$

Let OT be a subset of T. OT corresponds to the set of all ordinal notations of normal form. OT is defined as:

- $\mathsf{0} \in OT$
- If $a_0, ..., a_k \in OT$ and $a_0, ..., a_k \in PT$, if $a_k \preceq ... \preceq a_0$, their sum $a_0 + ... + a_k \in OT$
- For a term $a \in T$, $D_\mu(a) \in OT$ if and only if $a \in OT$ and $a \in \mathcal{C}_\mu(a)$

The definition of OT basically restricts all the non-standard notation that are possible but causes headaches. Examples of non-standard notation include:

- $D_0(\mathsf{0}) + D_0(D_1(\mathsf{0})) \sim \psi_0(0) + \psi_0(\psi_1(0)) = 1 + \varepsilon_0 = \varepsilon_0$
- $D_0(D_0(D_1(\mathsf{0}))) \sim \psi_0(\psi_0(\psi_1(0))) = \psi_0(\varepsilon_0) = \varepsilon_0$ (not normal form, $\varepsilon_0 \notin C_0(\varepsilon_0)$)

We can now define a map to associate each ordinal notation to an ordinal below $\psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$:

Definition 9.2.

The map $o : OT \mapsto C_0\left(\varepsilon_{\Omega_\omega+1}\right)$ is defined recursively as such:

1. $o(\mathsf{0}) := 0$
2. $o(a_0 + ... + a_k) = o(a_0) + ... + o(a_k)$
3. $o(D_\nu(a)) = \psi_\nu(o(a))$

This should be fairly self-explanatory with the notation that has been chosen essentially being red versions of the usual ordinals. OT here reprsents all ordinal notations that recursively satisfy the normal form, i.e. $D_a(b)$ always satisfies $b \in \mathcal{C}_a(b)$.

From this, Buchholz showed that:

Theorem 9.3 (Correspondence of Ordinal Notation with Buchholz's OCF).

The set $\{a \in OT \mid a \prec D_1(0)\}$ under $\prec$ is :

- order isomorphic to $\left(C_0\left(\varepsilon_{\Omega_\omega+1}\right), \in\right)$
- a well-ordered set
- has order type $\psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$

Additionally, for every ordinal notation $a \prec D_1(0)$, the ordinal $o(a)$ is the order type of the set $\{x \in OT \mid x \prec a\}$ under $\prec$.

Proof. WIP: His proofs is not that long, maybe we can break it down here? Maybe move it to an appendix or something ■

Now just like how the fundamental sequence of an ordinal $\alpha[n] < \alpha$, we can similarly define a fundamental sequence $a[n]$ for each ordinal notation a, where $a[n] \prec a$. This requires defining a computable version of *cofinality* (denoted dom in Buchholz's paper)

Definition 9.4 (Computable Cofinality and Fundamental Sequences).

We will define "cofinality" $\mathrm{dom}()$ of each ordinal notation term, and the fundamental sequence of an ordinal notation term $t[z]$. This is essentially a 1-to-1 copy of the fundamental sequence rules, except that since we are dealing with finite strings, they are recursive and computable.

- $\mathrm{dom}(0) := \emptyset$
- $\mathrm{dom}(a_0 + \ldots + a_k) = \mathrm{dom}(a_k)$ $(a_i \in OT)$
  $(a_0 + \ldots + a_k)[z] = a_0 + \ldots + (a_k[z])$
- $\mathrm{dom}(D_0(0)) := \{0\}$, and $D_0(0)[0] = 0$ ($\{0\}$ is analogous with 1)
- $\mathrm{dom}(D_{n+1}(0)) := T_u := \{t \in OT \mid t \prec D_{u+1}(0)\}$ for $n \in \mathbb{N}$ (T_u is analogous with Ω_{u+1}), $D_{n+1}(0)[z] = z$
- $\mathrm{dom}(D_\omega(b)) := \mathbb{N}$ (i.e. ω), $D_\omega(b)[n] = D_n(b)$
- $\mathrm{dom}(D_v(b))$ where $b \neq 0$ and $v \in \mathbb{N}$
  - $\mathrm{dom}(b) = \{0\} \Rightarrow \mathrm{dom}(D_v(b)) = \mathbb{N}$ (Analogous with rule 5) $D_v(b)[n] = D_v(b[0]) \cdot n$
  - If $\mathrm{dom}(b) = \mathbb{N}$ or $\mathrm{dom}(b) = T_u$ for some $u < v$ then $\mathrm{dom}(D_v(b)) = \mathrm{dom}(b)$ (Analogous with rule 6a), $D_v(b)[z] = D_v(b[z])$
  - If $\mathrm{dom}(b) = T_u$ for some $u \geq v$ then $\mathrm{dom}(D_v(b)) = \mathbb{N}$ (Analogous with rule 6b), $D_v(b)[n] = D_v(b[G[n]])$, where $G[0] = 0, G[n+1] = D_u(b[G[n]])$

Another thing to note about the fundamental sequences is that for a term $t \in OT$, all terms in its fundamental sequence $t[n] \in OT$. (Also is Buchholz's paper)

Lemma 9.5.

If $a, z \in OT$, and $z \in \mathrm{dom}(a)$, then $a[z] \in OT$ and $a[z] \prec a$

Recall that when defining ordinal notations of order type $< \varepsilon_0$, we mainly build up in layers from the bottom-up with Cantor Normal Form, but for PrSS we introduced a "top-down" approach, where each standard form sequence was nested fundamental sequences of the limit of PrSS. Here, we go for a similar “top-down” approach, where every ordinal notation $a \prec D_\omega(0)$ can be expressed as taking fundamental sequences:

$$a = D_\omega(0)[n_1][n_2]...[n_k]$$

For example, to get $D_0(D_1(0) + D_1(0)) \sim \varepsilon_1$:

- $D_0(D_\omega(0)) \sim \psi_0(\Omega_\omega)$
  - ⋮
  - [2] $D_0(D_2(0)) \sim$ BHO
    - ⋮
    - [3] $D_0(D_1(D_1(D_1(0)))) \sim \Gamma_0$
    - [2] $D_0(D_1(D_1(0))) \sim \zeta_0$
      - ⋮
      - [1] $D_0(D_1(D_0(D_1(0)))) \sim \varepsilon_{\varepsilon_0}$
        - ⋮
        - [1] $D_0(D_1(D_0(D_0(0)))) \sim \varepsilon_{\omega^\omega}$
        - [0] $D_0(D_1(D_0(0))) \sim \varepsilon_\omega$
          - ⋮
          - [2] $D_0(D_1(0) + D_1(0)) \sim \varepsilon_1$
          - [1] $D_0(D_1(0)) \sim \varepsilon_0$
          - [0] $D_0(0) \sim 1$
      - [0] $D_0(D_1(0)) \sim \varepsilon_0$
    - [1] $D_0(D_1(0)) \sim \varepsilon_0$
    - [0] $D_0(0) \sim 1$
  - [1] $D_0(D_1(0)) \sim \varepsilon_0$
  - [0] $D_0(D_0(0)) \sim \omega$

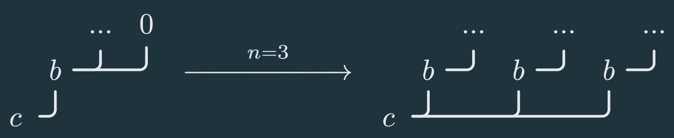
9.2 Buchholz’s Hydra

Definition 9.6.

This is somewhat similar to the Kirby-Paris hydra game we have earlier, but each node is labelled with an ordinal $\leq \omega$, the child nodes of the root must be labelled 0.

At each step, we choose a leaf node a to chop off. The hydra chooses some $n \in \mathbb{N}$.

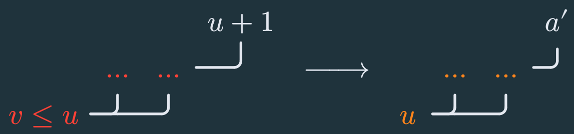
1. If a has label 0, we proceed as in Kirby-Paris’ game. Call the node’s parent b, and its grandparent c (if it exists). First we delete a. If c exists (i.e. b is not the root), we make n copies of b and all its children and attach them to c:



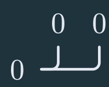
2. If a has label $u + 1$, we go down the tree looking for a node b whose label is $v \leq u$:

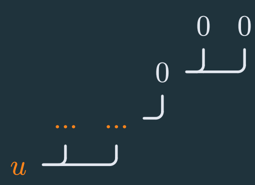
Let the subtree rooted in b be called S. The nodes highlighted in red (and node a highlighted in blue) make up S.

Let’s make a copy of S called S', and re-label the b inside S' to u.



Now let’s define the function $S(T)$ that takes in a tree T, and replaces a' with T

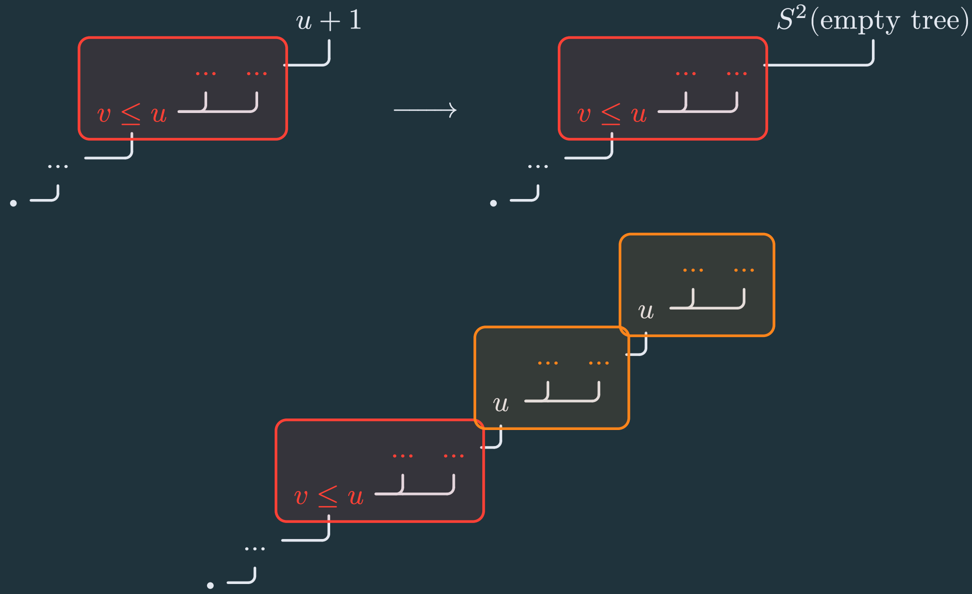
For example, if T is 

then $S(T)$ is 

Similarly, if T is the empty tree then $S(T)$ just becomes 

When we cut off a head of label $u + 1$, we replace it with $S^n(\text{empty tree})$ where S^n denotes function iteration.

An illustration of the whole process with $n = 2$ is as such:



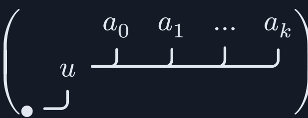
3. If a has label ω, replace the label ω with $n + 1$

Similarly to the Kirby-Paris Hydra, we can associate each hydra with an ordinal notation term T by the following rules:

1. For each leaf node with a label u, we replace it with the ordinal notation $D_u(\mathsf{0})$

$o\Big($  $\Big) \longrightarrow D_u(\mathsf{0})$

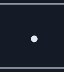
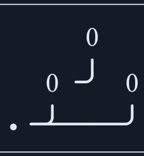
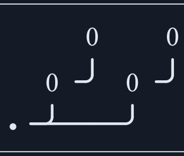
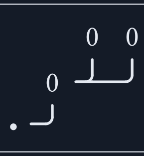
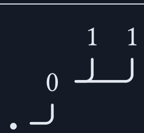
2. For a node with label u with children of ordinal notations $a_0, \ldots, a_k$, we replace it with the ordinal notation $D_u(a_0 + \ldots + a_k)$

$o\Bigg($  $\Bigg) \longrightarrow D_u(o(a_0) + o(a_1) + \ldots + o(a_k))$

3. At the root node, the resulting ordinal expressions of each of the root's children $a_0, \ldots, a_k$ are summed up: $a_0 + \ldots + a_k$

Note that case 2 in the definition above is different from the original definiton that Buchholz gave. This difference basically corresponds to the differences in the system of fundamental sequence (Specifically "Rule 6(b)").

Here are some examples of Buchholz's hydra and the associated ordinal notations and ordinals. The hydras have been rotated to save space.

Buchholz Hydra	Associated Ordinal Notation	Associated Ordinal
	$\mathsf{0}$	0
	$D_0(\mathsf{0})$	1
	$D_0(\mathsf{0}) + D_0(\mathsf{0})$	2
	$D_0(D_0(\mathsf{0}))$	ω
	$D_0(D_0(\mathsf{0})) + D_0(\mathsf{0})$	$\omega + 1$
	$D_0(D_0(\mathsf{0})) + D_0(D_0(\mathsf{0}))$	$\omega 2$
	$D_0(D_0(\mathsf{0}) + D_0(\mathsf{0}))$	ω^2
	$D_0(D_0(D_0(\mathsf{0})))$	ω^ω
	$D_0(D_0(D_0(D_0(\mathsf{0}))))$	ω^{ω^ω}
	$D_0(D_1(\mathsf{0}))$	$\psi_0(\Omega) = \varepsilon_0$
	$D_0(D_1(\mathsf{0}) + D_1(\mathsf{0}))$	$\psi_0(\Omega 2) = \varepsilon_1$
	$D_0(D_1(D_1(\mathsf{0})))$	$\psi_0(\Omega^2) = \zeta_0$

Buchholz Hydra	Associated Ordinal Notation	Associated Ordinal
	$D_0(D_1(D_1(D_1(\mathbf{0}))))$	$\psi_0(\Omega^\Omega) = \Gamma_0$
	$D_0(D_1(D_1(D_1(D_1(\mathbf{0})))))$	$\psi_0\left(\Omega^{\Omega^\Omega}\right) = \text{LVO}$
	$D_0(D_2(\mathbf{0}))$	$\psi_0(\Omega_2) = \text{BHO}$
	$D_0(D_\omega(\mathbf{0}))$	$\psi_0(\Omega_\omega) = \text{BO}$

By restricting Buchholz Hydra to only those corresponding to OT (rather than all of T), we can prove the termination of Buchholz Hydras.

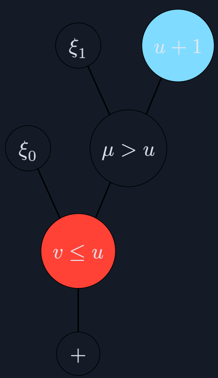
Standardness

OT is defined as all $D_v(a)$ such that $a \in \mathcal{C}_v(a)$ (and $v \leq \omega$),

Termination

Again, we show that "expansion" of the hydra always decreases the associated ordinal notation.

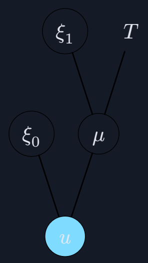
It is obvious how case 1 would cause termination is just Kirby-Paris hydra all over again. The main complication comes from case 2:



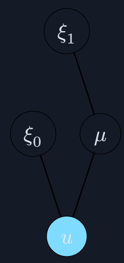
$$
\begin{aligned}
&= \psi_v(\xi_0 + \psi_\mu(\xi_1 + \psi_{u+1}(0))) \\
&= \psi_v(\xi_0 + \psi_\mu(\xi_1 + \Omega_{u+1}))
\end{aligned}
$$

And we have $S(T)$ as:

$S(T) =$

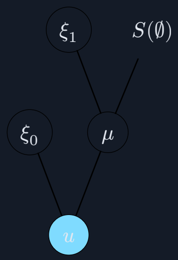


$S(\emptyset) =$

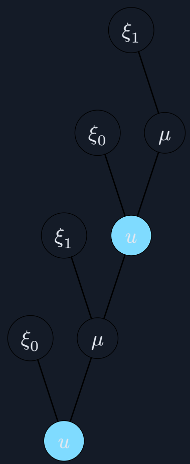


$$= \psi_u(\xi_0 + \psi_\mu(\xi_1))$$

$S^2(\emptyset) =$



$=$



$$\begin{aligned} &= \psi_u(\xi_0 + \psi_\mu(\xi_1 + S(\emptyset))) \\ &= \psi_u(\xi_0 + \psi_\mu(\xi_1 + \psi_u(\xi_0 + \psi_\mu(\xi_1)))) \end{aligned}$$

Since $\psi_u(\alpha) < \Omega_{u+1} = \psi_{u+1}(0)$, which is the ordinal notation corresponding to our original $u+1$ leaf node that we cut, it stands to reason that cutting off the head decreases the associated ordinal notation of the whole tree as a result.

9.3 The Hyper Primitive Sequence System (HPrSS)

The Hyper Primitive Sequence System (HPrSS) is an extension of the Primitive Sequence System (PrSS), and much like how PrSS can be thought of as a linearized version of the Kirby-Paris Hydra, HPrSS can be thought of as a linearized version of Buchholz's Hydra.

Definition 9.7.

A sequence $s = (s_1, s_2, ..., s_m)$ is a valid HPrSS sequence (i.e. $s \in T_{\text{HPrSS}}$) if s is either empty $()$ or starts with 0: $(0, ...)$.

We then define its fundamental sequence $s[n]$, which can be used to define a fast-growing function similarly to PrSS.

We then take the last element of the sequence, s_m, and find its **parent**, the rightmost s_i such that $s_i < s_m$ (same as PrSS). We then keep finding that parent's parent, and so on until we can no longer find a parent.

We let $m = m_0$, then denote the index of s_{m_0}'s parent as m_1, the index s_{m_1}'s parent as m_2, and so on.

We end up with a table like this:

k	m_k	s_{m_k}
0	$m_0 = m$	s_{m_0}
1	m_1	s_{m_1}
$\vdots$	$\vdots$	$\vdots$
k	m_k	s_{m_k}

Now let's define a **difference sequence** $N = (N_0, N_1, ..., N_{k-1})$, where $N_i = S_{m_i} - S_{m_{i+1}}$.

1. If $N_0 = 1$, then we proceed as normal PrSS, with **good root** being $\left(s_1, ..., s_{m_1 - 1}\right)$ and **bad root** being $\left(s_{m_1}, ..., s_{m_0 - 1}\right)$. and if $m_1 = 0$ then good root is empty.
2. If $N_0 \neq 1$, then we take the difference sequence $N = (N_0, N_1, ..., N_{k-1})$, and try to find the first element in the rest of the sequence N_i such that $N_i < N_0$. Put more formally,

$$p = \min\{i \mid i \in N \text{ and } 0 < i \leq k - 1 \text{ and } N_i < N_0\}$$

   We then define r and **ascension factor** δ as such
   1. If p doesn't exist/no such i exists (i.e., $N_i \nless N_0$ for all $0 < i \leq k - 1$), then we have $r = 1$, and ascension factor $\delta = S_{m_0} - 1$
   2. If p does exist, then $r = m_p$ and ascension factor $\delta = S_{m_0} - S_{m_p} - 1$.

   We define **good root** $G = (s_1, ..., s_{r-1})$ for $r > 1$, or just empty $G = ()$ if $r = 1$. We then define **bad root** as a **function** $B(n) = \left(s_r + n \cdot \delta, ..., s_{m_0 - 1} + n \cdot \delta\right)$, i.e. $\left(s_r, ..., s_{m_0 - 1}\right)$ but add $n \cdot \delta$ to each element.

   Finally, we define $s[n]$ as the concatenation of $G, B(0), ..., B(n)$.

In the original $N_0 = 1$ case, where normal PrSS rules apply, we can think of it as $r = m_1$ and $\delta = 0$.

Just like in PrSS, we can define a fast-growing function from these fundamental sequences also labelled $s[n]$:

$$s[n] = (G, B_0, ..., B_n)[f(n)]$$

where $f(n)$ is a fast-growing function, usually $f(n) = n + 1$.

Example 9.8.

Expand the following HPrSS sequences:

1. $(0, 2, 1)[n]$
2. $(0, 2)[n]$
3. $(0, 2, 1, 3)[n]$

Answers:

1. For $s = (0, 2, 1)$, the parent of $s_3 = 1$ is $s_1 = 0$, so we have:

k	m_k (indices of parents)	s_{m_k} (values of parents)
0	$m_0 = 3$	$s_{m_0} = s_3 = 1$
1	$m_1 = 1$	$s_{m_1} = s_1 = 0$

Our difference sequence is just $N = (1)$, so by normal PrSS expansion we have good root $G = ()$ and bad root $B = (0, 2)$, and as such

$$(0, 2, 1)[n] = \underbrace{(0, 2, 0, 2, \ldots)}_{(0,2) \text{ repeated } n \text{ times}}$$

Now using our new system, we have $r = m_1 = 1$ so we still have empty good root $G = ()$, and an ascension factor of $\delta = 0$, so our bad root function

$$\begin{aligned} B(n) &= \left(s_r + n \cdot 0, \ldots, s_{m_0 - 1} + n \cdot 0\right) \\ &= \left(s_1, \ldots, s_{m_0 - 1}\right) \\ &= (0, 2) \end{aligned}$$

So we end up with the same result.

2. For $s = (0, 2)$, the parent of $s_2 = 2$ is $s_1 = 0$, so we have

k	m_k (indices of parents)	s_{m_k} (values of parents)
0	$m_0 = 2$	$s_{m_0} = s_2 = 2$
1	$m_1 = 1$	$s_{m_1} = s_1 = 0$

Our difference sequence is $N = (2)$, so we fall under the sub-case 1, where there exists no i such that $N_i < N_0$. We then have $r = 1$ which leads to an empty good root $G = ()$, and an ascension factor of $\delta = s_{m_0} - 1 = 2 - 1 = 1$. So our bad root function is:

$$\begin{aligned} B(n) &= \left(s_r + n \cdot \delta, \ldots, s_{m_0 - 1} + n \cdot \delta\right) \\ &= (s_1 + n \cdot 1) \\ &= (0 + n) \\ &= (n) \end{aligned}$$

So concatenating $G, B(0), \ldots, B(n)$, we concatenate $(), (0), (1), \ldots, (n)$:

$$(0, 2)[n] = (0, 1, 2, 3, \ldots, n)$$

3. For $s = (0, 2, 1, 3)$, the parent of $s_4 = 3$ is $s_3 = 1$, whose parent is $s_1 = 0$, so we have

k	m_k (indices of parents)	s_{m_k} (values of parents)
0	$m_0 = 4$	$s_{m_0} = s_4 = 3$
1	$m_1 = 2$	$s_{m_1} = s_2 = 1$
2	$m_2 = 1$	$s_{m_2} = s_1 = 0$

Our difference sequence is $N = (2, 1)$. Since for $i = 1$, $N_i < N_0$, we have $p = 1$ can use the second sub-case, and we have $r = m_p = m_1$ and $\delta = s_{m_0} - s_{m_p} - 1 = s_4 - s_1 - 1 = 1$.

You should be able to perform good root and bad root calculations by now, so concatenating $G = (0, 2)$ and $B(n) = (n + 1)$ we have:

$$(0, 2, 1, 3)[n] = (0, 2, 1, 2, 3, \ldots)$$

Definition 9.9.

The set of **standard form** of sequences in HPrSS, OT_{HPrSS} is a subset of T_{HPrSS} defined as such;

1. For any $n \in \mathbb{N}$, $(0, n) \in OT_{\text{HPrSS}}$.
2. For any $S \in OT \setminus \{()\}$ and $n \in \mathbb{N}$, $s[n] \in OT_{\text{HPrSS}}$

Similar to our section on formalizing the order isomorphism of PrSS to ε_0, the OT here is built in a "top-down" approach. We re-use the definition of $\sqsubset$ from our PrSS section:

Definition 9.10.

For two sequences s, t, we define the relation $s \sqsubset t$ if and only if there exists $n_1, n_2, \ldots, n_k \in \mathbb{N}$ such that

$$s = t[n_1][n_2]\ldots[n_k]$$

Then we define $s \sqsupset t$ if and only if $t \sqsubset s$.

And as such we can re-define OT_{HPrSS}:

Definition 9.11.

The set of **standard form** of sequences in HPrSS, OT_{HPrSS} is a subset of T_{HPrSS} defined as such;

$$OT_{\text{HPrSS}} = \{s \in T_{\text{HPrSS}} \mid \text{There exists an } n \in \mathbb{N} \text{ such that } s \sqsubset (0, n)\}$$

Order Isomorphism to $\psi_0(\Omega_\omega)$

We can prove that $\left(OT_{\text{HPrSS}}, \underset{\text{lex}}{\prec}\right)$ is order isomorphic to $(\psi_0(\Omega_\omega), \in)$ in a similar way as we did for PrSS.

[WIP] How to prove this is beyond me

As we saw earlier in PrSS, $(0, 1, 2, 3, \ldots)$ is equivalent to $\omega^{\omega^{\cdot^{\cdot^{\cdot}}}} = \varepsilon_0$. Since we now have $(0, 2)[n]$ which expands to $(0, 1, 2, 3, \ldots)$, we have $(0, 2)$ corresponding to ε_0. In fact, we can expand upon this and define larger and larger ordinals:

We denote W_v as $D_v(0)$ for brevity.

HPrSS	Ordinal Notation	Ordinal
$()$	0	0
(0)	$D_0(0) = W_0$	1
$(0, 0)$	$D_0(0) + D_0(0)$	2
$(0, 1)$	$D_0(D_0(0))$	ω
$(0, 1, 2)$	$D_0(D_0(D_0(0)))$	ω^ω
$(0, 1, 2, 3)$	$D_0(D_0(D_0(D_0(0))))$	ω^{ω^ω}
$(0, 2)$	$D_0(W_1)$	ε_0
$(0, 2, 1)$	$D_0(W_1 + D_0(0))$	$\varepsilon_0 \cdot \omega$
$(0, 2, 1, 2)$	$D_0(W_1 + D_0(D_0(0)))$	$\varepsilon_0 \cdot \omega^\omega$
$(0, 2, 1, 2, 3)$	$D_0(W_1 + D_0(D_0(D_0(0))))$	$\varepsilon_0 \cdot \omega^{\omega^\omega}$
$(0, 2, 1, 3)$	$D_0(W_1 + D_0(W_1))$	ε_0^2
$(0, 2, 1, 3, 2, 4)$	$D_0(W_1 + D_0(W_1 + D_0(W_1)))$	$\varepsilon_0^{\varepsilon_0}$
$(0, 2, 2)$	$D_0(W_1 + W_1)$	ε_1

HPrSS	Ordinal Notation	Ordinal
$(0,2,2,2)$	$D_0(W_1 + W_1 + W_1)$	ε_2
$(0,2,3)$	$D_0(D_1(W_0))$	$\psi_0(\Omega \cdot \omega) = \varepsilon_\omega$
$(0,2,3,5)$	$D_0(D_1(D_0(W_1)))$	$\psi_0(\Omega \cdot \psi_0(\Omega)) = \varepsilon_{\varepsilon_0}$
$(0,2,4)$	$D_0(D_1(W_1))$	$\psi_0(\Omega^2) = \zeta_0$
$(0,2,4,4)$	$D_0(D_1(W_1 + W_1))$	$\psi_0(\Omega^3) = \eta_0 = \varphi(3,0)$
$(0,2,4,5)$	$D_0(D_1(D_1(W_0)))$	$\psi_0(\Omega^\omega) = \varphi(\omega,0)$
$(0,2,4,5,7)$	$D_0(D_1(D_1(D_0(W_1))))$	$\psi_0(\Omega^{\psi_0(\Omega)}) = \psi_0(\Omega^{\varepsilon_0}) = \varphi(\varepsilon_0,0)$
$(0,2,4,6)$	$D_0(D_1(D_1(W_1)))$	$\psi_0(\Omega^\Omega) = \Gamma_0$
$(0,2,4,6,8)$	$D_0(D_1(D_1(D_1(W_1))))$	$\psi_0\left(\Omega^{\Omega^\Omega}\right) = \text{LVO}$
$(0,3)$	$D_0(W_2)$	$\psi_0(\Omega_2) = \psi_0(\varepsilon_{\Omega+1}) = \text{BHO}$
$(0,3,1)$	$D_0(W_2 + W_0)$	$\psi_0(\Omega_2 + 1) = \psi_0(\varepsilon_{\Omega+1}) \cdot \omega$
$(0,3,2)$	$D_0(W_2 + W_1)$	$\psi_0(\Omega_2 + \Omega)$
$(0,3,3)$	$D_0(W_2 + W_2)$	$\psi_0(\Omega_2 \cdot 2)$
$(0,3,4)$	$D_0(D_2(W_0))$	$\psi_0(\Omega_2 \cdot \omega)$
$(0,3,5)$	$D_0(D_2(W))$	$\psi_0(\Omega_2 \cdot \Omega)$
$(0,3,6)$	$D_0(D_2(W_2))$	$\psi_0(\Omega_2^2)$
$(0,4)$	$D_0(W_3)$	$\psi_0(\Omega_3)$
$(0,n+1)$	$D_0(W_n)$	$\psi_0(\Omega_n)$

Growth rate

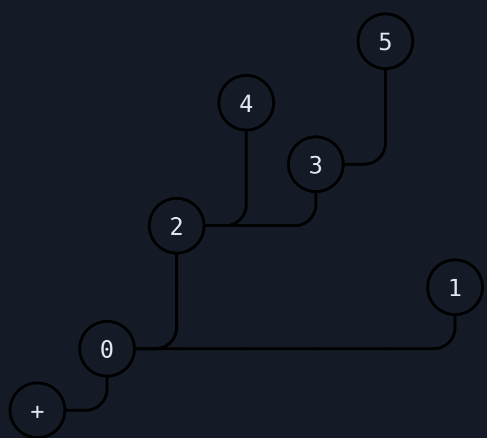
Association with Buchholz Hydras

There is a somewhat weird way I came up with to associate HPrSS with Buchholz Hydras (unproven)

Suppose we have a sequence: $(0,2,4,3,5,1)$. We label each element's parent:

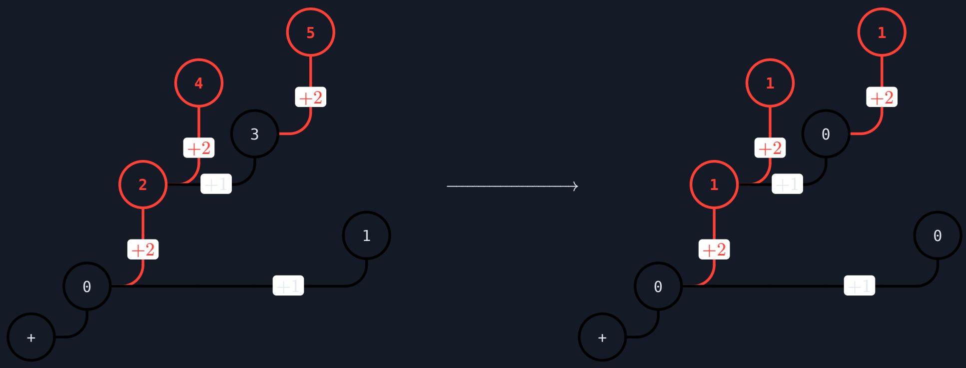
$$(0,2,4,3,5,1)$$

Then we draw a tree representing the parent relationships:



So for example, 5's parent is 3, whose parent is 2, whose parent is 0.

Now let's consider the different in values. If the a node's parent's value is 1 less than its own value, we label the node 0. If a node's parent's value is 2 less than its own value, we label the node 1. And in general, if a node's parent's value is n less than its own value, we label the node $n-1$. This results in the following Buchholz Hydra:



9.4 The Pair Sequence System (PSS)

The Pair Sequence System (PSS) can also be considered an extension to the primitive sequence system, where instead of it being a sequence of natural numbers, the Pair Sequence System can be thought og as a sequence of *pairs* of natural numbers. There are two common ways to write PSS, either inline, like $(0,0)(1,1)(2,1)(2,0)$ or as a matrix like so:

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Definition 9.12.

The **Pair Sequence System** is a two-row matrix M, which can also be thought of as a sequence of pairs.

$$M = \begin{pmatrix} a_1 & a_2 & \dots & a_m \\ b_1 & b_2 & \dots & b_m \end{pmatrix}$$

The first row $(a_1, a_2, \dots, a_m)$ must satisfy PrSS rules, i.e.:

1. the row must start with 0 (i.e. $a_1 = 0$)
2. for each element in a row, the value of its parent must be 1 less than its own value

For the second row, every element in the second row must be lesser than or equal to the corresponding value in the first row, i.e. $a_i \geq b_i$ for all $1 \leq i \leq m$.

It's fundamental sequence $M[n]$ is defined as such:

1. If $\begin{pmatrix} a_m \\ b_m \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, then we just remove $\begin{pmatrix} a_m \\ b_m \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

$$M[n] = \begin{pmatrix} a_1 & a_2 & \dots & a_{m-1} \\ b_1 & b_2 & \dots & b_{m-1} \end{pmatrix}$$

   This is similar to PrSS when a sequence ends with a 0.

2. If $\begin{pmatrix} a_m \\ b_m \end{pmatrix}$ if of the form $\begin{pmatrix} a_m \\ 0 \end{pmatrix}$, where $a_m > 0$, then within row one we find the parent of a_m. Let's denote the index of the parent as m_1, and the value of the parent becomes a_{m_1}. We define **Good Root** G and **Bad Root** B as:

$$G = \begin{pmatrix} a_1 & \dots & a_{m_1-1} \\ b_1 & \dots & b_{m_1-1} \end{pmatrix}, \quad B = \begin{pmatrix} a_{m_1} & \dots & a_{m_1-1} \\ b_{m_1} & \dots & b_{m_1-1} \end{pmatrix}$$

   Then $s[n]$ becomes $G {}^\frown \underbrace{B {}^\frown \dots {}^\frown B}_{n \text{ times}}$, where $^\frown$ represents "matrix concatenation". This case, along with the previous case, basically the same mechanism as PrSS.

3. If $a_m > 0$ and $b_m > 0$, we first focus only on the first row of the matrix. Like HPrSS, we find the parent of a_m, then its parent, then its parent, all the way until 0. Let's label $m_0 = m$, m_1 to be the index of a_{m_0}'s parent, etc..., similar to how we did HPrSS.

$$M = \begin{pmatrix} a_{m_k} & \dots & a_{m_2} & \dots & a_{m_1} & \dots & a_{m_0} \\ b_{m_k} & \dots & b_{m_2} & \dots & b_{m_1} & \dots & b_{m_0} \end{pmatrix}$$

Then now within the sequence $\left(b_{m_k}, \dots, b_{m_2}, b_{m_1}, b_{m_0}\right)$, we find the rightmost element such that $b_{m_i} < b_{m_0}$. The column that this element is part of, $\left(\begin{smallmatrix} a_{m_i} & b_{m_i} \end{smallmatrix}\right)$, will serve as our separator between good root and bad root. Note how you first search for all of a_m's "ancestors", then exclusively within those columns, we search for b_m's parent.

Continuing, the **Good Root** is then defined as

$$G = \begin{pmatrix} a_0 & \dots & a_{m_i - 1} \\ b_0 & \dots & b_{m_i - 1} \end{pmatrix}$$

and the **Bad Root** is a function, with a similar definition to HPrSS. We define the **ascension factor** δ and **ascension matrix** Δ as:

$$\delta = a_m - a_{m_i}$$

$$\Delta = \begin{pmatrix} \delta \\ 0 \end{pmatrix} = \begin{pmatrix} a_m - a_{m_i} \\ 0 \end{pmatrix}$$

Then, $B(n)$ is essentially adding Δ to each pair in B element-wise, i.e.,

$$B(0) = \begin{pmatrix} a_{m_i} & \dots & a_{m-1} \\ b_{m_i} & \dots & b_{m-1} \end{pmatrix}$$

$$B(1) = \begin{pmatrix} a_{m_i} + \delta & \dots & a_{m-1} + \delta \\ b_{m_i} & \dots & b_{m-1} \end{pmatrix}$$

$$B(n) = \begin{pmatrix} a_{m_i} + \delta \cdot n & \dots & a_{m-1} + \delta \cdot n \\ b_{m_i} & \dots & b_{m-1} \end{pmatrix}$$

Case 2 can be thought of as a special case of this rule where $\Delta = \left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right)$.

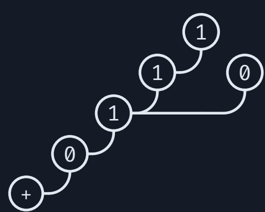
That might have been too complex, so here are some examples:

Note that when the bottom row is entirely 0s, we just get regular PrSS.

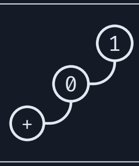
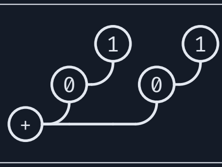
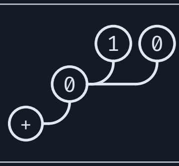
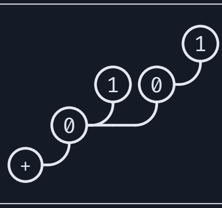
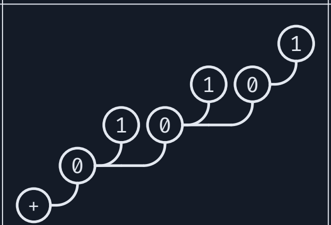
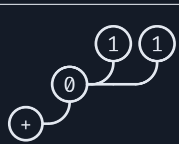
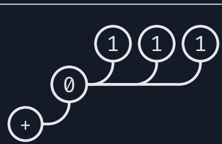
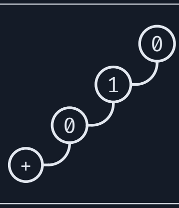
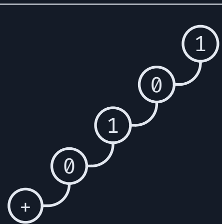
PSS Hydra

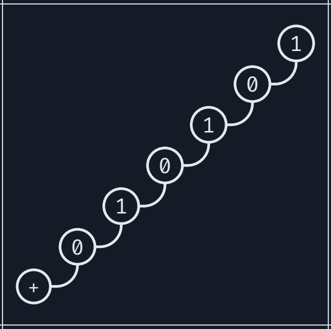
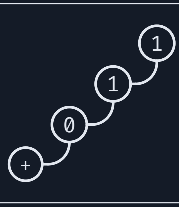
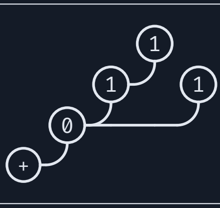
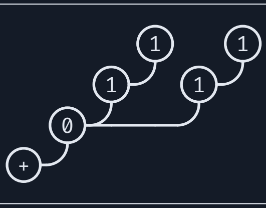
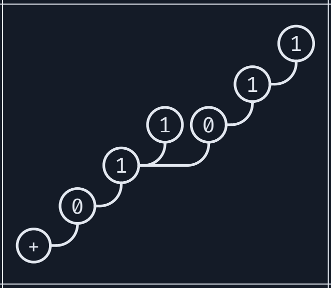
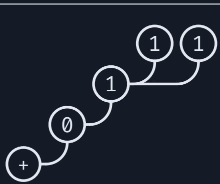
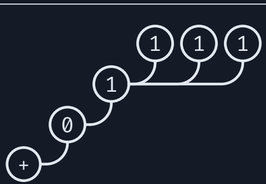
We can correspond a PSS into its hydra form, by constructing a hydra such that for each pair, the top row represents the node's height, and the bottom row represents the node's value:

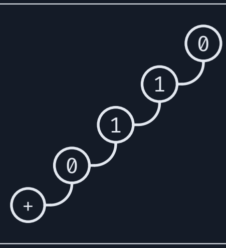
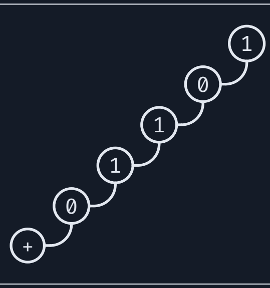
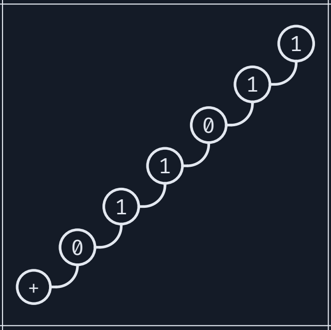
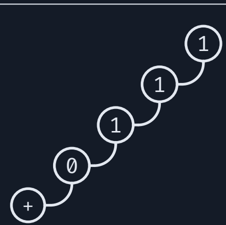
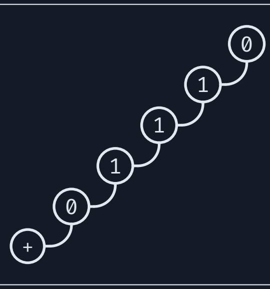
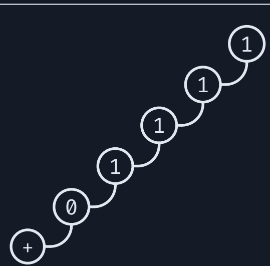
$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix} \longrightarrow$$



Now if we were to interpret it as a Buchholz Hydra, we get $D_0(D_1(D_1(D_1(0)) + D_0(0)))$, which suggests it corresponds to the ordinal $\psi_0(\Omega^\Omega \cdot \omega) = \Gamma_\omega$. This method will produce standard-form ordinal notation for notations less than $D_0(D_2(0)) \sim$ BHO

PSS	Hydra	Ordinal Notation	Ordinal
$\begin{pmatrix}0&1\\0&1\end{pmatrix}$		$D_0(D_1(\mathbf{0}))$	$\psi_0(\Omega)=\varepsilon_0$
$\begin{pmatrix}0&1&0&1\\0&1&0&1\end{pmatrix}$		$D_0(D_1(\mathbf{0}))+D_0(D_1(\mathbf{0}))$	$\psi_0(\Omega)\cdot 2=\varepsilon_0\cdot 2$
$\begin{pmatrix}0&1&1\\0&1&0\end{pmatrix}$		$D_0(D_1(\mathbf{0})+D_0(\mathbf{0}))$	$\psi_0(\Omega+1)=\varepsilon_0\cdot\omega$
$\begin{pmatrix}0&1&1&2\\0&1&0&1\end{pmatrix}$		$D_0(D_1(\mathbf{0})+D_0(D_1(\mathbf{0})))$	$\psi_0(\Omega+\psi_0(\Omega))=\varepsilon_0^2$
$\begin{pmatrix}0&1&1&2&2&3\\0&1&0&1&0&1\end{pmatrix}$		$D_0(D_1(\mathbf{0})+D_0(D_1(\mathbf{0})))$	$\psi_0(\Omega+\psi_0(\Omega))=\varepsilon_0^{\varepsilon_0}$
$\begin{pmatrix}0&1&1\\0&1&1\end{pmatrix}$		$D_0(D_1(\mathbf{0})+D_1(\mathbf{0}))$	$\psi_0(\Omega 2)=\varepsilon_1$
$\begin{pmatrix}0&1&1&1\\0&1&1&1\end{pmatrix}$		$D_0(D_1(\mathbf{0})\cdot 3)$	$\psi_0(\Omega 3)=\varepsilon_2$
$\begin{pmatrix}0&1&2\\0&1&0\end{pmatrix}$		$D_0(D_1(D_0(\mathbf{0})))$	$\psi_0(\Omega\cdot\omega)=\varepsilon_\omega$
$\begin{pmatrix}0&1&2&3\\0&1&0&1\end{pmatrix}$		$D_0(D_1(D_0(D_1(\mathbf{0}))))$	$\psi_0(\Omega\cdot\psi_0(\Omega))=\varepsilon_{\varepsilon_0}$

PSS	Hydra	Ordinal Notation	Ordinal
$\begin{pmatrix} 0 & 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$		$D_0(D_1(D_0(D_1(D_0(D_1(0))))))$	$\psi_0(\Omega \cdot \psi_0(\Omega \cdot \psi_0(\Omega))) = \varepsilon_{\varepsilon_{\varepsilon_0}}$
$\begin{pmatrix} 0 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}$		$D_0(D_1(D_1(0)))$	$\psi_0(\Omega^2) = \zeta_0$
$\begin{pmatrix} 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$		$D_0(D_1(D_1(0)) + D_1(0))$	$\psi_0(\Omega^2 + \Omega) = \varepsilon_{\zeta_0 + 1}$
$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 \end{pmatrix}$		$D_0(D_1(D_1(0)) \cdot 2)$	$\psi_0(\Omega^2 \cdot 2) = \zeta_1$
$\begin{pmatrix} 0 & 1 & 2 & 2 & 3 & 4 \\ 0 & 1 & 1 & 0 & 1 & 1 \end{pmatrix}$		$D_0(D_1(D_1(0) + D_0(D_1(D_1(0)))))$	$\psi_0(\Omega^2 \cdot \psi_0(\Omega^2)) = \zeta_{\zeta_0}$
$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 \end{pmatrix}$		$D_0(D_1(D_1(0) + D_1(0)))$	$\psi_0(\Omega^3) = \varphi(3,0) = \eta_0$
$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 \end{pmatrix}$		$D_0(D_1(D_1(0) \cdot 3))$	$\psi_0(\Omega^4) = \varphi(4,0)$

PSS	Hydra	Ordinal Notation	Ordinal
$\begin{pmatrix}0&1&2&3\\0&1&1&0\end{pmatrix}$		$D_0(D_1(D_1(D_0(0))))$	$\psi_0(\Omega^\omega) = \varphi(\omega, 0)$
$\begin{pmatrix}0&1&2&3&4\\0&1&1&0&1\end{pmatrix}$		$D_0(D_1(D_1(D_0(D_1(0)))))$	$\psi_0\left(\Omega^{\psi_0(\Omega)}\right)$ $= \varphi(\varepsilon_0, 0)$
$\begin{pmatrix}0&1&2&3&4&5\\0&1&1&0&1&1\end{pmatrix}$		$D_0(D_1(D_1(D_0(D_1(D_1(0))))))$	$\psi_0\left(\Omega^{\psi_0(\Omega^2)}\right)$ $= \varphi(\zeta_0, 0)$
$\begin{pmatrix}0&1&2&3\\0&1&1&1\end{pmatrix}$		$D_0(D_1(D_1(D_1(0))))$	$\psi_0(\Omega^\Omega) = \Gamma_0$
$\begin{pmatrix}0&1&2&3&4\\0&1&1&1&0\end{pmatrix}$		$D_0(D_1(D_1(D_1(D_0(0)))))$	$\psi_0\left(\Omega^{\Omega^\omega}\right) = \text{SVO}$
$\begin{pmatrix}0&1&2&3&4\\0&1&1&1&1\end{pmatrix}$		$D_0(D_1(D_1(D_1(D_1(0)))))$	$\psi_0\left(\Omega^{\Omega^\Omega}\right) = \text{LVO}$

Standardization

But the problem begins at the Bachmann-Howard Ordinal. If you think of it as $\psi_0(\Omega^{\Omega^{\cdot^{\cdot}}})$, in PSS that translates to:

$$\psi_0(\Omega) \sim \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\psi_0(\Omega^2) \sim \begin{pmatrix} 0 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\psi_0(\Omega^\Omega) \sim \begin{pmatrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

$$\psi_0\left(\Omega^{\Omega^\Omega}\right) \sim \begin{pmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$$\psi_0(\Omega^{\Omega^{\cdot^{\cdot}}}) \sim \begin{pmatrix} 0 & 1 & 2 & 3 & 4 & \ldots \\ 0 & 1 & 1 & 1 & 1 & \ldots \end{pmatrix}$$

So we need a matrix with good root $G = \binom{0}{0}$, base bad root $B = \binom{1}{0}$, ascension $\Delta = \binom{1}{0}$, which leads us to:

$$\text{BHO} \sim \begin{pmatrix} 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix} \sim \psi_0(\psi_1(\psi_2(0)))$$

which as we have already seen is non-standard.

The standardization algorithm is as such:

- Find the innermost non-standard term $D_v(D_{v+1}(a_0 + \ldots + a_k))$, such that $D_{v+1}(a_0 + \ldots + a_k) \notin \mathcal{C}_v(D_{v+1}(a_0 + \ldots + a_k))$
- Collect all terms $a_0, \ldots, a_i \geq W_{v+2}$ (There will be such terms because otherwise it would be standard form)
  - If all terms $a_0, \ldots, a_k \geq W_{v+2}$, then replace
  $$D_v(D_{v+1}(a_0 + \ldots + a_k)) \to D_v(a_0 + \ldots + a_k)$$
  - If only some terms $a_0, \ldots, a_i \geq W_{v+2}$, then replace
  $$D_v(D_{v+1}(a_0 + \ldots + a_k)) \to D_v(a_0 + \ldots + a_i + D_{v+1}(a_0 + \ldots + a_k))$$

Example 9.13.

Standardize the following:

1. $D_0(D_1(W_2 + W_1))$

   $D_1(W_2 + W_1)$ is standard form since $W_2 + W_1 \in C_1(W_2 + W_1)$
   $D_0(D_1(W_2 + W_1))$ is **not** standard form since $W_2 + W_1 \notin C_0(D_1(W_2 + W_1))$ as $W_2 > D_1(x)$, so $D_1(W_2 + W_1) \notin C_0(D_1(W_2 + W_1))$.

   So we have $W_2 \geq W_2$, while $W_1 < W_2$

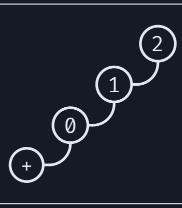
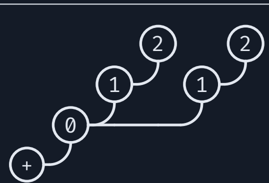
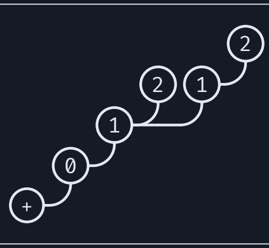
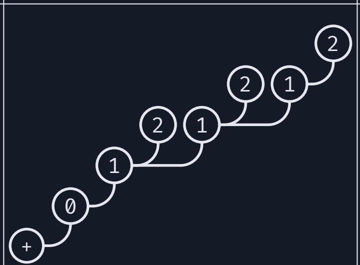
   We convert our term to $D_0(W_2 + D_1(W_2 + W_1))$
2. $D_0(D_1(D_2(D_3(D_4(W_5) + W_4) + D_3(D_4(W_5) + W_3) + W_2)))$

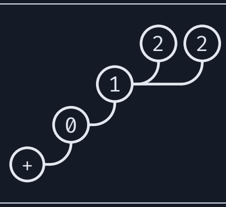
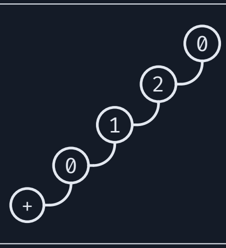
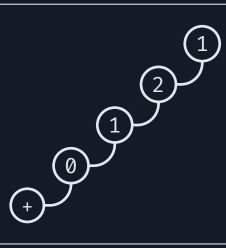
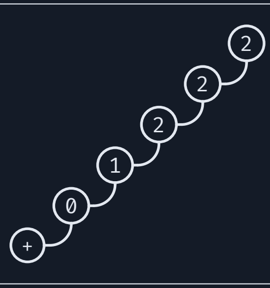
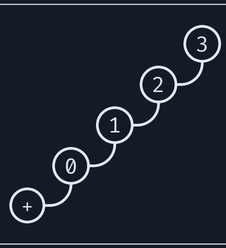
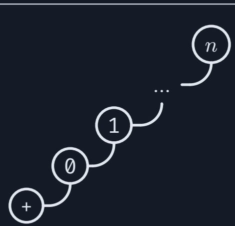
   Terms highlighted represent the a's that get modified

$$
\begin{aligned}
&D_0(D_1(D_2(D_3(D_4(W_5) + W_4) + D_3(D_4(W_5) + W_3) + W_2)))\\
&D_0(D_1(D_2(D_3(W_5 + W_4) + D_3(D_4(W_5) + W_3) + W_2)))\\
&D_0(D_1(D_2(D_3(W_5 + W_4) + D_3(D_4(W_5) + W_3) + W_2)))\\
&D_0(D_1(D_2(W_5 + W_4 + D_3(D_4(W_5) + W_3) + W_2)))\\
&D_0(D_1(D_2(W_5 + W_4 + D_3(D_4(W_5) + W_3) + W_2)))\\
&D_0(D_1(D_2(W_5 + W_4 + D_3(W_5 + W_3) + W_2)))\\
&D_0(D_1(D_2(W_5 + W_4 + D_3(W_5 + W_3) + W_2)))\\
&D_0(D_1(W_5 + W_4 + D_3(W_5 + W_3) + D_2(W_5 + W_4 + D_3(W_5 + W_3) + W_2)))\\
&D_0(D_1(W_5 + W_4 + D_3(W_5 + W_3) + D_2(W_5 + W_4 + D_3(W_5 + W_3) + W_2)))\\
&D_0(W_5 + W_4 + D_3(W_5 + W_3) + D_2(W_5 + W_4 + D_3(W_5 + W_3) + W_2))
\end{aligned}
$$

This "raw" form (i.e. $D_0(D_1(D_2(0)))$ instead of $D_0(D_2(0))$) is also known as PSS Hydra since its basically another way to write the hydra diagram.

For post-BHO expressions we will include both the "raw" PSS hydra form and the standardized form, with PSS Hydra on top and standardized below.

PSS	Hydra	Ordinal Notation	Ordinal
$\begin{pmatrix} 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix}$		$D_0(D_1(W_2))$ $D_0(W_2)$	$\psi_0(\Omega_2) = \text{BHO}$
$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 1 & 2 \end{pmatrix}$		$D_0(D_1(W_2) + D_1(W_2))$ $D_0(W_2 + D_1(W_2))$	$\psi_0(\Omega_2 + \psi_1(\Omega_2))$
$\begin{pmatrix} 0 & 1 & 2 & 2 & 3 \\ 0 & 1 & 2 & 1 & 2 \end{pmatrix}$		$D_0(D_1(W_2 + D_1(W_2)))$ $D_0(W_2 + D_1(W_2 + D_1(W_2)))$	$\psi_0(\Omega_2 + \psi_1(\Omega_2 + \psi_1(\Omega_2)))$ $= \psi_0\left(\Omega_2 + \psi_1(\Omega_2)^2\right)$
$\begin{pmatrix} 0 & 1 & 2 & 2 & 3 & 3 & 4 \\ 0 & 1 & 2 & 1 & 2 & 1 & 2 \end{pmatrix}$		$D_0(D_1(W_2 + D_1(W_2 + D_1(W_2))))$ $D_0(W_2 + D_1(W_2 + D_1(W_2 + D_1(W_2))))$	$\psi_0(\Omega_2 + \psi_1(\Omega_2 + \psi_1(\Omega_2 + \psi_1(\Omega_2))))$ $= \psi_0\left(\Omega_2 + \psi_1(\Omega_2)^{\psi_1(\Omega_2)}\right)$

PSS	Hydra	Ordinal Notation	Ordinal
$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 2 \end{pmatrix}$		$D_0(D_1(W_2+W_2))$ $D_0(W_2+W_2)$	$\psi_0(\Omega_2 \cdot 2)$
$\begin{pmatrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 0 \end{pmatrix}$		$D_0(D_1(D_2(W_0)))$ $D_0(D_2(W_0))$	$\psi_0(\Omega_2 \cdot \omega)$
$\begin{pmatrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 1 \end{pmatrix}$		$D_0(D_1(D_2(W_1)))$ $D_0(D_2(W_1))$	$\psi_0(\Omega_2 \cdot \Omega)$
$\begin{pmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 2 & 2 \end{pmatrix}$		$D_0(D_1(D_2(D_2(W_2))))$ $D_0(D_2(D_2(W_2)))$	$\psi_0\left(\Omega_2^{\Omega_2}\right)$
$\begin{pmatrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{pmatrix}$		$D_0(D_1(D_2(W_3)))$ $D_0(W_3)$	$\psi_0(\Omega_3)$
$\begin{pmatrix} 0 & 1 & \ldots & n \\ 0 & 1 & \ldots & n \end{pmatrix}$		$D_0(D_1(\ldots D_{n-1}(W_n)\ldots))$ $D_0(W_n)$	$\psi_0(\Omega_n)$

We see that the limit of PSS is also $\psi_0(\Omega_\omega)$.

9.5 Patcail's Hydra/Mini-Nuclear Array Notation [WIP]

```
# FSes FOR PATCAIL NOTATION
## Normalization function
The function Norm(x) is defined as follows:
1. Norm(0) = 0
2. If x = [[0,a],b]:
   Norm(x) = [c,[A,b]], where
   A = Norm(a)
```

```
   c = A but all instances of b are replaced with [A,b].
3. Otherwise, Norm([a,b]) = [Norm(a),b]
## FSes
The function FS(x,k) is defined as follows:
1. FS(0,k) = 0
2. FS([0,a],c) = a
3. FS(n,k) = k
4. FS([a,b],k) = FS(Norm(x),k) if Norm(a) is of the form [0,c].
5. Otherwise, FS([a,b],k) = [FS(a,k),b].
Note that n is a term, equivalent to ω.
```

WIP:

- Explore the hydra a little
- sgh-mgh relation to the hydra
- sgh-mgh catching point at BO

sgh-mgh catching point

We earlier showed in chapter 6, that $H_{\varepsilon_0}(n) \approx f_{\varepsilon_0}(n)$. We will now related sgh and mgh to the rest of these hierarchies.

10 Extended Buchholz's OCF [WIP]

Why do we have to stop at Ω_ω? What about $\Omega_{\omega+1}$? We can extend Buchholz's function as such to allow for even larger ordinals:

Definition 10.1 (Extended Buchholz's function).

Using the same definition of Ω_ν, we define the **Extended Buchholz's function** ψ and the set $C_\nu(\alpha)$ as such:

1. $\Omega_\nu \subseteq C_\nu(\alpha)$
2. For any two ordinals $\xi, \eta \in C_\nu(\alpha)$, their sum $\xi + \eta \in C_\nu(\alpha)$
3. For any ordinal $\xi \in C_\nu(\alpha)$, as long as $\xi < \alpha$, then $\psi_\mu(\xi) \in C_\nu(\alpha)$ for all $\mu \in C_\nu(\alpha)$
4. $\psi_\nu(\alpha) = \min\{\gamma \in \text{Ord} \mid \gamma \notin C_\nu(\alpha)\}$, i.e., the smallest ordinal not inside $C_\nu(\alpha)$

It has the same properties as the original, just that instead of being restricted to ψ_μ with $\mu \leq \omega$, we now have $\mu \in C_\nu(\alpha)$ as highlighted in red. This is often abbreviated in the community as **EBOCF** (**E**xtended **B**uchholz's **O**rdinal **C**ollapsing **F**unction)

With this, we can go beyond, letting $\text{TFBO} = \psi_0(\Omega_{\omega+1})$. We can make $\psi_0\left(\Omega_{\varepsilon_0}\right)$), $\psi_0\left(\Omega_{\Gamma_0}\right)$, or even $\psi_0(\Omega_\Omega)$.

The limit of this notation is the **Extended Buchholz Ordinal (EBO)** $\psi_0\left(\Omega_{\Omega_{\cdot_{\cdot}}}\right)$. The infinitely nested Ω -subscript tower is the **Omega fixed point** where $\Omega_\bullet = \bullet$, and is commonly denoted as Λ. Therefore the EBO can also be written as $\psi_0(\Lambda)$.

10.1 Normal Form and Fundamental Sequences

The normal form of EBOCF is very similar to that of BOCF:

Theorem 10.2.

Every ordinal $\alpha < \psi_0\left(\Omega_{\Omega_{\cdot_{\cdot}}}\right) = \psi_0(\Lambda)$ can be uniquely expressed in the form:

$$\alpha = \psi_{\nu_1}(\alpha_1) + \ldots + \psi_{\nu_n}(\alpha_n)$$

Where:

- $\alpha_1, \ldots, \alpha_n, \nu_1, \ldots, \nu_n \in C_0(\Lambda)$
- $\psi_{\nu_1}(\alpha_1) \geq \ldots \geq \psi_{\nu_n}(\alpha_n)$
- $\alpha_1 \in C_{\nu_1}(\alpha_1), \ldots, \alpha_n \in C_{\nu_n}(\alpha_n)$

The fundamental sequence rules are also very similar, the only change being Rule 4, highlighted in fuchsia.

Definition 10.3 (System of fundamental sequences for ψ).

For ordinals $\alpha \leq \psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$ (expressed in normal form), the η^{th} term of the fundamental sequence of α, $\alpha[\eta]$ is defined as such:

1. If $\alpha = \psi_{k_1}(\alpha_1) + \ldots + \psi_{k_n}(\alpha_n)$, then $\text{cof}(\alpha) = \text{cof}\left(\psi_{k_n}(\alpha_n)\right)$, and $\alpha[\eta] = \psi_{k_1}(\alpha_1) + \ldots + \left(\psi_{k_n}(\alpha_n)[\eta]\right)$
2. If $\alpha = \psi_0(0) = 1$ then $\text{cof}(\alpha) = 1$. We need a sequence of length 1, so we set $\alpha[0] = 0$
3. If $\alpha = \psi_{\nu+1}(0) = \Omega_{\nu+1}$, then $\text{cof}(\alpha) = \Omega_{\nu+1}$. We need a sequence of length $\Omega_{\nu+1}$, so we set $\alpha[\eta] = \Omega_{\nu+1}[\eta] = \eta$
4. If $\alpha = \psi_\nu(0)$, and $\text{cof}(\nu) \geq \omega$ and then $\text{cof}(\alpha) = \text{cof}(\nu)$ and $\alpha[\eta] = \psi_{\nu[\eta]}(0) = \Omega_{\nu[\eta]}$

5. If $\alpha = \psi_\nu(\beta + 1)$ then $\operatorname{cof}(\alpha) = \omega$ and $\alpha[\eta] = \psi_\nu(\beta) \cdot \eta$
6. If $\alpha = \psi_\nu(\beta)$ where β is a limit ordinal,
   (a) If $\operatorname{cof}(\beta) = \omega$ or $\operatorname{cof}(\beta) \leq \Omega_\nu$, then we have $\operatorname{cof}(\alpha) = \operatorname{cof}(\beta)$, so we need a sequence of length $\operatorname{cof}(\beta)$, so $\alpha[\eta] = \psi_\nu(\beta[\eta])$
   (b) Otherwise if $\operatorname{cof}(\beta) > \Omega_\nu$ and $\operatorname{cof}(\beta) \neq \omega$ then $\operatorname{cof}(\alpha) = \omega$. This is where the definitions diverge.

   We define μ as $\operatorname{cof}(\beta) = \Omega_{\mu+1}$.

   $\alpha[n] = \psi_\nu(\beta[\gamma[n]])$, where $\gamma[0] = 0$, and $\gamma[\eta + 1] = \psi_\mu(\beta[\gamma[\eta]])$. To illustrate this nesting process:

$$\alpha[0] = \psi_\nu(\beta[\gamma[0]]) = \psi_\nu(\beta[0])$$
$$\alpha[1] = \psi_\nu(\beta[\gamma[1]]) = \psi_\nu(\beta[\psi_\mu(\beta[0])])$$
$$\alpha[2] = \psi_\nu(\beta[\gamma[2]]) = \psi_\nu(\beta[\psi_\mu(\beta[\psi_\mu(\beta[0])])])$$

   In the case of $\mu = \nu$ however, we can use a shortcut: $\alpha[0] = \psi_\nu(\beta[0])$, and $\alpha[\eta + 1] = \psi_\nu(\beta[\alpha[\eta]])$

   However there are many other systems out there:

   - Dennis Maksudov:

     Same definition, except $\gamma[0] = \Omega_\mu$.
   - David_Exmachina:

     $\alpha[0] = 0$, and $\gamma[1] = 0$, otherwise $\gamma[\eta + 1]$ unchanged.

The final wrinkle is to define a fundamental sequence of $\alpha = \psi_0\left(\Omega_{\Omega_{\ddots}}\right) = \psi_0(\Lambda)$:
- $\alpha[0] = \psi_0(\Omega)$
- $\alpha[1] = \psi_0(\Omega_\Omega)$
- $\alpha[2] = \psi_0\left(\Omega_{\Omega_\Omega}\right)$
- ...

Example 10.4.

Evaluate the fundamental sequences of the following:
1. $\psi_0(\Omega_{\omega+1})$
2. $\psi_0(\Omega_{\omega+1}^2)$
3. $\psi_0\left(\Omega_{\varepsilon_0}\right)$
4. $\psi_0(\Omega_\Omega)$

Answers:
1. $\psi_0(\Omega_{\omega+1})[n]$

   $\operatorname{cof}(\Omega_{\omega+1}) = \Omega_{\omega+1}$ so rule 6(b), and $\operatorname{cof}(\psi_0(\Omega_{\omega+1})) = \omega$.

   Then $\mu = \omega$ and

$$\psi_0(\Omega_{\omega+1})[0] = \psi_0(\Omega_{\omega+1}[0]) = \psi_0(0)$$
$$\psi_0(\Omega_{\omega+1})[1] = \psi_0(\Omega_{\omega+1}[\psi_\omega(\Omega_{\omega+1}[0]]) = \psi_0(\psi_\omega(0))$$
$$\psi_0(\Omega_{\omega+1})[2] = \psi_0(\Omega_{\omega+1}[\psi_\omega(\Omega_{\omega+1}[\psi_\omega(\Omega_{\omega+1}[0])]]) = \psi_0(\psi_\omega(\psi_\omega(0)))$$

   and the general pattern is that:

$$\psi_0(\Omega_{\omega+1})[0] = \psi_0(0)$$
$$\psi_0(\Omega_{\omega+1})[1] = \psi_0(\Omega_\omega)$$
$$\psi_0(\Omega_{\omega+1})[2] = \psi_0(\Omega_\omega^{\Omega_\omega})$$
$$\psi_0(\Omega_{\omega+1})[3] = \psi_0\left(\Omega_\omega^{\Omega_\omega^{\Omega_\omega}}\right)$$
$$\vdots$$
$$\psi_0(\Omega_{\omega+1})[n] = \psi_0\left(\Omega_\omega^{\Omega_\omega^{\Omega_\omega^{\cdot^{\cdot^{\cdot}}}}}\right)$$

This is the same mechanism as to how $\psi_0(\Omega_2)[n] = \psi_0\left(\Omega^{\Omega^{\cdot^{\cdot^{\cdot}}}}\right)$ in Buchholz OCF

2. $\psi_0(\Omega_{\omega+1}^2)$ - Firstly, $\Omega_{\omega+1}^2 = \psi_{\omega+1}(\Omega_{\omega+1})$. Then we have $\mathrm{cof}(\Omega_{\omega+1}^2) = \Omega_{\omega+1}$ and

$$\psi_{\omega+1}(\Omega_{\omega+1})[\eta] = \psi_{\omega+1}(\Omega_{\omega+1}[\eta]) = \psi_{\omega+1}(\eta) = \Omega_{\omega+1} \cdot \eta$$

Skipping the calculations, we get:

$$\psi_0(\Omega_{\omega+1}^2)[0] = \psi_0(\Omega_{\omega+1})$$
$$\psi_0(\Omega_{\omega+1}^2)[1] = \psi_0(\Omega_{\omega+1} \cdot \psi_0(\Omega_{\omega+1}))$$
$$\psi_0(\Omega_{\omega+1}^2)[2] = \psi_0(\Omega_{\omega+1} \cdot \psi_0(\Omega_{\omega+1} \cdot \psi_0(\Omega_{\omega+1})))$$
$$\vdots$$
$$\psi_0(\Omega_{\omega+1}^2)[n] = \psi_0(\Omega_{\omega+1} \cdot \psi_0(\Omega_{\omega+1} \cdot \psi_0(\Omega_{\omega+1} \cdot \psi_0(\ldots))))$$

3. $\psi_0\left(\Omega_{\varepsilon_0}\right)$ - Note that $\mathrm{cof}\left(\Omega_{\varepsilon_0}\right) \neq \varepsilon_0$, but instead $\mathrm{cof}\left(\Omega_{\varepsilon_0}\right) = \omega$, as theres a sequence of length ω:

$$(\Omega_\omega, \Omega_{\omega^\omega}, \Omega_{\omega^{\omega^\omega}}, \ldots)$$

that reaches Ω_{ε_0}. So this is rule 4:

$$\psi_0\left(\Omega_{\varepsilon_0}\right)[0] = \psi_0\left(\Omega_{\varepsilon_0[0]}\right) = \psi_0(\Omega_1)$$
$$\psi_0\left(\Omega_{\varepsilon_0}\right)[1] = \psi_0\left(\Omega_{\varepsilon_0[1]}\right) = \psi_0(\Omega_\omega)$$
$$\psi_0\left(\Omega_{\varepsilon_0}\right)[2] = \psi_0\left(\Omega_{\varepsilon_0[2]}\right) = \psi_0(\Omega_{\omega^\omega})$$
$$\psi_0\left(\Omega_{\varepsilon_0}\right)[3] = \psi_0\left(\Omega_{\varepsilon_0[3]}\right) = \psi_0(\Omega_{\omega^{\omega^\omega}})$$
$$\vdots$$

4. $\psi_0(\Omega_\Omega)$ - This time $\mathrm{cof}(\Omega_\Omega) = \Omega > \omega$. So by rule 4, $\Omega_\Omega[\eta] = \psi_\Omega(0)[\eta] = \psi_{\Omega[\eta]}(0) = \psi_\eta(0)$ Then by rule 6b, $\mu = 0$ since $\Omega = \Omega_{0+1}$, then $\mathrm{cof}(\psi_0(\Omega_\Omega)) = \omega$, and:

$$\psi_0(\Omega_\Omega)[\eta] = \psi_0(\Omega_\Omega[\gamma[\eta]]) = \psi_0\left(\Omega_{\gamma[\eta]}\right)$$

, and since $\mu = \nu$:

$$\psi_0(\Omega_\Omega)[0] = \psi_0\left(\Omega_{\gamma[0]}\right) = \psi_0(\Omega_0) = \omega$$
$$\psi_0(\Omega_\Omega)[\eta+1] = \psi_0\left(\Omega_{\gamma[0]}\right) = \psi_0\left(\Omega_{\psi_0(\Omega_\Omega)[\eta]}\right)$$

So we get:

$$\psi_0(\Omega_\Omega)[0] = \psi_0(\Omega_0) = \omega$$
$$\psi_0(\Omega_\Omega)[1] = \psi_0(\Omega_\omega)$$
$$\psi_0(\Omega_\Omega)[2] = \psi_0\left(\Omega_{\psi_0(\Omega_\omega)}\right)$$
$$\psi_0(\Omega_\Omega)[3] = \psi_0\left(\Omega_{\psi_0\left(\Omega_{\psi_0(\Omega_\omega)}\right)}\right)$$
$$\vdots$$

10.2 Ordinal notation associated with EBOCF

Once again, very similar to normal Buchholz Ordinal Notation:

Definition 10.5.

Let T be the set of terms and PT be the set of all prinicpal terms (terms whose associated ordinal are additively principal ordinals).

- $0 \in T$
- Given a principal term $s \in PT$ and a term $t \in T$, $s + t \in T$
- Given terms $s, t \in T$, $D_s(t) \in T$ and PT

Let $\prec$ be a binary relation on T.

For terms $s, t \in T$, $s \prec t$ is defined as:

- If $s = 0$ then $s \prec t$ if $t \neq 0$
- If $t = 0$ then $s \prec t$ is false
- If $s = a + b$ where $a \in PT$ and $b \in T \setminus \{0\}$, then
  - If $t = c + d$ for some $c \in PT$ and $d \in T \setminus \{0\}$
    - If $a \neq c$ then $s \prec t$ if and only if $a \prec c$
    - If $a = c$ then $s \prec t$ if and only if $b \prec d$
  - If $t \in PT$ then $s \prec t$ if and only if $a \prec t$
- If $s = D_a(b)$ for some $a, b \in T$ then:
  - If $t = c + d$ for some $c \in PT$ and $d \in T \setminus \{0\}$, then $s \prec t$ if and only if $s \preceq c$
  - If $t = D_c(d)$ for some $c, d \in T$ then:
    - If $a \neq c$ then $s \prec t$ if and only if $a \prec c$
    - If $a = c$ then $s \prec t$ if and only if $b \prec d$

For terms s, t, u, we define the ternary relation $s \in \mathcal{C}_t(u)$ if and only if:

- $s \prec D_t(0)$
- $s = a + b$ for some $a \in PT$ and $b \in T \setminus \{0\}$ and $a \in \mathcal{C}_t(u)$ and $b \in \mathcal{C}_t(u)$
- $s = D_a(b)$ for some $a, b \in T$ and $a \in \mathcal{C}_t(u)$ and $b \in \mathcal{C}_t(u)$ and $b \prec u$

Let OT be a subset of T. OT corresponds to the set of all ordinal notations of normal form. For a term s, $s \in OT$ is defined as:

- If $s = 0$ then $s \in OT$
- If $s = a + b$ for some $a, b \in PT$, then $s \in OT$ if and only if $a, b \in OT$ and $b \preceq a$
- If $s = a + b + c$ for some $a, b \in PT$ and $c \in T \setminus \{0\}$ then $s \in OT$ if and only if $a, (b + c) \in OT$ and $b \preceq a$
- If $s = D_a(b)$ for some $a, b \in T$ then $s \in OT$ if and only if $a, b \in OT$ and $b \in \mathcal{C}_a(b)$

10.3 Extended Weak Buchholz/Nothing OCF

This is identical to the definition of EBOCF, except you **can't use addition**. The interesting about this function is that despite this apparent limitation, it can still reach the Extended Buchholz Ordinal.

Like most other OCFs, this is usually denoted with ψ, but when comparing with regular Buchholz it gets very confusing very quickly, so I'll denote it with p instead:

> **Definition 10.6 (Extended Weak Buchholz function).**
>
> Using the same definition of Ω_ν, we define the **Weak** Extended Buchholz's function p and the set $c_\nu(\alpha)$ as such:
>
> 1. $\Omega_\nu \subseteq c_\nu(\alpha)$
> 2. For any ordinal $\xi \in c_\nu(\alpha)$, as long as $\xi < \alpha$, then $p_\mu(\xi) \in c_\nu(\alpha)$ for all $\mu \in c_\nu(\alpha)$
> 3. $p_\nu(\alpha) = \min\{\gamma \in \text{Ord} \mid \gamma \notin c_\nu(\alpha)\}$, i.e., the smallest ordinal not inside $c_\nu(\alpha)$

We start with $p_0(0) = 1$, $p_0(1) = 2$, and in general, for $n \in \mathbb{N}$, $p_0(n) = n + 1$. However, we can't even reach ω, as $p_0(p_0(p_0(...))) = 1 + 1 + ...$ will always be a finite number.

Similarly to how ψ_0 gets "stuck" at ε_0, our weak p_0 gets "stuck" at ω, and the way to get "unstuck" is to find the smallest ordinal α that is greater than all natural numbers and satisfies $\alpha \in c_0(\alpha)$, which in this case is also $p_1(0) = \Omega$. So $p_0(\Omega) = \omega$.

But before we continue, lets iron out some properties about this new function

Properties

> I cooked all of this up myself so it might be jank

> **Proposition 10.7.**
>
> 1. $p_\nu(0) = \Omega_\nu$
> 2. $\Omega_\nu \leq p_\nu(\alpha) < \Omega_{\nu+1}$

Proof.

1. Follows from the definition
2. $\Omega_\nu \subseteq c_\nu(\alpha) \Rightarrow \Omega_\nu \leq p_\nu(\alpha)$, and the cardinality of $\Omega_{\nu+1}$ is greater than that of $c_\nu(\alpha)$. Hence there exists a $\gamma < \Omega_{\nu+1}$ such that $\gamma \notin c_\nu(\alpha)$, and therefore $\psi_\nu(\alpha) < \Omega_{\nu+1}$.

■

> **Proposition 10.8.**
>
> If $\alpha < \beta$ and $\alpha \in c_\nu(\beta)$, then $p_\nu(\alpha) < p_\nu(\beta)$.

Proof. We first note that if $\alpha \leq \beta$, we trivially have $c_\nu(\alpha) \subseteq c_\nu(\beta)$ and $p_\nu(\alpha) \leq p_\nu(\beta)$. Since $\alpha \in c_\nu(\alpha) \subseteq c_\nu(\beta)$, $p_\nu(\alpha) \in c_\nu(\beta)$ and $p_\nu(\alpha) < p_\nu(\beta)$. ■

> **Proposition 10.9 (Uniqueness of ordinal representation in p).**
>
> If $\gamma = p_{\nu_1}(\xi_1) = p_{\nu_2}(\xi_2)$, then $\nu_1 = \nu_2$ and $\xi_1 = \xi_2$

Proof. Follows from Proposition 10.7 ■

> **Proposition 10.10.**
>
> If $\alpha < \omega^{\Omega_\nu + 1}$ then $\alpha \in c_\nu(\alpha)$ and $p_\nu(\alpha) = \Omega_\nu + \alpha$.

Proof. By induction:

1. Base case: $p_\nu(0) = \Omega_\nu$

2. Successor case: Suppose $\alpha \in c_\nu(\alpha)$ and $p_\nu(\alpha) = \Omega_\nu + \alpha$ (and obviously since $\alpha < \omega^{\Omega_\nu+1} \Rightarrow \alpha + 1 < \omega^{\Omega_\nu+1}$)

   If $\alpha \in c_\nu(\alpha)$, then since $\alpha < \alpha + 1$, we have $p_\nu(\alpha) \in c_\nu(\alpha + 1)$.

   Nothing else new can be produced (yes we can have $p_{\nu+1}(\alpha) \in c_\nu(\alpha + 1)$ but there exists smaller ordinals not contained within $c_\nu(\alpha)$)

   $p_\nu(\alpha + 1)$ is the smallest ordinal not in $c_\nu(\alpha + 1)$, which is $\psi_\nu(\alpha) + 1 = \Omega_\nu + \alpha + 1$ from the induction hypothesis.

   Now we show that: $\alpha + 1 \in c_\nu(\alpha + 1)$

   Since $\alpha > 0$ (otherwise refer to the base case), $\alpha = p_\mu(\xi)$ for some $\mu, \xi \in c_\nu(\alpha)$. Since $\xi < \alpha$, we have $\alpha = p_\mu(\xi) = \Omega_\mu + \xi$ and $\xi + 1 \in c_\nu(\xi + 1)$ from the induction hypothesis, and $\alpha + 1 = \Omega_\mu + \xi + 1 = p_\mu(\xi + 1) \in c_\nu(\alpha)$.
3. Limit case: for some limit ordinal $\alpha < \omega^{\Omega_\nu+1}$, suppose $\beta \in c_\nu(\beta)$ and $p_\nu(\beta) = \Omega_\nu + \beta$ for all $\beta < \alpha$.
   - If $\alpha < \Omega_\nu$, $\alpha \in \Omega_\nu \subseteq c_\nu(\alpha)$
   - If $\alpha = \Omega_\nu > 0$, $p_\nu(0) \in c_\nu(\alpha)$ since $0 < \alpha$
   - If $\Omega_\nu < \alpha < \omega^{\Omega_\nu+1}$:

     For all $\beta \in c_\nu(\beta) \subseteq c_\nu(\alpha)$, and as such for all $\beta < \alpha$ we have $p_\nu(\beta) = \Omega_\nu + \beta \in c_\nu(\alpha)$.

   Then

$$\begin{aligned} p_\nu(\alpha) &= \overset{+}{\sup}\{p_\nu(\beta) \mid \beta < \alpha \text{ and } \beta \in c_\nu(\beta) \subseteq c_\nu(\alpha)\} \\ &= \overset{+}{\sup}\{\Omega_\nu + \beta \mid \beta < \alpha \text{ and } \beta \in c_\nu(\beta) \subseteq c_\nu(\alpha)\} \\ &= \Omega_\nu + \alpha \end{aligned}$$

■

Conjecture 10.11.
$p_0(\Omega_\xi) = \psi_0(\xi)$

Analysis

Weak Buchholz	**Regular Buchholz**
$p_0(0)$	$1 = \psi_0(0)$
$p_0(1) = p_0(p_0(0))$	$2 = \psi_0(0) + \psi_0(0)$
$p_0(2) = p_0(p_0(p_0(0)))$	$3 = \psi_0(0) + \psi_0(0) + \psi_0(0)$
$p_0(\Omega) = p_0(p_1(0))$	$\omega = \psi_0(1)$
$p_0(\Omega + 1) = p_0(p_1(p_0(0))$	$\omega + 1 = \psi_0(1) + \psi_0(0)$
$p_0(\Omega 2) = p_0(p_1(p_1(0))$	$\omega 2 = \psi_0(1) + \psi_1(0)$
$p_0(\Omega 3) = p_0(p_1(p_1(p_1(0)))$	$\omega 3 = \psi_0(1) + \psi_1(0) + \psi_1(0)$
$p_0(\Omega_2) = p_0(p_2(0))$	$\omega^2 = \psi_0(2)$
$p_0(\Omega_\omega) = p_0\left(p_{p_0(\Omega)}(0)\right) = p_0\left(p_{p_0(p_1(0))}(0)\right)$	$\omega^\omega = \psi_0(\omega) = \psi_0(\psi_0(1))$
$p_0(\Omega_{\omega+1}) = p_0\left(p_{p_0(\Omega+1)}(0)\right) = p_0\left(p_{p_0(p_1(p_0(0)))}(0)\right)$	$\omega^{\omega+1} = \psi_0(\omega + 1)$

Weak Buchholz	**Regular Buchholz**
$p_0(\Omega_{\omega 2}) = p_0\left(p_{p_0(\Omega 2)}(0)\right) = p_0\left(p_{p_0(p_1(p_1(0)))}(0)\right)$	$\omega^{\omega 2} = \psi_0(\omega 2)$
$p_0(\Omega_{\omega^2}) = p_0\left(p_{p_0(\Omega_2)}(0)\right) = p_0\left(p_{p_0(p_2(0))}(0)\right)$	$\omega^{\omega^2} = \psi_0(\omega^2)$
$p_0(\Omega_{\omega^\omega}) = p_0\left(\Omega_{p_0(\Omega_\omega)}\right) = p_0\left(\Omega_{p_0\left(\Omega_{p_0(\Omega)}\right)}\right)$	$\omega^{\omega^\omega} = \psi_0(\omega^\omega)$
$p_0(\Omega_{\omega^{\omega^\omega}}) = p_0\left(\Omega_{p_0(\Omega_{\omega^\omega})}\right) = p_0\left(\Omega_{p_0\left(\Omega_{p_0\left(\Omega_{p_0(\Omega)}\right)}\right)}\right)$	$\omega^{\omega^{\omega^\omega}} = \psi_0(\omega^{\omega^\omega})$
$p_0(\Omega_\Omega) = p_0\left(\Omega_{p_1(0)}\right)$	$\varepsilon_0 = \psi_0(\Omega)$
$p_0(\Omega_{\Omega+1}) = p_0\left(\Omega_{p_1(0)}\right)$	$\varepsilon_0 = \psi_0(\Omega)$

10.4 Address Notation

10.5 Sudden Sequence System